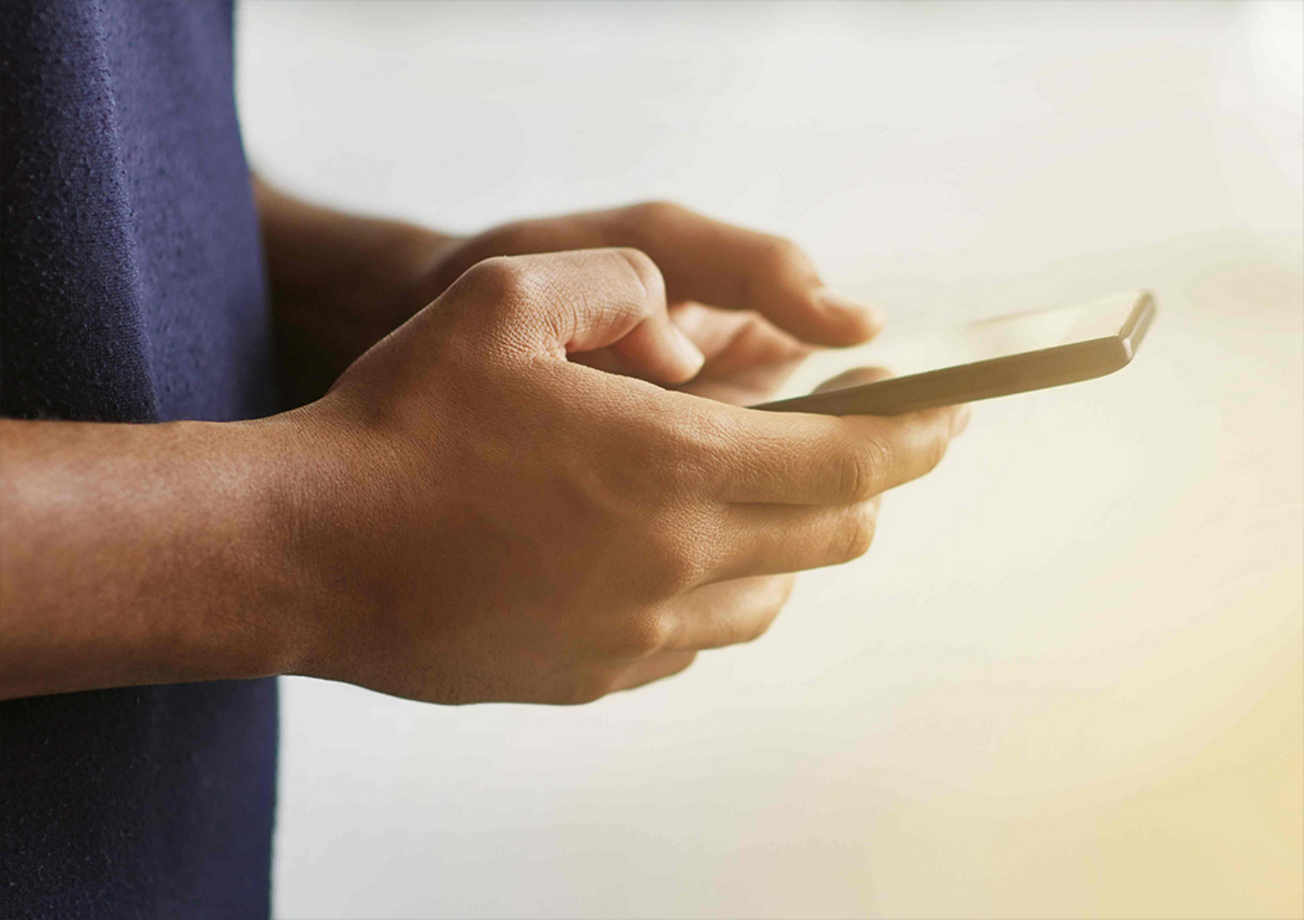


PUBLIC
2026-10-13

SAP AI Core

© 2026 SAP SE or an SAP affiliate company. All rights reserved.

Content

1 What Is SAP AI Core?. . . . 4
1.1 Metering and Pricing for SAP AI Core. . . . 7
Metering and Pricing for Generative AI. . . . 8
Metering and Pricing for Predictive AI. . . . 12

2 What's New for SAP AI Core 16

3 Concepts. . . . 113
3.1 AI API Overview. . . . 120
AI API Runtime Implementations. . . . 120
3.2 Resource Groups. . . . 123
Scope of Resources. . . . 124
3.3 Generative AI Hub in SAP AI Core Overview. . . . 125

4 Initial Setup. . . . 127
4.1 Enabling the Service in Cloud Foundry. . . . 127
Create a Subaccount. . . . 128
Enable Cloud Foundry. . . . 130
Create a Space. . . . 131
Add a Service Plan. . . . 132
Create a Service Instance. . . . 136
Create a Service Key. . . . 141
Use a Service Key. . . . 144
4.2 Enabling the Service in the Kyma Environment. . . . 147

5 Tutorials. . . . 148

6 APIs and API Extensions. . . . 149

7 Libraries and SDKs. . . . 150

8 Content Packages. . . . 152

9 Advanced Features. . . . 153
9.1 AI Content as a Service. . . . 153
Service Custom Resource. . . . 155
Getting Started as a Service Provider. . . . 157
Metering. . . . 159
Offboarding. . . . 159
Shared Resource Group. . . . 160

10 Security. 163
10.1 Security Features of Data, Data Flow, and Processes. 163
10.2 Encryption in Transit. 163
10.3 Encryption of Data. 164
10.4 Authentication and Administration. 164
10.5 Docker Images. 165
10.6 AI Content Security. 165
10.7 Kubernetes Security. 166
10.8 Configuration Data and Secrets. 167
10.9 Output Encoding. 167
10.10 Multitenancy. 168
10.11 Auditing and Logging Information. 169
10.12 Data Protection and Privacy. 171
Data Storage and Processing. 172
Change Logging and Read-Access Logging. 172
Consent. 172
Deletion. 173
Security and Customer Data Protection. 173

11 Accessibility Features in SAP AI Core. 174

12 Monitoring and Troubleshooting. 175
12.1 Troubleshooting. 175
Repository. 176
Configuration. 177
Artifacts. 179
Application. 181
Execution. 186
Docker. 188
Deployment. 189
Miscellaneous. 190

13 Support Process. 193

14 Service Offboarding. 194

1 What Is SAP AI Core?

Learn more about the SAP AI Core service on SAP Business Technology Platform (SAP BTP). Build a platform for your artificial intelligence solutions.

SAP AI Core is a service within the SAP Business Technology Platform. It's designed to manage the execution and operations of AI assets in a standardized, scalable, and hyperscaler-agnostic manner. It seamlessly integrates with SAP solutions, allowing any AI function to be easily implemented using open-source frameworks. SAP AI Core supports full lifecycle management of AI scenarios. Users can access generative AI capabilities and prompt lifecycle management through the generative AI hub.

SAP AI Core lets you make data-driven decisions confidently and efficiently, tailored to address business challenges. It handles large volumes of data and offers scalable machine learning capabilities to automate tasks like triaging customer feedback or tickets and performing classification tasks. SAP AI Core includes preconfigured SAP solutions and supports open-source machine learning frameworks. It integrates with Argo Workflow and KServe, and can be embedded into other applications.

SAP AI Core allows you to experiment with and utilize natural language prompts with a variety of generative AI models in the generative AI hub.

> ⓘ Note
>
> SAP AI Core, including generative AI hub must **not** be used to generate synthetic data for training or fine-tuning models. Any exceptions must be explicitly approved and documented.

> → Tip
>
> The English version of this guide is open for contributions and feedback using GitHub. This allows you to get in contact with responsible authors of SAP Help Portal pages and the development team to discuss documentation-related issues. To contribute to this guide, or to provide feedback, choose the corresponding option on SAP Help Portal:
>
> - *Feedback* > *Create issue*: Provide feedback about a documentation page. This option opens an issue on GitHub.
> - *Feedback* > *Edit page*: Contribute to a documentation page. This option opens a pull request on GitHub.
>
> You need a GitHub account to use these options.
>
> More information:
>
> - Contribution Guidelines
> - Introduction Video
> - Introduction Blog Post

Features

Generative AI hub	Access generative AI models for prompt development and experimentation. Manage the lifecycle and evaluation of prompts and generative AI workflows.
Execute pipelines	Execute pipelines as a batch job, for example, to preprocess or train your models, or perform batch inference.
Serve inference requests	Deploy a trained machine learning model as a Web service to serve inference requests of trained models with high performance.
Manage the AI scenario lifecycle	Manage your ML artifacts and workflows, such as model training, metrics tracking, data, models, and model deployments via a uniform API lifecycle.
Benefit from multitenancy support	Use this service in tenant-aware applications. Implement multi-tenant services to segregate your AI assets and executions to isolate your tenants within SAP AI Core.
Integrate your cloud infrastructure	Register your Docker registry, synchronize your AI content from your git repository, and register your object store for training data and trained models. Productize your AI content and expose it as a service to consumers in the SAP BTP marketplace.

Environment

This service is available in the following environments:

- Cloud Foundry
- Kyma
- Kubernetes

Multitenancy

This service supports multitenancy. It can be used in tenant-aware applications. For more information, see Multitenancy [page 168].

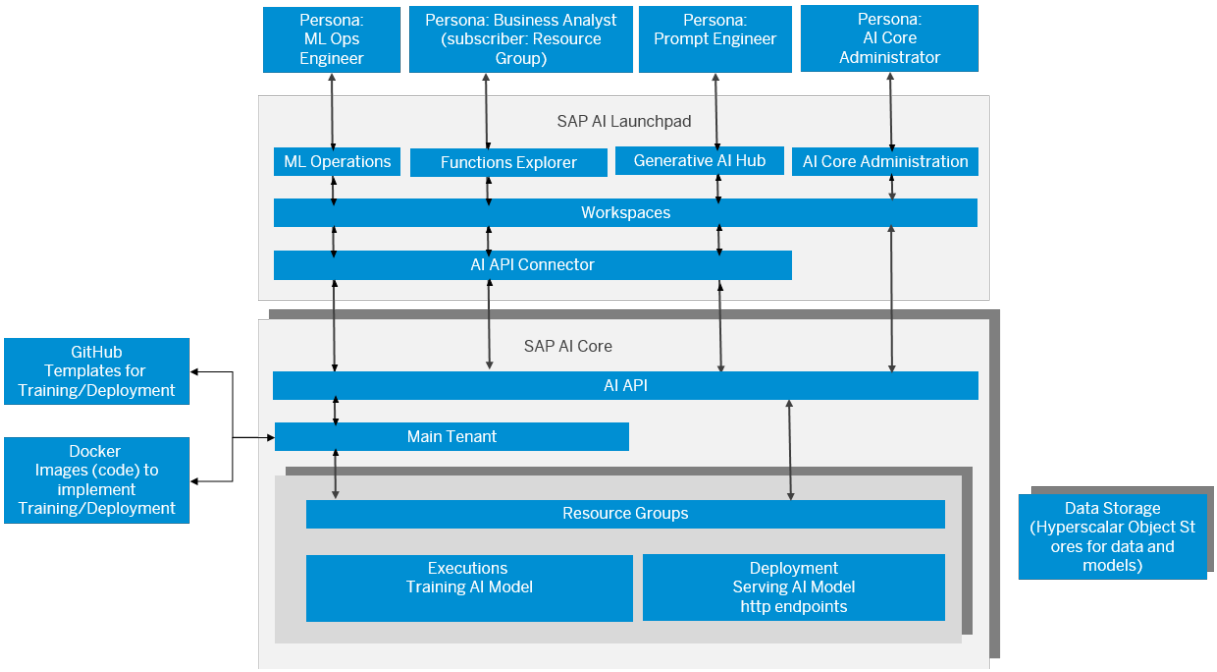
Architectural Overview

The SAP AI Core service works in conjunction with SAP AI Launchpad and the `AI API`. The key components are SAP AI Core, SAP AI Launchpad, and the `AI API`.

- **SAP AI Core** provides an engine that lets you run AI workflows and model serving workloads.
- **SAP AI Launchpad** manages a number of AI runtimes. It allows various user groups to access and manage their AI scenarios.

- **AI API** provides a standard way of managing the AI scenario lifecycle on different runtimes, regardless of whether they're provided on SAP technology (such as SAP S/4HANA) or on partner technology (such as Amazon Web Services). When the AI API is deployed on runtimes other than SAP AI Core, the runtimes have to provide a runtime adapter.

Architectural Overview [page 6] shows these three main components in an overview architecture diagram.



Architectural Overview

Tools

SAP AI Core connects with various internal and external tools. You interact with different repositories, systems, and objects. Some of these objects come from SAP, while others must be provided by you. This setup enables enhanced control through authorizations, and supports continuous integration and continuous deployment (CI/CD). The following table lists the key repositories, systems, and objects:

What	Why
AI API	For managing your artifacts and workflows (such as training scripts, data, models, and model servers) across multiple runtimes **Note** The AI API can also be used to integrate other machine learning platforms, engines, or runtimes into the AI ecosystem.
Argo Workflows	A container native, workflow engine for Kubernetes.
Docker repo	For custom Docker images referenced in the templates
Git repo	For storing training and serving workflows and templates

What	Why
Hyperscaler storage	For storage of input and output artifacts, such as training data and models (for example, SAP BTP Object Store Service)
KServing (K)	For optimized deployments of machine learning models. Deployment templates use KServe notation.
Kubernetes (K8s)	The K8s cluster orchestrates and scales the pods, which are used in AI pipelines. Resource group isolation is based on a K8s namespace.
SAP AI Launchpad	SAP AI Launchpad is a multitenant software as a service (SaaS) application on SAP Business Technology Platform (SAP BTP). Customers and partners can use SAP AI Launchpad to manage AI use cases (scenarios) across multiple instances of AI runtimes (such as SAP AI Core). SAP AI Launchpad also provides generative AI capabilities via the Generative AI Hub.

Related Information

Generative AI Hub
AI API Overview [page 120]
SAP AI Launchpad

1.1 Metering and Pricing for SAP AI Core

SAP AI Core provides a scalable infrastructure for AI model management, with usage-based pricing that lets you pay only for the resources you use.

The service offers various plans, including a standard, and extended plan. If you want to access all capabilities, we recommend the extended plan.

We offer the following model options:

- Generative AI with SAP-hosted, SAP-managed, and remote foundation models, and pipelines such as grounding
- Predictive AI with your custom AI models

You have the flexibility to select the model options that best suit your needs.

Custom AI models incur compute, storage, and baseline costs. The system automatically applies baseline charges every hour, up to a maximum of 730 hours.

If you use foundation models and grounding, the platform waives compute, storage, and baseline costs. Instead, charges accrue based on the number of tokens used by the foundation models.

More Information

For more information about the potential costs associated with using foundation models and grounding in SAP AI Core, including metering information, example pricing calculations, and typical consumption patterns, see Metering and Pricing for Generative AI [page 8].

For more information about the potential costs associated with using custom AI models in SAP AI Core, including metering information, example pricing calculations, and typical consumption patterns, see Metering and Pricing for Predictive AI [page 12].

Consumption Information

To see detailed consumption information in SAP BTP cockpit, open your global account and click *Usage* from the navigation panel. Filter by AI Core and click Export. The report contains generative AI hub model consumption under *Applications* and resource group level consumption under *instance*.

For more information, see Monitoring Usage and Consumption Costs in Your Global Account in SAP BTP.

1.1.1 Metering and Pricing for Generative AI

The use of generative AI in SAP AI Core is measured using tokens, which are converted into capacity units based on input and output volume. Billing is calculated according to these capacity units and varies by model type and usage pattern.

> **Note**
>
> The generative AI hub is available only as part of the extended service plan.

Metering for Generative AI

Use of large language models (LLMs) is metered in tokens. Tokens are the fundamental building blocks of text that a language model processes. They represent single characters, parts of words, entire words, or even punctuation marks. They're the unit into which the LLM breaks down text input and output.

It's important to distinguish between input and output tokens. Input tokens refer to the text provided in the prompt, while output tokens represent the response generated by the LLM. The ratio between input and output tokens varies depending on the prompt and the use case. For example, in summarization tasks, the ratio of input to output tokens could be 3:1 or even 4:1. In contrast, tasks like generating explanations or drafting longer pieces of text could have an inverse ratio, with output tokens significantly outnumbering input tokens.

The number of tokens required depends on the content and complexity of your prompt and the specific task assigned to the LLM. Token consumption varies with the capabilities of the LLM being used, with more advanced models typically requiring and processing a higher number of tokens.

Consumption of generative AI in SAP AI Core is measured in GenAI tokens. GenAI tokens are a virtual unit used to represent usage across different models. The actual number of processed tokens (input and output) that corresponds to one GenAI token varies by model.

Generative AI resource requirements vary based on the use case. Available resources are shown in the table:

Resource Type	SAP AI Core Resources	Unit of Measure (UoM)
Provisioning of foundation models	Access to (LLMs) and other generative AI capabilities	Tokens
Baseline	Grounding Service	Gigabyte day
Inference Observability	Storing inference records	DataVolume

For more information on inference observability, including conversion rates between inference records and DataVolume units, see 3720903.

Pricing for Generative AI

The billing metric for SAP BTP is referred to as “capacity units” (CUs). Capacity units are calculated using conversion factors applied to GenAI tokens.

In general, output tokens tend to be slightly more expensive than input tokens.

For more information about the supported models, including the conversion rates between model tokens and GenAI tokens, see SAP Note 3437766.

Here's an example estimation, using fictitious values:

Example

This example contains:

- A retrieval-augmented generation (RAG) system handling 25,000 requests in a single month results in the following costs:
- Language model use: The total of 25,000 requests, each with 3,500 input tokens and 300 output tokens.
- Grounding service: The RAG pipeline retrieves and processes context using the following orchestration modules:
  - Storage in Grounding (for storing relevant documents)
  - Retrieval in Grounding (for fetching context during inference)
  - Content Filtering
  - Data Masking (for privacy and compliance)
- Inference Observability: The 25,000 inference requests and responses are stored with associated metadata, labels and feedback for later analysis.

The following orchestration services are essential to support the grounding step and introduce additional capacity usage beyond the core LLM calls.

- The Grounding storage module stores contextual documents in the vector engine.
- The Grounding retrieval module retrieves context documents during inference.
- Content Filtering via Meta Llama Guard helps to ensure content safety and relevance.
- Data Masking, handled by SAP Data Privacy Integration, applies masking at the API level.

Each module uses its own metric (storage size, text blocks, API calls) multiplied by monthly usage, then converted to capacity units to estimate resource consumption and cost.

Calculate the calls incurred through LLM calls:

This example uses a model with an input GenAI token conversion rate of 0.00112 per 1,000 input tokens. The output GenAI token conversion rate is 0.00320 per 1,000 output tokens.

1. Calculate Total GenAI tokens
   Formula: Total GenAI tokens = (x/1000 × conversion rate per 1,000 input tokens) + (y/1000 × conversion rate per 1000 output tokens)
   In this formula, "x" is the number of input tokens and "y" is the number of output tokens. The conversion rates per 1,000 input and output tokens are given constants.
   This example includes 3,500 input tokens and 300 output tokens. The conversion rate is 0.00112 per 1,000 input tokens and 0.00320 per 1,000 output tokens.
   Values: Total GenAI Tokens = (3500/1000 × 0.00112) + (300/1000 × 0.00320) = 0.00488
2. Calculate capacity units
   Formula: capacity units = Total GenAI Tokens × 1.90385
   In this formula, "Total GenAI Tokens" is the value calculated in the previous step and 1.90385 the capacity unit value, given as a constant.
   Values: capacity units = 0.00488 × 1.90385 = 0.00929
3. Calculate Total Usage for handling 25,000 requests
   Formula: Total capacity units = Requests × capacity units per request
   In this formula, "Requests" is the number of requests to be handled (in this case, 25,000) and "capacity units per request" is the value calculated in the previous step.
   Values: Total capacity units = 25,000 × 0.00929 = 232.25 CU

Calculate Grounding Module Costs:

The RAG architecture needs various supporting services to handle contextual data for the language model. These services use resources measured by specific metrics, which are converted into capacity units based on usage volume and internal conversion factors

Each module uses its own metric (storage size, text blocks, API calls) multiplied by monthly usage. The values are converted to capacity units to quantify resource consumption and estimate cost.

Module	Service	Metric	Metric Value	Occurrences / Month	Capacity Units / Month
Grounding	Storage	Storage in GB / Day	1	30	6.2 CU
Grounding	Retrieval	Text Blocks	1	25.000	30 CU
Content Filter	Meta Llama Guard	Text Blocks	5	25.000	84 CU

Module	Service	Metric	Metric Value	Occurrences / Month	Capacity Units / Month
Data Masking	SAP Data Privacy Integration	API Calls	1	25.000	66.1 CU
Total					186.3 CU

Total consumption = 232.25 + 186.3 = 418.55 CU

Calculate Inference Observability Costs:

This example assumes that an inference record consumes 0.002048 MB per day. We assume all 25,000 records are created at day one and are kept for a month.

1. Calculate total storage consumption
   Formula: Total storage consumption per day = Number of inference records * 0.002048 MB
   Values: Total storage consumption per day = 25,000 * 0.002048 MB = 51.2 MB
2. Calculate data volume units per day
   Formula: data volume units per day = total storage consumption per day (in MB) * 0.00089
   Values: data volume units per day = 51.2 *0.00089 = 0.045568
3. Calculate capacity units per month
   Formula: capacity units per month = data volume units per day * 30
   Values: capacity units per month = 0.045568 * 30 = 1.36704 CU

Images

For image use with Mistral Small Instruct, images are converted into tokens in batches of 14*14 pixels. The formula for the number of image tokens is: (ResolutionX/14) * (ResolutionY/14). Images with a resolution higher than 1540*1540 are downscaled, while maintaining their aspect ratio.

Example

This example uses fictitious values.

Given an image of resolution 720*512, the number of GenAI tokens consumed through the image is:

GenAI Image tokens = (720/14)*(512/14) = 1,880

The calculation for image tokens also includes additional sequence start and end tokens, as well as tokens for image breaks.

Typical Consumption Patterns

The following table displays API request counts for productive IT use cases based on the consumption patterns “small”, “medium”, and “large”. It also specifies the average distribution of input and output tokens, which can help you estimate potential costs for consuming foundation models.

Use Case	Input Tokens	Output Tokens	Tokens Occurrences / Request per Month: Small	Tokens Occurrences / Request per Month: Medium	Tokens Occurrences / Request per Month: Large
RAG Chat	3500	300	1000	25k	300k
Basic Chat	500	100	1000	30k	300k
Summarization	5000	300	1000	25k	300k
Classification	3800	10	1000	5k	50k
Generation	500	3500	1000	2k	50k

The generative AI hub provides additional modules through the Orchestration API, including:

- Grounding
- Content Filtering
- Data Masking

Charges associated with the use of other SAP AI Core components can also apply. For more information, see Metering and Pricing for Predictive AI [page 12].

> → Recommendation
>
> For an estimate of projected costs, use the SAP AI Core Cost Calculator.

1.1.2 Metering and Pricing for Predictive AI

Metering for Predictive AI

Custom AI workloads involve diverse resource types, each with varying performance and efficiency levels. Measuring key resources like compute power, storage, and baseline costs is crucial. Non-billable units of measure (UoM) are used to represent the consumption of these resources.

or example, compute-heavy AI models on high-performance instances consume more resources per hour than smaller, less demanding workloads. This results in proportionally higher costs. Pricing reflects true resource consumption.

Custom AI development requirements vary significantly based on the use case. A range of compute infrastructure resources is available, differing in CPU cores, memory (GB), and GPU presence for GPU-powered workloads like model training. Available resources are shown in the table.

> ⓘ Note
>
> Infrastructure specifications depend on the chosen hyperscaler. For more information, see Choose an Instance.

Resource Type		SAP AI Core Resources	Unit of Measure (UoM)
Compute	Instance Types	For information about available instance types, see SAP Note 3660109.	Node hour
	Resource Plans	Starter Instance (min. 2.5 GB, 1 CPU core)	
		Basic Instance (min. 10 GB, 3 CPU cores)	
		Basic.8x Instance (min. 115 GB, 31 CPU cores)	
		Infer-S Instance (min. 10 GB, 3 CPU cores, standard GPU)	
		Infer-M Instance (min. 25 GB, 7 CPU cores, standard GPU)	
		Infer-L Instance (min. 55 GB, 15 CPU cores, standard GPU)	
		Train-L Instance (min. 47 GB, 5 CPU cores, advanced GPU)	
Storage		Standard SSD Volume	Gigabyte hour
Baseline		Fixed cluster resources (instances, storage, SAP HANA, Kubernetes, DevOps, support, and so on)	Tenant hour

Custom AI models can incur compute, storage, and baseline costs. You have the flexibility to select the components that best suit your needs. If you use compute or storage resources, the system automatically applies baseline charges every hour, up to a maximum of 730 hours.

> **Note**
>
> You can process data in parallel by using multiple nodes. Costs are calculated in node hours. For example, processing data for 1 hour on 2 nodes equals 2 node hours. However, it incurs only 1 hour of baseline costs.

Pricing for Predictive AI

The billing metric for SAP BTP is referred to as “capacity units”. Capacity units are calculated using conversion factors applied to compute, storage, and baseline intermediary UoM values. This allows you to calculate a specific price for the infrastructure used based on your compute and storage requirements.

Here's an example calculation, using fictitious values:

Example

Training a small model on a GPU node and serving it on a CPU node later in the same month results in the following costs:

- Baseline: Total node hours of compute instances across the full month
- Storage: Temporary high-speed storage (for example, SSD) used during training and inference to stream or copy data
- Compute: Compute consumption based on node hours and chosen compute instances across the full month

Resource Type	Unit of Measure	Capacity Unit (CU) Value	Fictional Usage	Conversion to CU
Compute	node hour	0.3900	300 node hours in Basic Instance	300 * 0.39 = 117 CU
Compute	node hour	1.552	100 node hours in Infer-M Instance	100 * 1.552 = 155.2 CU
Storage	gigabyte hour	0.0003	10.000 gigabyte hours	0.0003*5,000 = 1.5 CU
Baseline	tenant hour	1.2241	400 baseline tenant hours	1.2241*400 = 489.64 CUs

Infrastructure specifications are combined minimum values; actual numbers vary by deployed hyperscaler and may be higher.

The CUs can be converted to monetary values in the BTP Discovery Center Estimator. For more information, see BTP Discovery Center Estimator.

→ Recommendation

For an estimate of projected costs, use the SAP AI Core Cost Calculator.

Typical Consumption Patterns

Depending on your infrastructure and your consumption pattern (S, M or L), you need on average the following node hours for compute, storage, and baseline hours.

T Shirt Size	Starter	Basic	Basic8x	Infer.S	Infer.M	Infer.L	Train.L	Storage	Baseline
S	50	30	0	300	0	0	50	0	430

T Shirt Size	Starter	Basic	Basic8x	Infer.S	Infer.M	Infer.L	Train.L	Storage	Baseline
M	0	900	0	160	1880	1440	120	0	730
L	0	0	0	800	10000	0	2300	0	730

2 What's New for SAP AI Core

Component	Environment	Title	Description	Action	Lifecycle	Type	Line of Business	Modular Business Process	Product	Latest Revision	Available as of	Version	Scope
SAP AI Core	Cloud Foundry	Generative AI hub	New models are supported, including Gemini 3.8 Flash. For more information, see SAP Note 3437766.	Info only	General Availability	Announcement	Technology	Not applicable	SAP Business Technology Platform	2026-09-29	2026-09-29	Unavailable	Not for sovereign cloud offerings

Component	Environment	Title	Description	Action	Lifecycle	Type	Line of Business	Modular Business Process	Product	Latest Revision	Available as of	Version	Scope
SAP AI Core	Cloud Foundry	Generative AI hub	`SAP-RPT-1.6` launched. SAP-RPT-1.6 is SAP's relational pretrained transformer model for tabular enterprise data that delivers accurate predictions using in-context learning without requiring model training. It supports classification and regression use cases with built-in explainability, governance, and enterprise-grade orchestration capabilities. RPT-1.6 supports Explainability feature. For more information, see SAP-RPT-1.6 and SAP RPT 1.6 Explainability.	Info only	General Availability	Announcement	Technology	Not applicable	SAP Business Technology Platform	2026-09-07	2026-09-07	Unavailable	Not for sovereign cloud offerings

Component	Environment	Title	Description	Action	Lifecycle	Type	Line of Business	Modular Business Process	Product	Latest Revision	Available as of	Version	Scope
SAP AI Core	Cloud Foundry	Generative AI hub	Azure Content Safety Protected Material Detection for Code has a minimum input length requirement. This limitation is defined by Azure and cannot be modified. For more information, see Content Filtering and Streaming.	Info only	General Availability	Announcement	Technology	Not applicable	SAP Business Technology Platform	2026-09-29	2026-09-29	Unavailable	Not for sovereign cloud offerings

Component	Environment	Title	Description	Action	Lifecycle	Type	Line of Business	Modular Business Process	Product	Latest Revision	Available as of	Version	Scope
SAP AI Core	Cloud Foundry	Generative AI hub	The following updates have been made to the V2 orchestration workflow: • The `provider_specific_fields` field includes information returned by providers that has no direct equivalent in the OpenAI Chat Completions schema. • You can use the `model.param.tools` field to implement dynamic tool assembly at runtime	Info only	General Availability	Announcement	Technology	Not applicable	SAP Business Technology Platform	2026-09-29	2026-09-29	Unavailable	Not for sovereign cloud offerings

Component	Environment	Title	Description	Action	Lifecycle	Type	Line of Business	Modular Business Process	Product	Latest Revision	Available as of	Version	Scope
			without requiring tools to be predefined in a prompt template. For more information, see Provider-Specific Fields and Defining Tools through Model Parameters.										
SAP AI Core	Cloud Foundry	Generative AI hub	Added support for data encryption to help protect sensitive information at rest and in transit, enhancing data security and compliance. For more information, see Encryption of Data [page 164].	Info only	General Availability	Announcement	Technology	Not applicable	SAP Business Technology Platform	2026-09-07	2026-09-07	Unavailable	Not for sovereign cloud offerings

Component	Environment	Title	Description	Action	Lifecycle	Type	Line of Business	Modular Business Process	Product	Latest Revision	Available as of	Version	Scope
SAP AI Core	Cloud Foundry	Generative AI hub	New models are supported, including Gemini 3.5 Flash Lite. For more information, see SAP Note 3437766.	Info only	General Availability	Announcement	Technology	Not applicable	SAP Business Technology Platform	2026-09-07	2026-09-07	Unavailable	Not for sovereign cloud offerings
SAP AI Core	Cloud Foundry	Generative AI hub	Azure OpenAI models use a global content filter that blocks content classified as medium or high severity. Improved descriptions of safety capabilities, and a more streamlined content structure.	Info only	General Availability	Announcement	Technology	Not applicable	SAP Business Technology Platform	2026-09-07	2026-09-07	Unavailable	Not for sovereign cloud offerings

Component	Environment	Title	Description	Action	Lifecycle	Type	Line of Business	Modular Business Process	Product	Latest Revision	Available as of	Version	Scope
SAP AI Core	Cloud Foundry	Generative AI hub	Tabular AI Orchestration enables the consumption of tabular AI models through a unified prediction endpoint. For more information, see Consume Tabular Orchestration	Info only	General Availability	Announcement	Technology	Not applicable	SAP Business Technology Platform	2026-09-07	2026-09-07	Unavailable	Not for sovereign cloud offerings
SAP AI Core	Cloud Foundry	Generative AI hub	Users with SFTP pipelines configured with `ssh-dss` keys must update their configuration with another key type. Key type `ED25519` is recommended. For more information, see Grounding Generic Secrets for SFTP.	Info only	General Availability	Announcement	Technology	Not applicable	SAP Business Technology Platform	2026-08-13	2026-08-13	Unavailable	Not for sovereign cloud offerings

Component	Environment	Title	Description	Action	Lifecycle	Type	Line of Business	Modular Business Process	Product	Latest Revision	Available as of	Version	Scope
SAP AI Core	Cloud Foundry	Generative AI hub	New models are supported, including Gemini Text Embedding 2 and GPT-5.6-Sol, Terra and Luna. For more information, see SAP Note 3437766.	Info only	General Availability	Announcement	Technology	Not applicable	SAP Business Technology Platform	2026-08-13	2026-08-13	Unavailable	Not for sovereign cloud offerings

Component	Environment	Title	Description	Action	Lifecycle	Type	Line of Business	Modular Business Process	Product	Latest Revision	Available as of	Version	Scope
SAP AI Core	Cloud Foundry	Generative AI hub	The orchestration content filtering module no longer includes the content of `system` and `developer` messages when sending requests to Azure Prompt Shield (prompt attack detection). As a result, requests that were previously filtered only because of prompt-attack-like text in a `system` or `developer` message are no longer filtered. Other content filters are unaffected. For more information, see Content Filtering.	Info only	General Availability	Changed	Technology	Not applicable	SAP Business Technology Platform	2026-08-13	2026-08-13	Unavailable	Not for sovereign cloud offerings

Component	Environment	Title	Description	Action	Lifecycle	Type	Line of Business	Modular Business Process	Product	Latest Revision	Available as of	Version	Scope
SAP AI Core	Cloud Foundry	Generative AI hub	Prompt templates and orchestration configs support pagination. This is useful when you have large numbers of templates and configs, and want to retrieve them in pages. For more information, see Get a Prompt Template, Get Prompt Template History, Get Orchestration Configsand Get Orchestration Config History.	Info only	General Availability	Announcement	Technology	Not applicable	SAP Business Technology Platform	2026-08-13	2026-08-13	Unavailable	Not for sovereign cloud offerings

Component	Environment	Title	Description	Action	Lifecycle	Type	Line of Business	Modular Business Process	Product	Latest Revision	Available as of	Version	Scope
SAP AI Core	Cloud Foundry	Dataset API	Alicloud OSS and OpenStack Ceph object stores are now supported by the Dataset API. For more information, see Register an Object Store Secret.	Info only	General Availability	Announcement	Technology	Not applicable	SAP Business Technology Platform	2026-08-13	2026-08-13	Unavailable	Not for sovereign cloud offerings
SAP AI Core	Cloud Foundry	Generative AI hub	New models are supported, including Mistral Medium 3.5. For more information, see SAP Note 3437766.	Info only	General Availability	Announcement	Technology	Not applicable	SAP Business Technology Platform	2026-07-27	2026-07-27	Unavailable	Not for sovereign cloud offerings

Component	Environment	Title	Description	Action	Lifecycle	Type	Line of Business	Modular Business Process	Product	Latest Revision	Available as of	Version	Scope
SAP AI Core	Cloud Foundry	Incremental improvements	We've made the following improvements • IP addresses cannot be used as Git repository URLs. • We've harmonized the error messages across orchestration.	Info only	General Availability	Announcement	Technology	Not applicable	SAP Business Technology Platform	2026-07-27	2026-07-27	Unavailable	Not for sovereign cloud offerings

Component	Environment	Title	Description	Action	Lifecycle	Type	Line of Business	Modular Business Process	Product	Latest Revision	Available as of	Version	Scope
SAP AI Core	Cloud Foundry	Generative AI hub	`SAP-RPT-1.5` launched. SAP-RPT-1.5 is a relational pre-trained transformer for prediction tasks on structured and relational data. It's developed and maintained by SAP. RPT-1.5 performs predictive tasks such as classification and regression out of the box through in-context learning, without requiring model training or fine-tuning. Its table-native architecture is designed to deliver high-quality predictions on enterprise tabular data.	Info only	General Availability	Announcement	Technology	Not applicable	SAP Business Technology Platform	2026-07-27	2026-07-27	Unavailable	Not for sovereign cloud offerings

Component	Environment	Title	Description	Action	Lifecycle	Type	Line of Business	Modular Business Process	Product	Latest Revision	Available as of	Version	Scope
			For more information, see SAP-RPT-1.5.										
SAP AI Core	Cloud Foundry	Generative AI hub	Prompt Optimization includes additional evaluation techniques for evaluations response formatting. For more information, see Response Formatting	Info only	General Availability	Announcement	Technology	Not applicable	SAP Business Technology Platform	2026-06-11	2026-06-11	Unavailable	Not for sovereign cloud offerings
SAP AI Core	Cloud Foundry	Generative AI hub	We now support new models, including Gemini-3.1-flash-image, Gemini-3-pro-image and Anthropic--claude-4.8-opus. For more information, see SAP Note 3437766.	Info only	General Availability	Announcement	Technology	Not applicable	SAP Business Technology Platform	2026-07-01	2026-07-01	Unavailable	Not for sovereign cloud offerings

Component	Environment	Title	Description	Action	Lifecycle	Type	Line of Business	Modular Business Process	Product	Latest Revision	Available as of	Version	Scope
SAP AI Core	Cloud Foundry	Generative AI hub	We now support new models, including GPT 5.1 and Gemini 3.5 flash. For more information, see SAP Note 3437766.	Info only	General Availability	Announcement	Technology	Not applicable	SAP Business Technology Platform	2026-06-11	2026-06-11	Unavailable	Not for sovereign cloud offerings

Component	Environment	Title	Description	Action	Lifecycle	Type	Line of Business	Modular Business Process	Product	Latest Revision	Available as of	Version	Scope
SAP AI Core	Cloud Foundry	Generative AI hub	For prompt optimizations, you can now reconcile mismatched variables between prompts and datasets by implementing variable mapping. Additional metrics have been added, including `EXACT_MATCH`: a Boolean indicating whether the output exactly matches the reference. For more information, see Variable Mapping and Create a Configuration for a Prompt Optimization.	Info only	General Availability	Announcement	Technology	Not applicable	SAP Business Technology Platform	2026-06-03	2026-06-03	Unavailable	Not for sovereign cloud offerings

Component	Environment	Title	Description	Action	Lifecycle	Type	Line of Business	Modular Business Process	Product	Latest Revision	Available as of	Version	Scope
SAP AI Core	Cloud Foundry	Generative AI hub	We now support new models, including Chat GPT 5.5. For more information, see SAP Note 3437766.	Info only	General Availability	Announcement	Technology	Not applicable	SAP Business Technology Platform	2026-06-03	2026-06-03	Unavailable	Not for sovereign cloud offerings

Component	Environment	Title	Description	Action	Lifecycle	Type	Line of Business	Modular Business Process	Product	Latest Revision	Available as of	Version	Scope
SAP AI Core	Cloud Foundry	Service Plans	The free service plan has been discontinued. For more information, see SAP Note 3735945. If you use the free plan, you can update your plan to the standard or extended plan. For more information, see Service Plans [page 134]. Alternatively, a 30-day free trial including Generative AI hub is available. For more information, see Trial.	Info only	General Availability	Announcement	Technology	Not applicable	SAP Business Technology Platform	2026-05-22	2026-05-22	Unavailable	Not for sovereign cloud offerings

Component	Environment	Title	Description	Action	Lifecycle	Type	Line of Business	Modular Business Process	Product	Latest Revision	Available as of	Version	Scope
SAP AI Core	Cloud Foundry	Service Plans	The free service plan will be discontinued on 2026-05-22. For more information, see SAP Note 3735945. If you use the free plan, you can update your plan to the standard or extended plan. For more information, see Service Plans [page 134]. Alternatively, a 30-day free trial including Generative AI hub is available. For more information, see Trial.	Info only	General Availability	Announcement	Technology	Not applicable	SAP Business Technology Platform	2026-05-19	2026-05-19	Unavailable	Not for sovereign cloud offerings

Component	Environment	Title	Description	Action	Lifecycle	Type	Line of Business	Modular Business Process	Product	Latest Revision	Available as of	Version	Scope
SAP AI Core	Cloud Foundry	Generative AI hub	New models are supported, including Gemini 3.1 Flash Lite. For more information, see SAP Note 3437766.	Info only	General Availability	Announcement	Technology	Not applicable	SAP Business Technology Platform	2026-05-12	2026-05-12	Unavailable	Not for sovereign cloud offerings
SAP AI Core	Cloud Foundry	Generative AI hub	For model consumption through foundation models, you can reduce costs by sending requests in batches. For more information, see Batch Consumption.	Info only	General Availability	Announcement	Technology	Not applicable	SAP Business Technology Platform	2026-05-08	2026-05-08	Unavailable	Not for sovereign cloud offerings
SAP AI Core	Cloud Foundry	Generative AI hub	`metaConfigId` has been added as a filter for pipeline queries in the pipelines API for the grounding module of orchestration. For more information, see Data Pipelines.	Info only	General Availability	Announcement	Technology	Not applicable	SAP Business Technology Platform	2026-05-08	2026-05-08	Unavailable	Not for sovereign cloud offerings

Component	Environment	Title	Description	Action	Lifecycle	Type	Line of Business	Modular Business Process	Product	Latest Revision	Available as of	Version	Scope
SAP AI Core	Cloud Foundry	Generative AI hub	The templating module of orchestration has been updated to include prompt caching for use with OpenAI and Gemini models only. Prompt caching improves performance by reusing common prompt sections, enabling faster and more consistent responses across requests. For more information, see Prompt Caching.	Info only	General Availability	Announcement	Technology	Not applicable	SAP Business Technology Platform	2026-05-08	2026-05-08	Unavailable	Not for sovereign cloud offerings

Component	Environment	Title	Description	Action	Lifecycle	Type	Line of Business	Modular Business Process	Product	Latest Revision	Available as of	Version	Scope
SAP AI Core	Cloud Foundry	Generative AI hub	The templating module of orchestration has been updated to include response formatting. You can define schemas for the LLM to follow to guide the structure of your outputs. For more information, see Response Formatting.	Info only	General Availability	Announcement	Technology	Not applicable	SAP Business Technology Platform	2026-05-08	2026-05-08	Unavailable	Not for sovereign cloud offerings

Component	Environment	Title	Description	Action	Lifecycle	Type	Line of Business	Modular Business Process	Product	Latest Revision	Available as of	Version	Scope
SAP AI Core	Cloud Foundry	Generative AI hub	You can now track the progress of long-running prompt optimizations in real-time. When viewing execution status, progress details are automatically displayed in a normalized format. You can also use additional metrics such as ROUGE, BLEU, COMET and metrics designed to evaluate tool calling. For more information, see Prompt Optimization.	Info only	General Availability	Announcement	Technology	Not applicable	SAP Business Technology Platform	2026-05-08	2026-05-08	Unavailable	Not for sovereign cloud offerings

Component	Environment	Title	Description	Action	Lifecycle	Type	Line of Business	Modular Business Process	Product	Latest Revision	Available as of	Version	Scope
SAP AI Core	Cloud Foundry	Generative AI hub	New models are supported, including Claude Opus 4.7, GPT Realtime and Mistral Small. Additionally, Gemini-2.5 Flash support image generation. For more information, see SAP Note 3437766.	Info only	General Availability	Announcement	Technology	Not applicable	SAP Business Technology Platform	2026-05-08	2026-05-08	Unavailable	Not for sovereign cloud offerings

Component	Environment	Title	Description	Action	Lifecycle	Type	Line of Business	Modular Business Process	Product	Latest Revision	Available as of	Version	Scope
SAP AI Core	Cloud Foundry	Generative AI hub	Inference observability allows you to store and retrieve payloads from foundation model inferences for analytical purposes. You can add custom labels for filtering, attach feedback after inference completion, and store data in a registered S3 object store or opt for metadata-only storage. Use inference headers to explicitly control which requests are recorded, and leverage the REST APIs to manage labels, feedback, and retrieve records for analysis. For more information, see In-	Info only	General Availability	Announcement	Technology	Not applicable	SAP Business Technology Platform	2026-04-27	2026-04-27	Unavailable	Not for sovereign cloud offerings

Component	Environment	Title	Description	Action	Lifecycle	Type	Line of Business	Modular Business Process	Product	Latest Revision	Available as of	Version	Scope
			ference Observability.										
SAP AI Core	Cloud Foundry	Generative AI hub	The Google Drive object store has been added to orchestration workflows. For more information, see Create a Generic Secret for Google Drive and Create a Pipeline with Google Drive.	Info only	General Availability	Announcement	Technology	Not applicable	SAP Business Technology Platform	2026-04-27	2026-04-27	Unavailable	Not for sovereign cloud offerings
SAP AI Core	Cloud Foundry	Custom AI Executions	You can now track the progress of long-running executions in real-time. When viewing execution status, progress details are automatically displayed in a normalized format. For more information, see Start Training.	Info only	General Availability	Announcement	Technology	Not applicable	SAP Business Technology Platform	2026-04-27	2026-04-27	Unavailable	Not for sovereign cloud offerings

Component	Environment	Title	Description	Action	Lifecycle	Type	Line of Business	Modular Business Process	Product	Latest Revision	Available as of	Version	Scope
SAP AI Core	Cloud Foundry	Administration	mTLS certificate secrets are supported, enabling your AI workloads to securely authenticate to external services using mutual TLS (mTLS) without manual certificate management. For more information, see Manage mTLS Certificate Secrets.	Info only	General Availability	Announcement	Technology	Not applicable	SAP Business Technology Platform	2026-04-27	2026-04-27	Unavailable	Not for sovereign cloud offerings
SAP AI Core	Cloud Foundry	Generative AI hub	New models are supported, including GPT 5.4, GPT 5.4-nano and GPT 5.3-Codex. For more information, see SAP Note 3437766.	Info only	General Availability	Announcement	Technology	Not applicable	SAP Business Technology Platform	2026-04-27	2026-04-27	Unavailable	Not for sovereign cloud offerings

Component	Environment	Title	Description	Action	Lifecycle	Type	Line of Business	Modular Business Process	Product	Latest Revision	Available as of	Version	Scope
SAP AI Core	Cloud Foundry	Generative AI hub	For security reasons, the system prompt for SAP-ABAP-1 is now predefined. The prompt is added automatically to all requests. If you include a system prompt in a request, the request fails and an error is returned. In most scenarios, a user prompt is sufficient. If you need additional instructions, include them at the beginning or end of the user prompt.	Info only	General Availability	Announcement	Technology	Not applicable	SAP Business Technology Platform	2026-03-01	2026-03-01	Unavailable	Not for sovereign cloud offerings

Component	Environment	Title	Description	Action	Lifecycle	Type	Line of Business	Modular Business Process	Product	Latest Revision	Available as of	Version	Scope
SAP AI Core	Cloud Foundry	Generative AI hub	The `masking_providers` property in the Orchestration V2 Data Masking module configuration is deprecated and has been replaced by the `providers` property. If you are currently using `masking_providers` in your masking module configurations, update your specifications to use the `providers` property instead. The deprecated `masking_providers` property will be removed on September 15, 2026. For more information, see	Info only	Deprecated	Announcement	Technology	Not applicable	SAP Business Technology Platform	2026-03-16	2026-03-16	Unavailable	Not for sovereign cloud offerings

Component	Environment	Title	Description	Action	Lifecycle	Type	Line of Business	Modular Business Process	Product	Latest Revision	Available as of	Version	Scope
			Enhancing Model Consumption with Data Masking.										
SAP AI Core	Cloud Foundry	Generative AI hub	New models are supported, including Amazon Nova Lite 2.0. For more information, see SAP Note 3437766.	Info only	General Availability	Announcement	Technology	Not applicable	SAP Business Technology Platform	2026-03-16	2026-03-16	Unavailable	Not for sovereign cloud offerings
SAP AI Core	Cloud Foundry	Generative AI hub	SAP RPT 1 supports `top_k` as an input parameter, and outputs now include confidence intervals. For more information, see Example Payloads for Inferencing: sap-rpt-1.	Info only	General Availability	Announcement	Technology	Not applicable	SAP Business Technology Platform	2026-03-01	2026-03-01	Unavailable	Not for sovereign cloud offerings

Component	Environment	Title	Description	Action	Lifecycle	Type	Line of Business	Modular Business Process	Product	Latest Revision	Available as of	Version	Scope
SAP AI Core	Cloud Foundry	Generative AI hub	PDF documents are supported in the Data Masking module in Orchestration, and ServiceNow has been added as a grounding repository. For more information, see Enhancing Model Consumption with Data Masking and Generic Secrets for Grounding.	Info only	General Availability	Announcement	Technology	Not applicable	SAP Business Technology Platform	2026-03-01	2026-03-01	Unavailable	Not for sovereign cloud offerings
SAP AI Core	Cloud Foundry	Generative AI hub	New models are supported, including Anthropic Claude Opus 4.6. For more information, see SAP Note 3437766.	Info only	General Availability	Announcement	Technology		SAP Business Technology Platform	2026-03-01	2026-03-01	Unavailable	Not for sovereign cloud offerings

Component	Environment	Title	Description	Action	Lifecycle	Type	Line of Business	Modular Business Process	Product	Latest Revision	Available as of	Version	Scope
SAP AI Core	Cloud Foundry	Generative AI hub	You can now manage metadata for your documents, collections and chunks created using the Vector API, to enable advanced filtering and organization of your content. For more information, see Metadata.	Info only	General Availability	Announcement	Technology	Not applicable	SAP Business Technology Platform	2026-02-16	2026-02-16	Unavailable	Not for sovereign cloud offerings
SAP AI Core	Cloud Foundry	Generative AI hub	You can merge and rank search results from multiple data repositories using post-processing capabilities of the Retrieval API. For more information, see Retrieval Search.	Info only	General Availability	Announcement	Technology	Not applicable	SAP Business Technology Platform	2026-02-16	2026-02-16	Unavailable	Not for sovereign cloud offerings

Component	Environment	Title	Description	Action	Lifecycle	Type	Line of Business	Modular Business Process	Product	Latest Revision	Available as of	Version	Scope
SAP AI Core	Cloud Foundry	Generative AI hub	Vector search supports advanced filtering capabilities including metadata filtering. For more information, see Vector Search.	Info only	General Availability	Announcement	Technology	Not applicable	SAP Business Technology Platform	2026-02-16	2026-02-16	Unavailable	Not for sovereign cloud offerings
SAP AI Core	Cloud Foundry	Generative AI hub	Custom metrics are supported in prompt optimizations, enabling you to define and optimize prompts based on your specific evaluation criteria. Only LLM-as-a-judge metrics with numerical or Boolean output types can be used in optimization tasks. For more information, see Create a Custom Metric and Create a Configuration for a Prompt Optimization.	Info only	General Availability	Announcement	Technology	Not applicable	SAP Business Technology Platform	2026-02-16	2026-02-16	Unavailable	Not for sovereign cloud offerings

Component	Environment	Title	Description	Action	Lifecycle	Type	Line of Business	Modular Business Process	Product	Latest Revision	Available as of	Version	Scope
SAP AI Core	Cloud Foundry	Generative AI hub	Custom metric creation has been enhanced. For more information, see Create a Custom Metric.	Info only	Deprecated	Announcement	Technology	Not applicable	SAP Business Technology Platform	2026-02-16	2026-02-16	Unavailable	Not for sovereign cloud offerings
SAP AI Core	Cloud Foundry	Generative AI hub	The Prompt Registry now enables you to create and manage orchestration configurations declaratively, allowing you to version and track complex AI workflows alongside your prompts for better governance and reproducibility. For more information, see Create an Orchestration Config (Imperative).	Info only	General Availability	Announcement	Technology	Not applicable	SAP Business Technology Platform	2026-02-16	2026-02-16	Unavailable	Not for sovereign cloud offerings

Component	Environment	Title	Description	Action	Lifecycle	Type	Line of Business	Modular Business Process	Product	Latest Revision	Available as of	Version	Scope
SAP AI Core	Cloud Foundry	Generative AI hub	New models are supported, including Open AI GPT 5.2.. For more information, see SAP Note 3437766.	Info only	General Availability	Announcement	Technology	Not applicable	SAP Business Technology Platform	2026-02-16	2026-02-16	Unavailable	Not for sovereign cloud offerings
SAP AI Core	Cloud Foundry	Generative AI hub	You can specify the `promptTemplateScope` access level in evaluation and prompt optimization configurations. For more information, see Create an Evaluation and Create a Configuration for a Prompt Optimization.	Info only	General Availability	Announcement	Technology	Not applicable	SAP Business Technology Platform	2026-01-30	2026-01-30	Unavailable	Not for sovereign cloud offerings

Component	Environment	Title	Description	Action	Lifecycle	Type	Line of Business	Modular Business Process	Product	Latest Revision	Available as of	Version	Scope
SAP AI Core	Cloud Foundry	Generative AI hub	You can provide separate test and train datasets for use in prompt optimizations. For more information, see Create a Configuration for a Prompt Optimization.	Info only	General Availability	Announcement	Technology	Not applicable	SAP Business Technology Platform	2026-01-19	2026-01-19	Unavailable	Not for sovereign cloud offerings
SAP AI Core	Cloud Foundry	Generative AI hub	When using the Vector API, you can assign a UUID during collection creation and bulk delete documents by ID. For more information, see Create a Collection and Delete Documents.	Info only	General Availability	Announcement	Technology	Not applicable	SAP Business Technology Platform	2026-01-19	2026-01-19	Unavailable	Not for sovereign cloud offerings

Component	Environment	Title	Description	Action	Lifecycle	Type	Line of Business	Modular Business Process	Product	Latest Revision	Available as of	Version	Scope
SAP AI Core	Cloud Foundry	Generative AI hub	You can now use the Claude Opus 4.5 model with Amazon Bedrock. The model is available with the extended service plan. For more information, see SAP Note 3437766.	Info only	General Availability	Announcement		Not applicable	SAP Business Technology Platform	2026-01-19	2026-01-19	Unavailable	Not for sovereign cloud offerings
SAP AI Core	Cloud Foundry	Generative AI hub	You can now use the `SAP-ABAP-1` model to better understand ABAP code. `SAP-ABAP-1` is a foundation model built by SAP and is fine-tuned on a large amount of ABAP code. For more information, see SAP-ABAP-1 and Example Payloads for Inferencing: SAP-ABAP-1.	Info only	General Availability	Announcement	Technology	Not applicable	SAP Business Technology Platform	2025-12-22	2025-12-22	Unavailable	Not for sovereign cloud offerings

Component	Environment	Title	Description	Action	Lifecycle	Type	Line of Business	Modular Business Process	Product	Latest Revision	Available as of	Version	Scope
SAP AI Core	Cloud Foundry	Generative AI hub	`SAP-RPT-1` launched. `SAP-RPT-1` is a relational pre-trained transformer for use on relational and structured data. It is developed and maintained by SAP. Relational Foundation Models (RFMs) are large-scale machine learning models designed to understand, process, and do predictions on tabular and relational data. RPT-1 solves predictive tasks such as classification and regression out-of-the-box without requiring any training or fine-tuning via in-con-	Info only	General Availability	Announcement	Technology	Not applicable	SAP Business Technology Platform	2025-12-08	2025-12-08	Unavailable	Not for sovereign cloud offerings

Component	Environment	Title	Description	Action	Lifecycle	Type	Line of Business	Modular Business Process	Product	Latest Revision	Available as of	Version	Scope
			text learning. Due to its table-native architecture, prediction quality on enterprise data is typically very high, ahead of state-of-the-art narrow AI models and LLMs employed for such tasks. For more information, see SAP-RPT-1 and Example Payloads for Inferencing: sap-rpt-1.										

Component	Environment	Title	Description	Action	Lifecycle	Type	Line of Business	Modular Business Process	Product	Latest Revision	Available as of	Version	Scope
SAP AI Core	Cloud Foundry	Generative AI hub	Evaluations added to Optimizations Evaluations provides tools for benchmarking large language models and prompts via orchestration configurations. Evaluations can be used for the following use cases: • Evaluating prompt templates and models as orchestration configurations to determine the most effective combination for a use case. • Use industry standard pre-de-	Info only	General Availability	Announcement	Technology	Not applicable	SAP Business Technology Platform	2025-12-08	2025-12-08	Unavailable	Not for sovereign cloud offerings

Component	Environment	Title	Description	Action	Lifecycle	Type	Line of Business	Modular Business Process	Product	Latest Revision	Available as of	Version	Scope
			fined metrics for model and prompt comparison with your use case specific dataset. • Use your own custom defined metrics for your prompt and model evaluation. For more information, see Evaluations.										

Component	Environment	Title	Description	Action	Lifecycle	Type	Line of Business	Modular Business Process	Product	Latest Revision	Available as of	Version	Scope
SAP AI Core	Cloud Foundry	Generative AI hub	Metrics added to Optimizations You can choose from a selection of system defined metrics, or define your own custom llm-as-a-judge metrics, including rating criteria. You can incorporate system defined and custom metrics into your evaluation workflows, and manage the lifecycle of your custom metrics. For more information, see Metrics.	Info only	General Availability	Announcement	Technology	Not applicable	SAP Business Technology Platform	2025-12-08	2025-12-08	Unavailable	Not for sovereign cloud offerings

Component	Environment	Title	Description	Action	Lifecycle	Type	Line of Business	Modular Business Process	Product	Latest Revision	Available as of	Version	Scope
SAP AI Core	Cloud Foundry	Generative AI hub	New models are supported, including Anthropic Claude 4.5 Haiku. For more information, see SAP Note 3437766.	Info only	General Availability	Announcement	Technology	Not applicable	SAP Business Technology Platform	2025-12-08	2025-12-08	Unavailable	Not for sovereign cloud offerings
SAP AI Core	Cloud Foundry	Generative AI hub	New models are supported, including Gemini 2.5 Flash Lite. For more information, see SAP Note 3437766.	Info only	General Availability	Announcement	Technology	Not applicable	SAP Business Technology Platform	2025-12-01	2025-12-01	Unavailable	Not for sovereign cloud offerings

Component	Environment	Title	Description	Action	Lifecycle	Type	Line of Business	Modular Business Process	Product	Latest Revision	Available as of	Version	Scope
SAP AI Core	Cloud Foundry	Generative AI hub	Orchestration V2 includes support for a unified embeddings endpoint that works consistently across different LLM providers. This endpoint can be used independently or combined with data anonymization capabilities. For more information, see Embeddings.	Info only	General Availability	Announcement	Technology	Not applicable	SAP Business Technology Platform	2025-11-17	2025-11-17	Unavailable	Not for sovereign cloud offerings
SAP AI Core	Cloud Foundry	Generative AI hub	You can define multiple model configuations in your orchestration workflow, enabling automatic fallbacks if processing fails (for example, due to unsupported models in a region). For more information, see Orchestration with Fallbacks.	Info only	General Availability	Announcement	Technology	Not applicable	SAP Business Technology Platform	2025-11-17	2025-11-17	Unavailable	Not for sovereign cloud offerings

Component	Environment	Title	Description	Action	Lifecycle	Type	Line of Business	Modular Business Process	Product	Latest Revision	Available as of	Version	Scope
SAP AI Core	Cloud Foundry	Generative AI hub	New models are supported, including Perplexity Sonar and Sonar Pro. For more information, see SAP Note 3437766.	Info only	General Availability	Announcement	Technology	Not applicable	SAP Business Technology Platform	2025-10-20	2025-10-20	Unavailable	Not for sovereign cloud offerings
SAP AI Core	Cloud Foundry	Generative AI hub	You can optimize prompts to by evaluating them against a specified metric. The optimized prompt is saved, improving model performance and output quality with prompt reuse. For more information, see Prompt Optimization.	Info only	General Availability	Announcement	Technology	Not applicable	SAP Business Technology Platform	2025-10-20	2025-10-20	Unavailable	Not for sovereign cloud offerings

Component	Environment	Title	Description	Action	Lifecycle	Type	Line of Business	Modular Business Process	Product	Latest Revision	Available as of	Version	Scope
SAP AI Core	Cloud Foundry	Generative AI hub	You can manage the lifecycle of your orchestration configs, including creating, saving, versioning, reuse. CRUD functionality, and import and export of orchestration configs are also supported. For more information, see Orchestration Config Management.	Info only	General Availability	Announcement	Technology	Not applicable	SAP Business Technology Platform	2025-10-20	2025-10-20	Unavailable	Not for sovereign cloud offerings
SAP AI Core	Cloud Foundry	Generative AI hub	You can use the quota management API endpoints to check and update your rate limits for models in the generative AI hub. For more information, see Rate Limit Management.	Info only	General Availability	Announcement	Technology	Not applicable	SAP Business Technology Platform	2025-10-06	2025-10-06	Unavailable	Not for sovereign cloud offerings

Component	Environment	Title	Description	Action	Lifecycle	Type	Line of Business	Modular Business Process	Product	Latest Revision	Available as of	Version	Scope
SAP AI Core	Cloud Foundry	Generative AI hub	SAP Document Management is supported as a repository type for the orchestration grounding module. For more information, see Grounding Generic Secrets for SAP Document Management Service.	Info only	General Availability	Announcement	Technology	Not applicable	SAP Business Technology Platform	2025-09-01	2025-09-01	Unavailable	Not for sovereign cloud offerings
SAP AI Core	Cloud Foundry	Generative AI hub	Orchestration model configurations have timeout and max tries parameters. For more information, see Templating.	Info only	General Availability	Announcement	Technology	Not applicable	SAP Business Technology Platform	2025-09-01	2025-09-01	Unavailable	Not for sovereign cloud offerings
SAP AI Core	Cloud Foundry	Generative AI hub	New models are supported, including GPT 5, GPT 5 Mini, and GPT 5 Nano. For more information, see SAP Note 3437766.	Info only	General Availability	Announcement	Technology	Not applicable	SAP Business Technology Platform	2025-08-18	2025-08-18	Unavailable	Not for sovereign cloud offerings

Component	Environment	Title	Description	Action	Lifecycle	Type	Line of Business	Modular Business Process	Product	Latest Revision	Available as of	Version	Scope
SAP AI Core	Cloud Foundry	Generative AI hub	New models are supported, including Gemini Embeddings. For more information, see SAP Note 3437766.	Info only	General Availability	Announcement	Technology	Not applicable	SAP Business Technology Platform	2025-08-06	2025-08-06	Unavailable	Not for sovereign cloud offerings

Component	Environment	Title	Description	Action	Lifecycle	Type	Line of Business	Modular Business Process	Product	Latest Revision	Available as of	Version	Scope
SAP AI Core	Cloud Foundry	Generative AI hub	This initial orchestration endpoint is deprecated and is scheduled for decommissioning on October 31, 2026. Following that date, the endpoint will no longer be available. We recommend that you create new orchestration workflows using version 2, and that you migrate existing workflows from version 1 to version 2. To use version 2, you'll need to update your endpoint from `/completion` to `/v2/completion` and modify your existing payloads. For more information, see SAP Note Orchestration Workflow V2 and 3634540.	Info only	Deprecated	Announcement	Technology	Not applicable	SAP Business Technology Platform	2025-07-21	2025-07-21	Unavailable	Not for sovereign cloud offerings

Component	Environment	Title	Description	Action	Lifecycle	Type	Line of Business	Modular Business Process	Product	Latest Revision	Available as of	Version	Scope
SAP AI Core	Cloud Foundry	Generative AI hub	New models are supported, including Nova Premier from AWS Bedrock. For more information, see SAP Note 3437766.	Info only	General Availability	Announcement	Technology	Not applicable	SAP Business Technology Platform	2025-07-21	2025-07-21	Unavailable	Not for sovereign cloud offerings
SAP AI Core	Cloud Foundry	Generative AI hub	Prompt shields are available as part of input filtering with Azure Content Safety. Prompt shields detect and mitigate prompt attacks, such as prompts designed to bypass safety mechanisms or override previous instruction. For more information, see Harm categories in Azure AI Content Safety and Input Filtering.	Info only	General Availability	Announcement	Technology	Not applicable	SAP Business Technology Platform	2025-06-23	2025-06-23	Unavailable	Not for sovereign cloud offerings

Component	Environment	Title	Description	Action	Lifecycle	Type	Line of Business	Modular Business Process	Product	Latest Revision	Available as of	Version	Scope
SAP AI Core	Cloud Foundry	Generative AI hub	GPT-4 versions 0613 and turbo-2024-04-09 and gpt-4-32k version 0613 have been deprecated. It's recommended that users of these models migrate to GPT-4.1 as a replacement. For more information, see SAP Note 3437766.	Info only	Deprecated	Announcement	Technology	Not applicable	SAP Business Technology Platform	2025-06-02	2025-06-02	Unavailable	Not for sovereign cloud offerings

Component	Environment	Title	Description	Action	Lifecycle	Type	Line of Business	Modular Business Process	Product	Latest Revision	Available as of	Version	Scope
SAP AI Core	Cloud Foundry	Generative AI hub	Gemini-1.5-pro and gemini-1.5-flash have been deprecated. It's recommended that users of these models migrate to gemini-2.0-flash, or gemini-2.0-flash-lite as replacements. For more information, see SAP Note 3437766.	Info only	Deprecated	Announcement	Technology	Not applicable	SAP Business Technology Platform	2025-06-02	2025-06-02	Unavailable	Not for sovereign cloud offerings
SAP AI Core	Cloud Foundry	Generative AI hub	Mistralai--mixtral08x7b-v01 has been deprecated. It's recommended that users of this model migrate to mistralai--mistral-small-instruct as a replacement. For more information, see SAP Note 3437766.	Info only	Deprecated	Announcement	Technology	Not applicable	SAP Business Technology Platform	2025-06-02	2025-06-02	Unavailable	Not for sovereign cloud offerings

Component	Environment	Title	Description	Action	Lifecycle	Type	Line of Business	Modular Business Process	Product	Latest Revision	Available as of	Version	Scope
SAP AI Core	Cloud Foundry	Generative AI hub	New models are supported, including OpenAI GPT, o3, o4-mini, 4.1, 4.1-mini, 4.1-nano and Mistral small instruct. For more information, see SAP Note 3437766.	Info only	General Availability	Announcement	Technology	Not applicable	SAP Business Technology Platform	2025-05-19	2025-05-19	Unavailable	Not for sovereign cloud offerings

Component	Environment	Title	Description	Action	Lifecycle	Type	Line of Business	Modular Business Process	Product	Latest Revision	Available as of	Version	Scope
SAP AI Core	Cloud Foundry	Generative AI hub	An additional module has been added to orchestration. The translation module translates text, and can be configured for input and output text. The input translation module helps improve answer quality when the configured model performs better when input is provided in a specific language, for example English. For more information, see Input Translation and Output Translation.	Info only	General Availability	Announcement	Technology	Not applicable	SAP Business Technology Platform	2025-04-24	2025-04-24	Unavailable	Not for sovereign cloud offerings

Component	Environment	Title	Description	Action	Lifecycle	Type	Line of Business	Modular Business Process	Product	Latest Revision	Available as of	Version	Scope
SAP AI Core	Cloud Foundry	SAP BTP Cockpit	A detailed consumption report is available in SAP BTP for Generative AI hub as the part of SAP AI Core service. You can see the consumption specific to an LLM model and Orchestration service, and consumption for each resource group. For more information, see Metering and Pricing for Generative AI [page 8].	Info only	General Availability	Announcement	Technology	Not applicable	SAP Business Technology Platform	2025-04-14	2025-04-14	Unavailable	Not for sovereign cloud offerings

Component	Environment	Title	Description	Action	Lifecycle	Type	Line of Business	Modular Business Process	Product	Latest Revision	Available as of	Version	Scope
SAP AI Core	Cloud Foundry	Generative AI hub	New models are supported, including Anthropic Claude 3.7 sonnet, GCP Vertex AI Gemini 2.0-flash and flash-lite, NVIDIA Llama 3.2 nv embedqa 1b, and OpenAI GPT 4o version 2024-11-20. For more information, see SAP Note 3437766.	Info only	General Availability	Announcement	Technology	Not applicable	SAP Business Technology Platform	2025-04-14	2025-04-14	Unavailable	Not for sovereign cloud offerings
SAP AI Core	Cloud Foundry	Generative AI hub	We now support the Anthropic models Claude Opus 4 and Claude Sonnet 4. For more information, see SAP Note 3437766.	Info only	General Availability	Announcement	Technology	Not applicable	SAP Business Technology Platform	2025-06-23	2025-06-23	Unavailable	Not for sovereign cloud offerings

Component	Environment	Title	Description	Action	Lifecycle	Type	Line of Business	Modular Business Process	Product	Latest Revision	Available as of	Version	Scope
SAP AI Core	Cloud Foundry	Generative AI hub	More endpoints have been added to the Pipeline API, so you can retrieve information about the documents within a pipeline, including processing status. For more information, see Data Pipelines.	Info only	General Availability	Announcement	Technology	Not applicable	SAP Business Technology Platform	2025-02-03	2025-02-03	Unavailable	Not for sovereign cloud offerings
SAP AI Core	Cloud Foundry	Generative AI hub	Selected models from Aleph Alpha are supported. For more information, see SAP Note 3437766.	Info only	General Availability	Announcement	Technology	Not applicable	SAP Business Technology Platform	2025-02-03	2025-02-03	Unavailable	Not for sovereign cloud offerings
SAP AI Core	Cloud Foundry	Generative AI hub	Additional selected models and model versions from Open AI are supported. For more information, see 3437766.	Info only	General Availability	Announcement	Technology	Not applicable	SAP Business Technology Platform	2025-03-03	2025-03-03	Unavailable	Not for sovereign cloud offerings

Component	Environment	Title	Description	Action	Lifecycle	Type	Line of Business	Modular Business Process	Product	Latest Revision	Available as of	Version	Scope
SAP AI Core	Cloud Foundry	Generative AI hub	You can now configure your orchestration workflow so that Llama Guard 3 filters input and output content. Llama Guard 3 provides several content categories for filtering content. Azure Content Safety is still supported. For more information, see Input Filtering and Output Filtering.	Info only	General Availability	Announcement	Technology	Not applicable	SAP Business Technology Platform	2025-01-19	2025-01-19	Unavailable	Not for sovereign cloud offerings

Component	Environment	Title	Description	Action	Lifecycle	Type	Line of Business	Modular Business Process	Product	Latest Revision	Available as of	Version	Scope
SAP AI Core	Cloud Foundry	Generative AI hub	Prompt registry Prompt registry integrates prompt templates into SAP AI Core, making them discoverable across your applications and orchestration It reduces the complexity of dealing with prompt templates and leveraging integration capabilities. For more information, see Prompt Registry.	Info only	General Availability	Announcement	Technology	Not applicable	SAP Business Technology Platform	2024-12-02	2024-12-02	Unavailable	Not for sovereign cloud offerings

Component	Environment	Title	Description	Action	Lifecycle	Type	Line of Business	Modular Business Process	Product	Latest Revision	Available as of	Version	Scope
SAP AI Core	Cloud Foundry	Generative AI hub	Retrieval and vector APIs added. The vector API can be used for managing collections and documents in the Vector database. The retrieval API searches data repositories and returns the relevant chunks for the user query. For more information, see Preparing Data Using the Vector API and Retrieval API.	Info only	General Availability	Announcement	Technology	Not applicable	SAP Business Technology Platform	2024-12-02	2024-12-02	Unavailable	Not for sovereign cloud offerings

Component	Environment	Title	Description	Action	Lifecycle	Type	Line of Business	Modular Business Process	Product	Latest Revision	Available as of	Version	Scope
SAP AI Core	Cloud Foundry	Generative AI hub	Additional selected models and model versions from Anthropic via AWS Bedrock are supported. For more information, see Foundation Models and SAP Note 3437766.	Info only	General Availability	Announcement	Technology	Not applicable	SAP Business Technology Platform	2024-12-02	2024-12-02	Unavailable	Not for sovereign cloud offerings
SAP AI Core	Cloud Foundry	Generative AI hub	Additional selected models and model versions from Open AI are supported. For more information, see SAP Note 3437766.	Info only	General Availability	Announcement	Technology	Not applicable	SAP Business Technology Platform	2024-12-02	2024-12-02	Unavailable	Not for sovereign cloud offerings
SAP AI Core	Cloud Foundry	Generative AI hub	Additional selected model versions from Vertex AI are supported. For more information, see SAP Note 3437766.	Info only	General Availability	Announcement	Technology	Not applicable	SAP Business Technology Platform	2024-12-02	2024-12-02	Unavailable	Not for sovereign cloud offerings

Component	Environment	Title	Description	Action	Lifecycle	Type	Line of Business	Modular Business Process	Product	Latest Revision	Available as of	Version	Scope
SAP AI Core	Cloud Foundry	Generic Secrets	Generic secrets are now available at the tenant-wide level. For more information, see Create a Generic Secret.	Info only	General Availability	Announcement	Technology	Not applicable	SAP Business Technology Platform	2024-12-02	2024-12-02	Unavailable	Not for sovereign cloud offerings
SAP AI Core	Cloud Foundry	Generative AI hub	Pipeline API added. This pipeline segments data into chunks and generates embeddings, which are multidimensional representations of textual information. The embeddings are stored in a vector database. For more information, see Create a Document Grounding Pipeline Using the Pipelines API.	Info only	General Availability	Announcement	Technology	Not applicable	SAP Business Technology Platform	2024-11-04	2024-11-04	Unavailable	Not for sovereign cloud offerings

Component	Environment	Title	Description	Action	Lifecycle	Type	Line of Business	Modular Business Process	Product	Latest Revision	Available as of	Version	Scope
SAP AI Core	Cloud Foundry	Generative AI hub	A new module has been added to the orchestration system. The grounding module enhances AI processes by integrating external data that is contextually relevant, domain-specific, or real-time. This additional data complements the natural language processing abilities of pretrained models, which are typically based on general information. For more information, see Grounding.	Info only	General Availability	Announcement	Technology	Not applicable	SAP Business Technology Platform	2024-11-04	2024-11-04	Unavailable	Not for sovereign cloud offerings
SAP AI Core	Cloud Foundry	Generative AI hub	Additional selected models from IBM are supported. For more information, see SAP Note 3437766.	Info only	General Availability	Announcement	Technology	Not applicable	SAP Business Technology Platform	2024-10-07	2024-10-07	Unavailable	Not for sovereign cloud offerings

Component	Environment	Title	Description	Action	Lifecycle	Type	Line of Business	Modular Business Process	Product	Latest Revision	Available as of	Version	Scope
SAP AI Core	Cloud Foundry	Generative AI hub	Additional selected models from Mistral AI are supported. For more information, see SAP Note 3437766.	Info only	General Availability	Announcement	Technology	Not applicable	SAP Business Technology Platform	2024-10-07	2024-10-07	Unavailable	Not for sovereign cloud offerings
SAP AI Core	Cloud Foundry	Generative AI hub	For selected models, response streaming is supported in the generative AI hub orchestration service. For more information, see Streaming.	Info only	General Availability	Announcement	Technology	Not applicable	SAP Business Technology Platform	2024-10-07	2024-10-07	Unavailable	Not for sovereign cloud offerings

Component	Environment	Title	Description	Action	Lifecycle	Type	Line of Business	Modular Business Process	Product	Latest Revision	Available as of	Version	Scope
SAP AI Core	Cloud Foundry	Generative AI hub	When creating a deployment for orchestration, you can restrict the choice of models using an allow or disallow list. You can use this to implement internal standards, for example where only certain LLMs are approved for use. For more information, see Create a Deployment for Orchestration.	Info only	General Availability	Announcement	Technology	Not applicable	SAP Business Technology Platform	2024-09-16	2024-09-16	Unavailable	Not for sovereign cloud offerings

Component	Environment	Title	Description	Action	Lifecycle	Type	Line of Business	Modular Business Process	Product	Latest Revision	Available as of	Version	Scope
SAP AI Core	Cloud Foundry	Generative AI hub	Additional module added to orchestration. The data masking module anonymizes or pseudonymizes personally identifiable information from input. For more information, see Orchestration Workflow V1 (Deprecated).	Info only	General Availability	Announcement	Technology	Not applicable	SAP Business Technology Platform	2024-09-16	2024-09-16	Unavailable	Not for sovereign cloud offerings
SAP AI Core	Cloud Foundry	Generative AI hub	Additional selected models from Meta are supported. For more information, see SAP Note 3437766.	Info only	General Availability	Announcement	Technology	Not applicable	SAP Business Technology Platform	2024-09-16	2024-09-16	Unavailable	Not for sovereign cloud offerings

Component	Environment	Title	Description	Action	Lifecycle	Type	Line of Business	Modular Business Process	Product	Latest Revision	Available as of	Version	Scope
SAP AI Core	Cloud Foundry	Generative AI hub	`tiiuae--falcon-40b-instruct` is deprecated and is unavailable from 2024-09-01. Users should choose another model. For more information, see SAP Note 3437766.	Info only	Deprecated	Changed	Technology	Not applicable	SAP Business Technology Platform	2024-09-01	2024-09-01	Unavailable	Not for sovereign cloud offerings
SAP AI Core	Cloud Foundry	Generative AI hub	Additional selected models from OpenAI are supported via Azure. For more information, see SAP Note 3437766.	Info only	General Availability	Announcement	Technology	Not applicable	SAP Business Technology Platform	2024-09-02	2024-09-02	Unavailable	Not for sovereign cloud offerings
SAP AI Core	Cloud Foundry	Generative AI hub	Additional selected models from Anthropic are supported via AWS Bedrock. For more information, see SAP Note 3437766.	Info only	General Availability	Announcement	Technology	Not applicable		2024-05-20	2024-05-20	Unavailable	Not for sovereign cloud offerings

Component	Environment	Title	Description	Action	Lifecycle	Type	Line of Business	Modular Business Process	Product	Latest Revision	Available as of	Version	Scope
SAP AI Core	Cloud Foundry	Generative AI hub	Orchestration combines content generation with a set of functions that are often required in business AI scenarios. Functions include templating, which lets you compose a prompt with placeholders that are filled during inference, and content filtering, which lets you restrict the type of content that is passed to and received from a generative AI model.	Info only	General Availability	Announcement	Technology	Not applicable	SAP Business Technology Platform	2024-08-05	2024-08-05	Unavailable	Not for sovereign cloud offerings

Component	Environment	Title	Description	Action	Lifecycle	Type	Line of Business	Modular Business Process	Product	Latest Revision	Available as of	Version	Scope
SAP AI Core	Cloud Foundry	Generative AI hub	Additional selected models from GCP Vertex AI are supported. For more information, see SAP Note 3437766.	Info only	General Availability	Announcement	Technology	Not applicable	SAP Business Technology Platform	2024-07-08	2024-07-08	Unavailable	Not for sovereign cloud offerings
SAP AI Core	Cloud Foundry	Resource limits per tenant	The maximum number of applications, git repository secrets, docker registry secrets, workflow templates and deployment templates is limited at tenant level to 50 per resource. If you reach this limit, you will receive an error message. You can free up space by deleting resources.	Info only	General Availability	Announcement	Technology	Not applicable	SAP Business Technology Platform	2024-06-24	2024-06-24	Unavailable	Not for sovereign cloud offerings

Component	Environment	Title	Description	Action	Lifecycle	Type	Line of Business	Modular Business Process	Product	Latest Revision	Available as of	Version	Scope
SAP AI Core	Cloud Foundry	Generative AI hub	Additional selected models from OpenAI are supported via Azure. For more information, see SAP Note 3437766.	Info only	General Availability	Announcement	Technology	Not applicable	SAP Business Technology Platform	2024-06-24	2024-06-24	Unavailable	Not for sovereign cloud offerings
SAP AI Core	Cloud Foundry	Generative AI hub	Selected models from Amazon and Anthropic are supported via AWS Bedrock. For more information, see SAP Note 3437766.	Info only	General Availability	Announcement	Technology	Not applicable	SAP Business Technology Platform	2024-06-03	2024-06-03	Unavailable	Not for sovereign cloud offerings
SAP AI Core	Cloud Foundry	Generative AI hub	Selected models from Meta are supported. For more information, see SAP Note 3437766.	Info only	General Availability	Announcement	Technology	Not applicable	SAP Business Technology Platform	2024-06-03	2024-06-03	Unavailable	Not for sovereign cloud offerings

Component	Environment	Title	Description	Action	Lifecycle	Type	Line of Business	Modular Business Process	Product	Latest Revision	Available as of	Version	Scope
SAP AI Core	Cloud Foundry	Generative AI hub	Selected models from Mistral AI are supported. For more information, see SAP Note 3437766.	Info only	General Availability	Announcement	Technology	Not applicable	SAP Business Technology Platform	2024-05-20	2024-05-20	Unavailable	Not for sovereign cloud offerings
SAP AI Core	Cloud Foundry	Deployments	When deployments are submitted, configurations are checked for errors, synchronously. For more information, see Deploy Models.	Info only	General Availability	Announcement	Technology	Not applicable	SAP Business Technology Platform	2024-04-08	2024-05-20	Unavailable	Not for sovereign cloud offerings
SAP AI Core	Cloud Foundry	Generative AI hub	Selected models from GCP Vertex AI are supported. For more information, see SAP Note 3437766.	Info only	General Availability	Announcement	Technology	Not applicable	SAP Business Technology Platform	2024-04-08	2024-04-08	Unavailable	Not for sovereign cloud offerings

Component	Environment	Title	Description	Action	Lifecycle	Type	Line of Business	Modular Business Process	Product	Latest Revision	Available as of	Version	Scope
SAP AI Core	Cloud Foundry	Resource groups limit per tenant	The maximum number of resource groups is limited at tenant level to 50. If you reach this limit, you receive an error message. To free up space, delete some resource groups. Alternatively, raise a ticket to increase your quota. For more information, see Delete a Resource Group.	Info only	General Availability	Announcement	Technology	Not applicable	SAP Business Technology Platform	2024-03-08	2024-03-08	Unavailable	Not for sovereign cloud offerings
SAP AI Core	Cloud Foundry	Authentication	Authentication through X.509 certificates is supported. For more information, see Create a Service Key [page 141] and Use a Service Key [page 144].	Info only	General Availability	New	Technology	Not applicable	SAP Business Technology Platform	2024-02-18	2024-02-18	Unavailable	Not for sovereign cloud offerings

Component	Environment	Title	Description	Action	Lifecycle	Type	Line of Business	Modular Business Process	Product	Latest Revision	Available as of	Version	Scope
SAP AI Core	Cloud Foundry	generative AI hub SDK	The generative AI hub SDK is now available. For more information, see generative AI hub SDK .	Info only	General Availability	New	Technology	Not applicable	SAP Business Technology Platform	2024-02-18	2024-02-18	Unavailable	Not for sovereign cloud offerings

Component	Environment	Title	Description	Action	Lifecycle	Type	Line of Business	Modular Business Process	Product	Latest Revision	Available as of	Version	Scope
SAP AI Core	Cloud Foundry	Generative AI hub	The generative AI hub includes prompt experimentation, prompt management, and administrative tools. Prompt experimentation includes creating and running natural language prompts with a choice of large language models and parameters. Prompt management includes saving prompts with collections, metadata in the form of tags and notes, versioning and deletion. For more information, see Generative AI Hub in SAP AI Core Overview [page 125].	Info only	General Availability	New	Technology	Not applicable	SAP Business Technology Platform	2023-12-20	2023-12-20	Unavailable	Not for sovereign cloud offerings

Component	Environment	Title	Description	Action	Lifecycle	Type	Line of Business	Modular Business Process	Product	Latest Revision	Available as of	Version	Scope
SAP AI Core	Cloud Foundry	Artifact signatures for artifact outputs from executions	Artifact signatures (hashes) can be generated and made available to other executions and deployments to verify the integrity of an artifact. For more information, see Using Artifact Signatures.	Info only	General Availability	New	Technology	Not applicable	SAP Business Technology Platform	2023-07-31	2023-07-31	Unavailable	Not for sovereign cloud offerings

Component	Environment	Title	Description	Action	Lifecycle	Type	Line of Business	Modular Business Process	Product	Latest Revision	Available as of	Version	Scope
SAP AI Core	Cloud Foundry	Template Generator	A wizard to generate workflow and serving templates in VS Code. Using user responses, it simplifies and automates the template writing process. For more information, see SAP AI Core toolkit documentation.	Info only	General Availability	New	Technology	Not applicable	SAP Business Technology Platform	2023-08-04	2023-08-04	Unavailable	Not for sovereign cloud offerings
SAP AI Core	Cloud Foundry	SAP AI Core Toolkit	SAP AI Core is available through the VS Code GUI, through the SAP AI Core toolkit extension. For more information, see SAP AI Core toolkit documentation.	Info only	General Availability	New	Technology	Not applicable	SAP Business Technology Platform	2023-08-04	2023-08-04	Unavailable	Not for sovereign cloud offerings

Component	Environment	Title	Description	Action	Lifecycle	Type	Line of Business	Modular Business Process	Product	Latest Revision	Available as of	Version	Scope
SAP AI Core	Cloud Foundry	LLM Package	The content package for large language models for SAP AI Core simplifies the deployment of large language models with integrated and automated workflows. For more information, see PyPi LLM .	Info only	General Availability	New	Technology	Not applicable	SAP Business Technology Platform	2023-05-31	2023-05-31	Unavailable	Not for sovereign cloud offerings
SAP AI Core	Cloud Foundry	Dataset API	You can upload, download, and delete artifacts using the SAP AI Core Dataset API, when direct access to files in the object store is not possible or desirable. Currently, *****Using a Third-Party API Platform***** and curl interfaces are supported.	Info only	General Availability	New	Technology	Not applicable	SAP Business Technology Platform	2023-04-24	2023-05-02	Unavailable	Not for sovereign cloud offerings

Component	Environment	Title	Description	Action	Lifecycle	Type	Line of Business	Modular Business Process	Product	Latest Revision	Available as of	Version	Scope
SAP AI Core	Cloud Foundry	Metadata in Response to List Executables	When you use the endpoint to list executables, the response body now contains metadata about the parameters and artifacts. For parameters, the `description` and `default` values are returned. For artifacts, the `kind`, `description`, and `labels` can be added using annotations. For more information, see List Executables, Workflow Templates, and Serving Templates.	Info only	General Availability	Changed	Technology	Not applicable	SAP Business Technology Platform	2023-04-02	2023-04-02	Unavailable	Not for sovereign cloud offerings

Component	Environment	Title	Description	Action	Lifecycle	Type	Line of Business	Modular Business Process	Product	Latest Revision	Available as of	Version	Scope
SAP AI Core	Cloud Foundry	Contribute to Our Documentation	At SAP, we endeavor to make sure that our documentation works for you. If you feel that something is missing or that something doesn't quite hit the mark, you can now provide feedback and suggest changes directly from SAP Help Portal. You can do so in one of two ways: • Click *Edit* in the toolbar to open the document in GitHub. There, you can suggest a change and submit a pull request for us to review.	Info only	General Availability	New	Technology	Not applicable	SAP Business Technology Platform	2023-03-13	2023-03-13	Unavailable	Not for sovereign cloud offerings

Component	Environment	Title	Description	Action	Lifecycle	Type	Line of Business	Modular Business Process	Product	Latest Revision	Available as of	Version	Scope
			• Click *Feedback* in the toolbar to create a GitHub issue and tell us how we can improve the documentation for you. For more information, including guidelines on how to contribute, see Open Documentation Initiative.										

Component	Environment	Title	Description	Action	Lifecycle	Type	Line of Business	Modular Business Process	Product	Latest Revision	Available as of	Version	Scope
SAP AI Core	Cloud Foundry	Bulk PATCH Endpoint to STOP or DELETE Multiple Executions or Deployments	Executions and Deployments can now receive PATCH requests for bulk adjustments, provided `bulkUpdates` is enabled in the relevant template. For more information, see Workflow Templates and Serving Templates.	Info only	General Availability	New	Technology	Not applicable	SAP Business Technology Platform	2023-02-27	2023-02-09	Unavailable	Not for sovereign cloud offerings

Component	Environment	Title	Description	Action	Lifecycle	Type	Line of Business	Modular Business Process	Product	Latest Revision	Available as of	Version	Scope
SAP AI Core	Cloud Foundry	Sync Endpoint for Argo CD	In addition to automatically syncing applications, you can request a sync manually by using an API endpoint. For more information on syncing applications, see Create an Application.	Info only	General Availability	New	Technology	Not applicable	SAP Business Technology Platform	2023-02-27	2023-02-09	Unavailable	Not for sovereign cloud offerings
SAP AI Core	Cloud Foundry	WebHDFS Artifacts	WebHDFS artifacts are now supported. For more information about WebHDFS artifacts on SAP AI Launchpad, see Register an Object Store Secret.	Info only	General Availability	New	Technology	Not applicable	SAP Business Technology Platform	2023-02-27	2023-02-09	Unavailable	Not for sovereign cloud offerings
SAP AI Core	Cloud Foundry	Periodic Scheduling	Executions can be run automatically, to a prepared schedule. For more information, see Training Schedules.	Info only	General Availability	New	Technology	Not applicable	SAP Business Technology Platform	2023-02-27	2023-02-09	Unavailable	Not for sovereign cloud offerings

Component	Environment	Title	Description	Action	Lifecycle	Type	Line of Business	Modular Business Process	Product	Latest Revision	Available as of	Version	Scope
SAP AI Core	Cloud Foundry	Deployment Duration can be Limited	You can now use the `ttl` parameter to limit the duration of deployments to hours, days, or weeks.	Info only	General Availability	New	Technology	Not applicable	SAP Business Technology Platform	2022-11-20	2022-11-20	Unavailable	Not for sovereign cloud offerings
SAP AI Core	Cloud Foundry	The YAML Files for Starter Tutorials Available Directly from GitHub	The relevant tutorial steps have been updated to include the link to the associated files. The YAML code can also be copied and pasted directly from the tutorials as before.	Info only	General Availability	Changed	Technology	Not applicable	SAP Business Technology Platform	2022-11-20	2022-11-20	Unavailable	Not for sovereign cloud offerings

Component	Environment	Title	Description	Action	Lifecycle	Type	Line of Business	Modular Business Process	Product	Latest Revision	Available as of	Version	Scope
SAP AI Core	Cloud Foundry	Enhancements to the `GET Deployments` API Call Response	The `GET Deployments` response provides the number of min and max and running replicas, and the resource plan name. For more information, see the SAP AI Core API specification.	Info only	General Availability	Changed	Technology	Not applicable	SAP Business Technology Platform	2022-10-18	2022-10-18	Unavailable	Not for sovereign cloud offerings
SAP AI Core	Cloud Foundry	Azure Blob Storage Supported	You can now register Azure Blob Storage secrets and use them for Model Serving. For more information about registering secret, see Register an Object Store Secret	Info only	General Availability	New	Technology	Not applicable	SAP Business Technology Platform	2022-10-18	2022-10-18	Unavailable	Not for sovereign cloud offerings

Component	Environment	Title	Description	Action	Lifecycle	Type	Line of Business	Modular Business Process	Product	Latest Revision	Available as of	Version	Scope
SAP AI Core	Cloud Foundry	Deployment Duration Variable	You can now limit the duration of a deployment by specifying the length of time that it should run for, in whole minutes hours or days.	Info only	General Availability	New	Technology	Not applicable	SAP Business Technology Platform	2022-10-04	2022-10-04	Unavailable	Not for sovereign cloud offerings

Component	Environment	Title	Description	Action	Lifecycle	Type	Line of Business	Modular Business Process	Product	Latest Revision	Available as of	Version	Scope
SAP AI Core	Cloud Foundry	Metrics Extension	If you have an AI API-enabled runtime, you can use the new “Metrics” extension to query which capabilities of the `metrics` endpoint are supported. You can specify the metrics capabilities to a very fine level of granularity, which lets you react in your client implementation accordingly. For more information, see Metrics Extension.	Info only	General Availability	New	Technology	Not applicable		2022-09-19	2022-09-19	Unavailable	Not for sovereign cloud offerings

Component	Environment	Title	Description	Action	Lifecycle	Type	Line of Business	Modular Business Process	Product	Latest Revision	Available as of	Version	Scope
SAP AI Core	Cloud Foundry	Git Repository Name No Longer Mandatory	When you register your git repository, you no longer have to enter the repository name. The field has been removed. Repositories already registered with names assigned will still work.	Info only	General Availability	Announcement	Technology	Not applicable	SAP Business Technology Platform	2022-09-19	2022-09-19	Unavailable	Not for sovereign cloud offerings

Component	Environment	Title	Description	Action	Lifecycle	Type	Line of Business	Modular Business Process	Product	Latest Revision	Available as of	Version	Scope
SAP AI Core	Cloud Foundry	API Deprecation and Decommissioning	API endpoints **`POST /lm/configurations/{configurationId}/executions`**, **`POST /lm/configurations/{configurationId}/deployments`**, **`GET lm/kpis`** and **`GET /analytics/resourceGroups`** are deprecated. Please update your existing API calls. For timetables and more information, see SAP Note 3239609.	Required	Deprecated	Changed	Technology	Not applicable	SAP Business Technology Platform	2022-09-05	2022-09-05	Unavailable	Not for sovereign cloud offerings

Component	Environment	Title	Description	Action	Lifecycle	Type	Line of Business	Modular Business Process	Product	Latest Revision	Available as of	Version	Scope
SAP AI Core	Cloud Foundry	Serving Component Upgraded to **KServe** Version 0.7 for Security Compliance	The upgrade means some changes to serving templates: the `spec.template.api` version changes from `serving.kubeflow.org/v1beta1` to `serving.kserve.io/v1beta1`, the `spec.template.spec.predictor.containers.name` changes from `kfserving-container` to `kserve-container`. There is no support for apiVersion: `serving.kubeflow.org/v1alpha2`. Please update your existing serving templates by Sep-	Required	General Availability	Changed	Technology	Not applicable	SAP Business Technology Platform	2022-08-22	2022-08-22	Unavailable	Not for sovereign cloud offerings

Component	Environment	Title	Description	Action	Lifecycle	Type	Line of Business	Modular Business Process	Product	Latest Revision	Available as of	Version	Scope
			tember 30, 2022.										
SAP AI Core	Cloud Foundry	Supported Object Stores	SAP AI Core supports multiple hyperscaler object stores, such as Amazon S3, OSS, and HANA Data Lake (HDL). For more information about registering secret, see Register an Object Store Secret	Info only	General Availability	New	Technology	Not applicable	SAP Business Technology Platform	2022-06-30	2022-06-30	Unavailable	Not for sovereign cloud offerings

Component	Environment	Title	Description	Action	Lifecycle	Type	Line of Business	Modular Business Process	Product	Latest Revision	Available as of	Version	Scope
SAP AI Core	Cloud Foundry	Enhanced Serving Template Parameters	The `executables.ai.sap.com/cascade-update-deployments` parameter can be used in the serving template, to update associated deployments automatically. For more information, Serving Templates and Change Serving Template and Update Deployments .	Info only	General Availability	New	Technology	Not applicable	SAP Business Technology Platform	2022-06-18	2022-06-18	Unavailable	Not for sovereign cloud offerings
SAP AI Core	Cloud Foundry	Tracking Service Improvement	The performance of tracking is improved.	Info only	General Availability	Announcement	Technology	Not applicable	SAP Business Technology Platform	2022-06-18	2022-06-18	Unavailable	Not for sovereign cloud offerings

Component	Environment	Title	Description	Action	Lifecycle	Type	Line of Business	Modular Business Process	Product	Latest Revision	Available as of	Version	Scope
SAP AI Core	Cloud Foundry	Documentation for the AI api client SDK has moved to PyPi.org.	Documentation for the AI api client SDK has moved to PyPi.org. For more information, Libraries and SDKs.	Info only	General Availability	Announcement	Technology	Not applicable	SAP Business Technology Platform	2022-06-07	2022-06-07	Unavailable	Not for sovereign cloud offerings

Component	Environment	Title	Description	Action	Lifecycle	Type	Line of Business	Modular Business Process	Product	Latest Revision	Available as of	Version	Scope
SAP AI Core	Cloud Foundry	Documentation for the sap-ai-sdk-core has moved to PyPi.org.	Documentation for the sap-ai-sdk-core has moved to PyPi.org. For more information, Libraries and SDKs .	Info only	General Availability	Announcement	Technology	Not applicable	SAP Business Technology Platform	2022-06-07	2022-06-07	Unavailable	Not for sovereign cloud offerings
SAP AI Core	Cloud Foundry	Additional Query Parameter for Metrics	The `$select` query parameter can be used to retrieve metric resource data, such as metrics or custom info selectively. For more information see, Querying Metric Data .	Info only	General Availability	New	Technology	Not applicable	SAP Business Technology Platform	2022-05-05	2022-05-05	Unavailable	Not for sovereign cloud offerings

Component	Environment	Title	Description	Action	Lifecycle	Type	Line of Business	Modular Business Process	Product	Latest Revision	Available as of	Version	Scope
SAP AI Core	Cloud Foundry	Horizontal Scaling for Resource Group Operator Enabled	The Resource Group Operator now allows horizontal scalability, improving performance.	Info only	General Availability	Changed	Technology	Not applicable	SAP Business Technology Platform	2022-04-09	2022-04-09	Unavailable	Not for sovereign cloud offerings

Component	Environment	Title	Description	Action	Lifecycle	Type	Line of Business	Modular Business Process	Product	Latest Revision	Available as of	Version	Scope
SAP AI Core	Cloud Foundry	Resource Group Id Length Extended	The resource group Id length was limited to 10 characters, however longer Ids are now supported. Ids must be of length minimum: 3, maximum: 253. The first and last characters must be either a lower case letter, an upper case letter or, a number. Character entries from the second to penultimate, must be either a lower case letter, an upper case letter, a number, a full stop, or a hyphen. No other special characters are permitted.	Info only	General Availability	Changed	Technology	Not applicable	SAP Business Technology Platform	2022-04-09	2022-04-09	Unavailable	Not for sovereign cloud offerings

Component	Environment	Title	Description	Action	Lifecycle	Type	Line of Business	Modular Business Process	Product	Latest Revision	Available as of	Version	Scope
SAP AI Core	Cloud Foundry	Generic Secret Support at Main Tenant Level	You can now store generic secrets at the main tenant level and use them with a generic service broker. For more information, see Register Generic Secrets.	Info only	General Availability	New	Technology	Not applicable	SAP Business Technology Platform	2022-02-19	2022-02-19	Unavailable	Not for sovereign cloud offerings

Component	Environment	Title	Description	Action	Lifecycle	Type	Line of Business	Modular Business Process	Product	Latest Revision	Available as of	Version	Scope
SAP AI Core	Cloud Foundry	AI API `meta` Endpoint	The new `meta` API endpoint lets clients query the capabilities of a runtime engine that implements AI API. Querying the `meta` endpoint returns information that the client can use to trigger an appropriate response. For example, SAP AI Launchpad could enable or disable certain features based on the capability of the runtime engine implementation of AI API. For more information, see AI API Runtime Implementations.	Info only	General Availability	New	Technology	Not applicable	SAP Business Technology Platform	2022-02-19	2022-02-19	Unavailable	Not for sovereign cloud offerings

3 Concepts

In this section, we'll explore some of the concepts surrounding both the SAP AI Core and SAP AI Launchpad services.

Term	Definition
AI API	An application programming interface that manages and organizes AI artifacts and workflows (such as training scripts, datasets, models, and model servers) across multiple AI runtimes (environments where AI models are executed, such as cloud platforms or edge devices) to facilitate the development, deployment, and monitoring of AI applications.
AI API connection	A link between SAP AI Launchpad and a runtime. This connection is established through the AI API. For more information, see About the AI API.
AI scenario consumer	A user, either paying or non-paying, who utilizes an AI-powered software application, API, or platform for tasks such as automation, prediction, or decision-making. The consumer may have varying levels of technical expertise, ranging from non-technical end-users to experienced developers.
AI scenario producer	The developer or provider responsible for creating, maintaining, and updating an AI scenario. AI scenarios can be managed in three ways: • SAP-managed: Developed and maintained by SAP • Partner-managed: Developed and maintained by SAP partners • In-house: Developed and maintained by the organization using the AI scenario
AI service	A Software as a Service (SaaS) offering provided by SAP Business Technology Platform that enables users to leverage artificial intelligence capabilities for specific use cases. It is typically accessed through an AI API and can be used to expose and deploy AI scenarios as consumable services.
AI use case	A specific application of AI technology designed to generate business value by improving efficiency, reducing costs, or enhancing customer satisfaction. Examples include invoice matching in finance, product recommendations in e-commerce, product review classification in customer service, fraud detection in banking, and predictive maintenance in manufacturing.
artifact	Any tangible by-product produced or consumed by an execution or deployment. Artifacts can include data, files, binaries, libraries, packages, or other resources. They serve as inputs or outputs for various stages of the software lifecycle and are often versioned to track changes and maintain a history of the development process.
configuration	A collection of parameters, artifact references (such as datasets or models), and environment settings that are used to instantiate and run an execution or deployment of an executable or template. A configuration binds specific values to input parameters and specifies the versions of artifacts to be used. It is required to run an execution or deployment, and multiple configurations can be associated with a single executable or template (a 1:n relationship). Configurations are used in both training and serving processes to train models or serve predictions, respectively. They provide a way to customize and control the behavior of the executable or template for different scenarios or use cases.

Term	Definition
dataset	A collection of data used for training, testing, or analysis in artificial intelligence and machine learning applications. A dataset has the following characteristics: • References a specific data source, such as a file, database, or API • Includes metadata describing the structure, format, and content of the data • May require specific tools, libraries, or credentials to access and process the data • Can be used as input to machine learning models, algorithms, or data processing pipelines • Is uniquely identified by an ID, which is used to reference the dataset in configurations and bind it to specific components or tasks • May be versioned to track changes and ensure reproducibility of results • Can be divided into subsets for training, validation, and testing purposes • May undergo preprocessing, cleaning, or augmentation steps to prepare the data for use in machine learning tasks
deployment	An instance of a model serving template (a serving executable or deployment template) that is configured to use a model artifact and apply it to data passed in a serving request (for example, for prediction). A successful deployment creates a model server and generates a deployment URL for inference. As input, a deployment takes one or more models and parameters from a configuration. The serving executable or deployment template defines the expected parameters and input dataset required for the serving process. Values for these parameters and input models are provided by a configuration. Deployments are implemented on a runtime that produces HTTPS endpoints, enabling secure access to the deployed models for inference.
deployment template	A template that defines the parameters and inputs required to serve or deploy one or more trained models. The template is instantiated with a specific configuration, which provides the necessary parameters and inputs. The resulting instance of the template is then used to serve the trained models. Deployment templates are used in the application to facilitate the deployment and serving of machine learning models. A deployment template is also known as a “serving executable”.
embedding	A dense, low-dimensional vector representation that captures the semantic and syntactic information of a discrete input item, such as a word, phrase, or other entity, learned from a large dataset. Embeddings are typically used as input features in machine learning models to represent the underlying meaning and relationships of the items in a continuous space.

Term	Definition
executable	A reusable template that defines a workflow or pipeline for tasks such as training a machine learning model or creating a deployment. It contains placeholders for input artifacts (datasets or models) and parameters (custom key-pair values) that enable the template to be reused in different scenarios. There are two types of executables: • Non-deployable executables: When instantiated, they result in executions that may produce output artifacts. • Deployable executables: When instantiated, they result in deployments that generate URLs for inference. Executables are referred to as "templates" within the application. They can have user-defined labels applied to them, which are listed with the executable's details. The code within an executable defines the training pipeline or model deployment pipeline, leveraging the placeholders for input artifacts and parameters to make the template reusable.
execution	A workflow execution, also known as a "run" in the app, is an instance of a non-deployable executable that represents a single run of a pipeline. There are two types of executions: • Training execution: A run of a training pipeline that takes input artifacts and parameters, and produces trained AI models as output artifacts. • Batch-inferencing execution: A run of a batch-inferencing pipeline that takes input artifacts and parameters, and produces result sets as output artifacts. Each execution is associated with metadata, including metrics, labels, tags, and custom information, which can be queried using the execution ID. For example, a training execution might have metrics like accuracy and loss, while a batch-inferencing execution might have metrics like processing time and number of processed records.
function	A modular, reusable piece of functionality that can be utilized across different scenarios or workflows. Functions are designed to perform specific tasks or computations and can be combined or integrated into larger scenarios to achieve complex business objectives. For example, a data preprocessing function can be used to clean and transform input data before feeding it into a machine learning model, while a model evaluation function can be used to assess the performance of a trained model. Functions promote code reusability, maintainability, and efficiency in building AI solutions.
input artifact	Placeholder in an executable or template that enables the attachment of datasets or models required for the execution of an AI workflow or pipeline. These artifacts can include training data, validation data, pre-trained models, or any other data or model assets needed for the AI process.
job executable	A simplified representation of a workflow executable. A job executable encapsulates the essential components and logic of a workflow, allowing users to easily understand and manage the execution of AI and machine learning tasks within the Functions Explorer interface of SAP AI Launchpad.
job template	A predefined configuration that specifies the parameters and resources required to execute a long-running AI process, such as model training, batch inference, or data preprocessing. It defines the Docker image, compute resources, environment variables, and other settings needed to run the job. Job templates act as blueprints that can be instantiated to create and execute specific job instances with customized parameters.

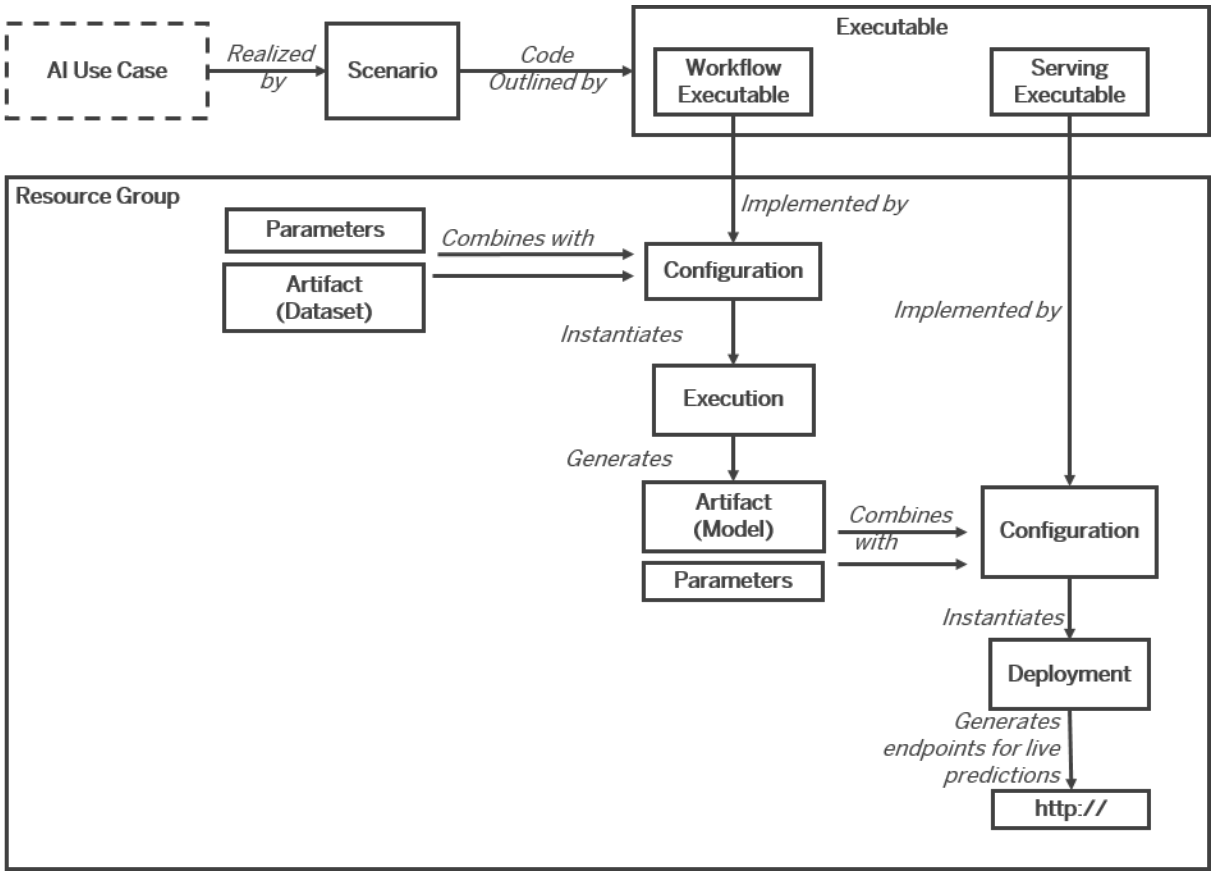
Term	Definition
knowledge graph	A structured representation of entities, their attributes, and the relationships between them, which can be inferred from various data sources, including structured, semi-structured, and unstructured data. Knowledge graphs enable machines to understand and reason about complex real-world concepts and their interconnections, facilitating tasks such as semantic search, recommendation systems, and question answering. Examples of knowledge graphs include ontologies, taxonomies, and knowledge bases.
label	A key-value pair attached to a metric to provide additional context and metadata. Labels are used to classify and categorize metrics, enabling users to filter, group, and aggregate data for better insights and analysis. A set of labels can be applied to each instance of a metric record, allowing for more granular and targeted monitoring.
metrics	A key/value pair where the value is numeric. Metrics are used to measure and monitor the performance, progress, and quality of a model during the training process. Common examples of metrics include accuracy, precision, recall, F1 score, and mean squared error. Metrics can have optional step, timestamp, and label fields, which provide additional information about the metric's context. Every metric (and associated labels), tag, and custom info must be associated with a specific training execution. This association allows for proper organization, tracking, and comparison of different training runs. Once the metric, tag, and custom info are saved, they can be queried using an execution ID. SAP AI Launchpad provides a user-friendly interface to visualize and analyze the captured metrics, enabling users to monitor the model's performance and make informed decisions based on the data.
model	An artifact that is the output of a machine learning training process, representing the learned patterns, parameters, and structure of a trained system. • A model is generated by a training process, which optimizes the model's parameters based on training data. • A model consists of one or more files stored in a hyperscaler storage system (such as SAP AI Core's connected data storage) or the data lake for SAP HANA Cloud. • Each model is uniquely identified by a model ID, which is used in a configuration to bind the model as an input artifact to the serving executable for deployment. • Models can be manually uploaded to the connected data storage or automatically generated and uploaded as the output of a training execution in SAP AI Launchpad. • The model encapsulates the learned knowledge of the AI system and is used for making predictions or decisions on new, unseen data during inference. • Different types of models exist, such as neural networks, decision trees, or clustering models, each suited for different AI tasks and data types. • In SAP AI Core, models are managed and versioned artifacts that can be deployed to serving endpoints for production inference.
model serving template	A predefined configuration that specifies how a trained machine learning model is to be deployed and served for inference in a production environment, typically including details such as the model framework, runtime environment, resources required, and API endpoints.
operations	All activities within the AI lifecycle, including data preparation, model training, model deployment, application integration, model monitoring, and continuous improvement.

Term	Definition
output artifact	Generated results, typically AI models, produced by executing or running a training process or pipeline. When AI models are generated from an execution, they are automatically uploaded and registered to the connected data lake for SAP HANA Cloud or to the hyperscaler data storage, which is a scalable cloud storage solution provided by major cloud service providers such as Amazon Web Services (AWS), Microsoft Azure, or Google Cloud Platform (GCP).
parameter	A placeholder in an executable (script, program, or model) to which a value of a specific data type (String, Integer, Float, Boolean, List, or Dictionary) is assigned during runtime. The values are provided by configurations, which are settings or files that define the parameter values. Parameters are used to provide input data, settings, or options to the executable for customizing its behavior in executions or deployments.
prompt	A natural language instruction or query given to a generative AI model to elicit a response. An AI platform like SAP AI Launchpad includes capabilities for prompt experimentation, management, and administration: • Prompt experimentation: Creating and running natural language prompts with a choice of large language models and adjustable parameters to test and optimize prompts. • Prompt management: Saving, organizing, and managing prompts using collections, metadata (tags and notes), versioning to track changes over time, and deletion when no longer needed. • Prompt administration: Tools for governing prompt usage, monitoring prompt performance, controlling access and permissions, and ensuring responsible AI practices.
resource group	A dedicated workspace for a specific AI scenario or use case. It allows users to organize, manage, and collaborate on related AI entities such as configurations, executions, deployments, and artifacts within a defined scope. Resource Groups ensure isolation and access control, enabling users to work on their AI projects independently while sharing common resources and artifacts as needed. Each Resource Group is associated with a unique subscription to an SAP AI Launchpad tenant.
result set	A data artifact or dataset that contains the results of a batch inference run (execution). A batch inference run uses a trained machine learning model to process an input dataset containing data points or instances and generates predictions or inferences for each data point. The result set is an output artifact that stores these predictions or inferences.
run	A training process that generates a model or models based on a run template. Also known as an "execution".
run template	A template that defines the components required for an AI pipeline within the ML Operations application. It specifies placeholders for parameters, input artifacts (such as datasets), and output artifacts (such as trained models) that are necessary to instantiate and run the executable successfully. The actual values for these placeholders are provided by a configuration, which is a set of settings and parameters specific to a particular use case. In SAP AI Launchpad, a run template is also referred to as a "workflow executable".
runtime	The environment or platform that provides the necessary processing resources to execute AI and machine learning workloads, such as training and inference. It enables businesses to make their applications intelligent by leveraging AI and machine learning technologies, and allows them to train AI services using their data to automate tasks and processes. SAP AI Core is an example of a runtime.

Term	Definition
SAP AI Core	A service in the SAP Business Technology Platform designed to handle the execution and operations of your AI assets (machine learning models, datasets, and other AI-related resources) in a standardized, scalable, and hyperscaler-agnostic way, meaning it can work with different cloud service providers such as AWS, Azure, and Google Cloud Platform. SAP AI Core provides seamless integration with your SAP solutions, enabling you to easily incorporate AI capabilities into applications like SAP S/4HANA, SAP SuccessFactors, or SAP Customer Experience. Any AI function can be realized using popular open-source frameworks such as TensorFlow, PyTorch, or scikit-learn. SAP AI Core supports the full lifecycle management of AI scenarios, including data preparation, model training, deployment, monitoring, and retraining.
SAP AI Launchpad	A multitenant software as a service (SaaS) application on SAP Business Technology Platform (SAP BTP). Customers and partners can use SAP AI Launchpad as a centralized platform to manage AI use cases (scenarios), which are specific applications or problems that can be solved using AI, across multiple instances of AI runtimes (such as SAP AI Core). SAP AI Launchpad also provides access to the generative AI hub, which offers a set of pre-built generative AI models and tools that users can leverage to create and deploy their own generative AI applications.
scenario	An implementation of a specific AI use case within a user's tenant. It consists of a pre-defined set of AI capabilities in the form of executables (pre-built AI models or services that can be directly deployed) and templates (outlines of AI models or services that need to be trained with customer data). A scenario can have multiple versions that correspond to different versions of its contained executables and templates. Placeholders within the executables and templates are populated with customer-specific values using a configuration. The purpose of a scenario is to group executables and templates related to an AI use case and make them available to all AI consumers in the tenant, while also enabling version tracking as the AI models and services evolve over time.
serving executable	A predefined AI pipeline that encapsulates the necessary components and logic to deploy an AI model or application. It serves as a blueprint for creating a specific deployment instance. A deployment template defines placeholders for parameters, input artifacts (such as datasets), and output artifacts (such as trained models). These placeholders are filled with actual values provided by a configuration when the template is instantiated. Instantiating a deployment template involves providing the required configuration, which specifies the concrete values for the parameters and input artifacts. This process creates a deployment instance that can be executed to train, evaluate, or serve an AI model or application. In the SAP AI Launchpad, deployment templates are referred to as “serving executables”.
tag	A name/value pair that is used to categorize and organize test executions. Tags allow for the segregation and grouping of test executions based on specific criteria, making it easier to manage, analyze, and report on the results. For example, you can assign tags to a group of selected test executions to indicate their purpose, priority, or other relevant characteristics. A set of tags can be associated with a MetricResource, which is an entity that represents a specific metric or measurement. The MetricResource, in turn, is linked to an execution, establishing a connection between the tags and the corresponding test run. This relationship enables the effective tracking, monitoring, and evaluation of test executions based on their assigned tags.

Term	Definition
tenant	A logically separated customer instance that represents an organization or a company. Each tenant has its own isolated collection of customized content and services, which are available only to that specific tenant, its users, and the service provider managing the multi-tenant environment.
training	The process of running a machine learning algorithm on a dataset to produce a trained model. The trained model is an artifact that captures the patterns and relationships learned from the data, which can then be used to make predictions or decisions on new, unseen data.
workflow executable	A template that defines the components required for an AI pipeline within the ML Operations application. It specifies placeholders for parameters, input artifacts (such as datasets), and output artifacts (such as trained models) that are necessary to instantiate and run the executable successfully. The actual values for these placeholders are provided by a configuration, which is a set of settings and parameters specific to a particular use case. In SAP AI Launchpad, a workflow executable is also referred to as a "run template".

You can see how these concepts interact from the following diagram:



3.1 AI API Overview

The AI API lets you manage your AI assets (such as training scripts, data, models, and model servers) across multiple runtimes.

Argo workflows and serving templates, as well as their execution and deployment, are managed using the SAP AI Core implementation of the AI API. In SAP AI Core, the Argo workflow and serving templates are mapped under the concept of `Executable`. For the mapping mechanism to work, the Argo workflows and serving templates require certain attributes in the metadata section of the `YAML` file. These attributes are shared by both template types.

SAP AI Core provides additional APIs that are runtime-specific. These are available in the AI Core API specification, which is an extension of the AI API specification.

Related Information

AI Core API
AI API

AI API Runtime Implementations

The AI API specification is a general specification for the lifecycle management of machine learning artifacts. SAP AI Core is one specific runtime implementation of the AI API specification. It is also possible to provide other runtime implementations of the AI API specification, independent of SAP AI Core. This section describes the necessary boundary conditions and implementation requirements.

The benefit of using AI API is that clients can integrate with all AI API-enabled runtime implementations. For example, SAP AI Launchpad can interact with custom runtime implementations as long as the same APIs are provided. Intelligent Scenario Lifecycle Management can also integrate with AI API-enabled runtimes. The SAP AI SDK Base (Python) can also be used (for more information, see sap-ai-sdk-core).

AI API Specification

The AI API specification comprises the following parts:

- A main specification
- Extensions:
  - Analytics extension
  - Resource group extension
  - Dataset management extension
    - Metrics extension

→ Recommendation

Implement at least the main specification, and then implement the extension specifications based on your use case.

AI API Runtime Capabilities Endpoint

`Meta API` is part of the AI API specification (endpoint `/lm/meta`). The implementation must return a configuration response that specifies the capabilities of the AI API runtime implementation.

`Meta API` allows AI API clients to query the capabilities of an AI API implementation so that they can select which commands or user interfaces are available. For example, some AI API runtimes may offer executions but not deployments. They may also offer logs for executions and not for deployments. As an example, if a client of SAP AI Core such as SAP AI Launchpad queries the `Meta API` endpoint of SAP AI Core, the response will be for example:

```
{
  "aiApi": {
    "capabilities": {
      "logs": {
        "deployments": true,
        "executions": true
      },
      "multitenant": true,
      "shareable": true,
      "staticDeployments": true,
      "timeToLiveDeployments": true,
      "userDeployments": true,
      "userExecutions": true,
      "executionSchedules": true
    },
    "limits": {
      "deployments": {
        "maxRunningCount": -1
      },
      "executions": {
        "maxRunningCount": -1
      },
      "minimumFrequencyHour": 1,
      "timeToLiveDeployments": {
        "minimum": "10m",
        "maximum": -1
      }
    },
    "version": "2.18.0"
  },
  "extensions": {
    "analytics": {
      "version": "1.0.0"
    },
    "metrics": {
      "capabilities": {
        "extendedResults": true
      },
      "version": "1.0.0"
    },
    "resourceGroups": {
      "version": "1.2.0"
    }
```

```
  },
  "runtimeApiVersion": "2.21.0",
  "runtimeIdentifier": "aicore"
}
```

SAP AI Launchpad and other clients can then react accordingly and hide the deployments on the user interface for this runtime implementation of AI API.

Capabilities include the following:

Capability	When true, allows users to:
`logs.executions`	View logs for an execution
`logs.deployments`	View logs for a deployment
`multitenant`	Use SAP AI Launchpad as a main tenant user (supports resource groups)
`shareable`	Clients can share one instance
`staticDeployments`	Static, always running endpoints for inference are available without the user having to start a deployment
`userDeployments`	Stop, update, or delete a deployment
`userExecutions`	Stop or delete an execution
`timeToLiveDeployments`	The runtime engine allows defining the time until a deployment is automatically deleted
`analytics`	Review summary information for all tenants
`bulkUpdates`	Stop or delete up to 100 executions or deployments at once
`executionSchedules`	Create schedules

Limits include the following:

Limit	Details
`deployments.maxRunningCount`	Limits the number of running concurrent deployments in a resource group, if any
`executions.maxRunningCount`	Limits the number of running concurrent executions in a resource group, if any
`timeToLiveDeployments.minimum`	The minimum possible value for the ttl parameter in a deployment, if supported
`timeToLiveDeployments.maximum`	The maximum possible value for the ttl parameter in a deployment, if supported
`minimumFrequencyHour`	The minimum possible value for schedule of an execution, if supported

In addition to the general AI API specification, there is also a number of extensions that cover additional use cases. These might not be implemented in all runtime engines.

The extensions are:

Extension	Details
`analytics`	The analytics extension contains endpoints for fetching analytical information of a resource group or tenant
`metrics`	The metrics extension contains endpoints for writing to and reading from metrics endpoints, to store and retrieve metrics generated during executions
`resourceGroups`	The resource group extension contains endpoints for managing resource groups
`dataset`	The dataset extension contains endpoints for uploading and downloading files

Related Information

Serving Templates
Workflow Templates
AI API Specification
Custom Runtime Capabilities Using the Meta API
Analytics Extension
Resource Groups Extension
Intelligent Scenario Lifecycle Management

3.2 Resource Groups

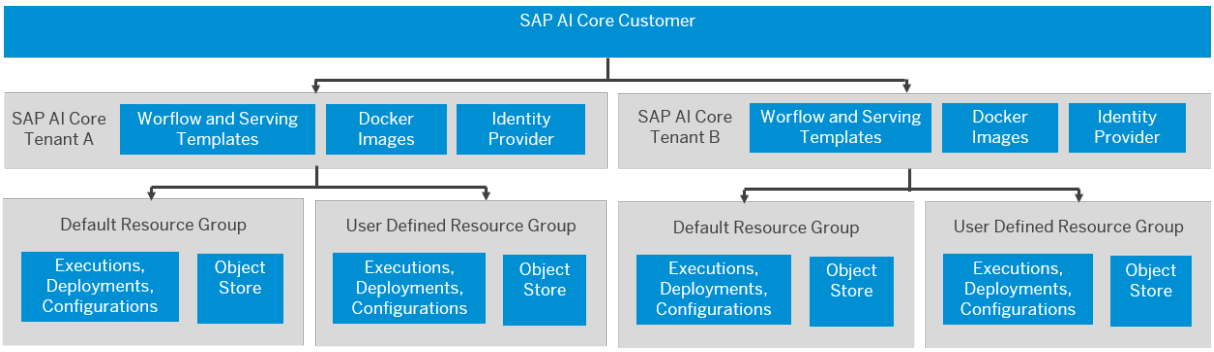
Tenants use resource groups to isolate related ML resources and workloads. Scenarios, executables, and Docker registry secrets are shared across all resource groups.

Resource groups represent a virtual collection of related resources within the scope of one tenant. When a tenant is onboarded, the system immediately creates a default resource group. Tenant administrators can create or delete additional resource groups using the AI API. Tenants can map resource groups based on corresponding usage scenarios.

If a tenant uses resource groups to isolate scenario consumer tenants and those resource groups are deleted, the scenario consumers are deprovisioned. The service does not recognize the scenario consumer of the tenant. The standard XSUAA multitenancy model is followed.

3.2.1 Scope of Resources

Resources that are available for tenants and resource groups differ based on the available scope.



Tenant-Level Resources

Tenant-level resources include:

- Workflow templates
- Serving templates
- Docker registry (containing the Docker images)
- User authentication and authorization (UAA)

User authentication and authorization is based on the SAP AI Core tenant. The tenant is the holder of the access token obtained using the SAP AI Core service key. The SAP AI Core tenant can set the resource group in the request header at runtime, or during lifecycle management, using the AI API. If the resource group is not set, the default resource group is used.

Resource Group-Level Resources

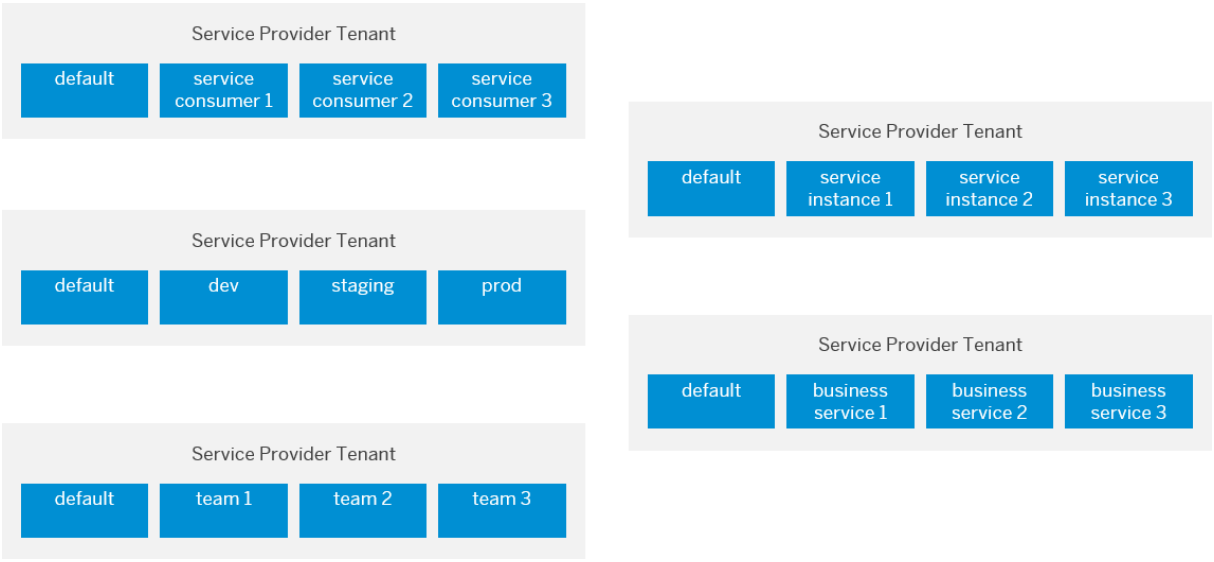
Executables at tenant level are shared across all of the resource groups. At resource group level, the object store is registered by setting the resource group header.

SAP AI Core tenants must consider security aspects in the design of AI functions.

> → Recommendation
>
> Do not use the same object store bucket with the same AWS IAM user for multiple resource groups.

Runtime entities such as executions, deployments, configurations, and artifacts belong to specific resource groups and cannot be shared across resource groups.

Examples of Resource Group Mapping



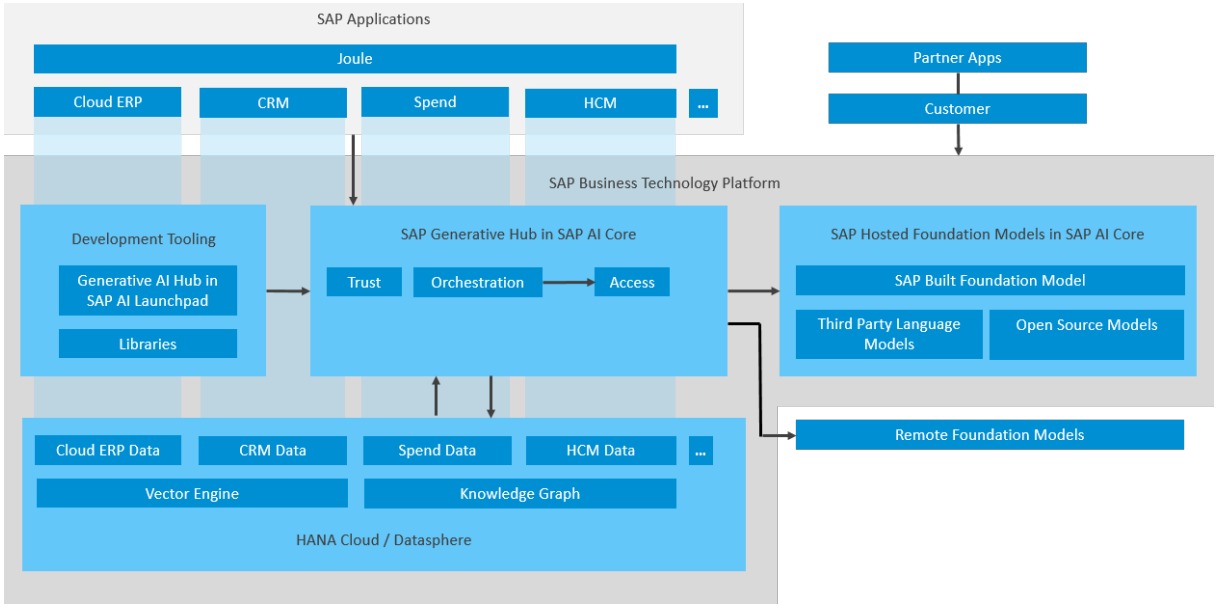
3.3 Generative AI Hub in SAP AI Core Overview

The generative AI hub incorporates generative AI into your AI activities in SAP AI Core and SAP AI Launchpad.

Generative AI models are self-supervised, deep learning models trained on vast amounts of unlabeled data. They use AI technology and industrial-scale computational resources to learn complex patterns and semantic knowledge bases. These models excel in tasks like natural language processing (NLP). By parsing inputs, such as prompts, and predicting target words, they return contextually relevant responses in natural language. A single model can handle multiple tasks by using different input formats and output modes.

Generative AI models are general by design, but you can fine-tune them with additional embeddings. In this way, you can make them suitable for specialized or domain-specific use cases.

SAP AI Core and the generative AI hub help you to integrate LLMs and AI into new business processes in a cost-efficient manner.



Generative AI Hub Architecture Overview

4 Initial Setup

Get started with SAP AI Core using the standard procedures for the SAP BTP, Cloud Foundry environment or Kyma environment.

Enabling the Service in Cloud Foundry [page 127]
Enable SAP AI Core using the standard procedures for the SAP BTP, Cloud Foundry environment.

Enabling the Service in the Kyma Environment [page 147]
Enable SAP AI Core using the standard procedures for the SAP BTP Kyma environment.

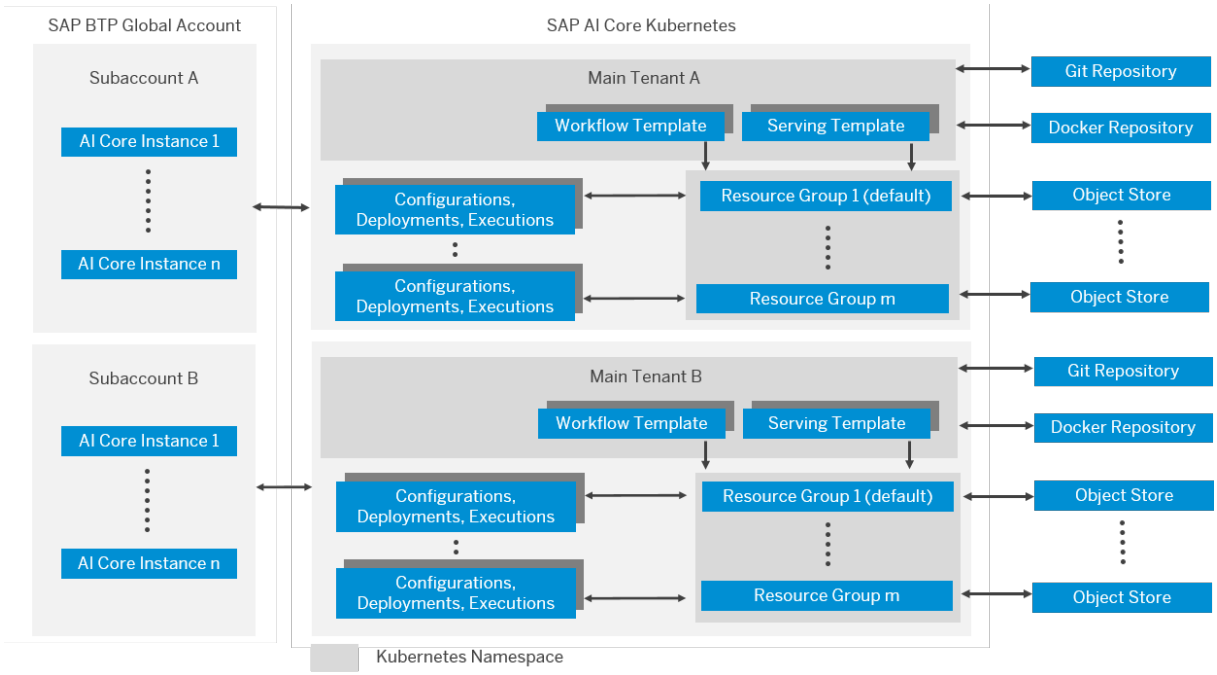
4.1 Enabling the Service in Cloud Foundry

Enable SAP AI Core using the standard procedures for the SAP BTP, Cloud Foundry environment.

When you provision SAP AI Core from the SAP BTP cockpit in SAP Business Technology Platform, the system generates a service key. This key contains the URLs and credentials you need to access the SAP AI Core instance.

SAP AI Core is a tenant-aware reuse service that isolates tenants based on the zone ID, which represents the subaccount. When you create an SAP AI Core service instance within a subaccount, it represents an SAP AI Core tenant.

> **Note**
>
> The SAP AI Core service doesn't isolate tenants based on the service instance ID. If you create multiple service instances within the same subaccount, they all reference the same SAP AI Core tenant.



Parent topic: Initial Setup [page 127]

Related Information

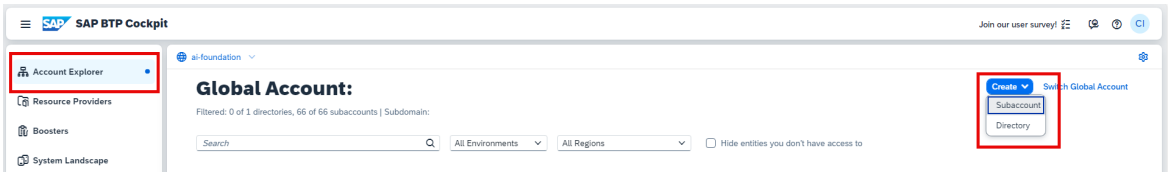
Enabling the Service in the Kyma Environment [page 147]

4.1.1 Create a Subaccount

Create a subaccount in your global account using the SAP BTP cockpit.

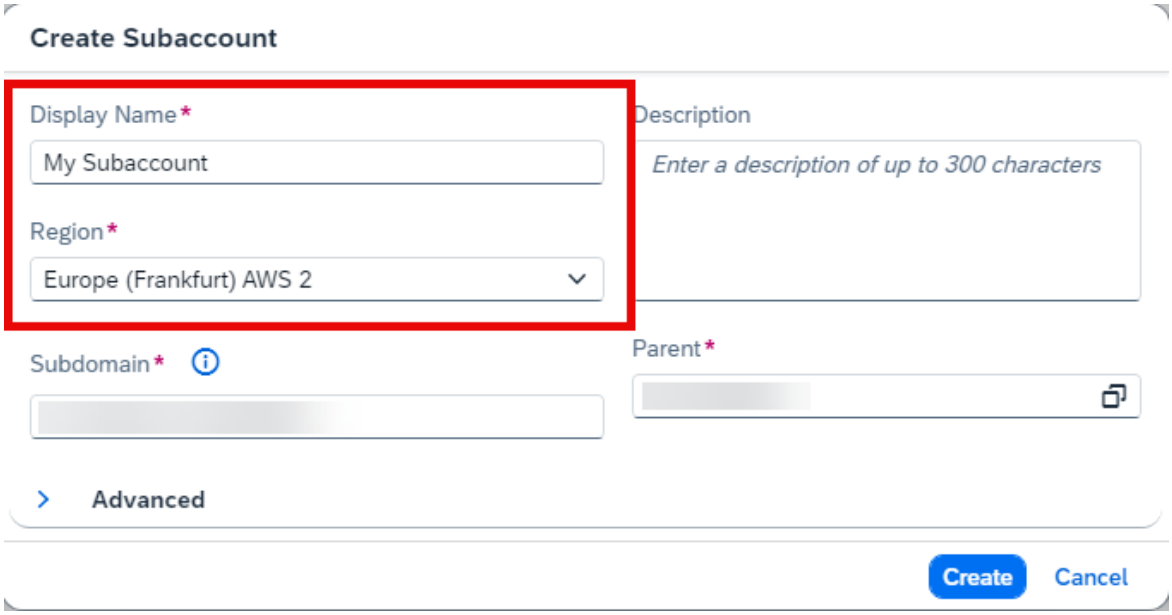
Procedure

1. In the SAP BTP cockpit, choose *Account Explorer* and then click *Create* > *Subaccount*.



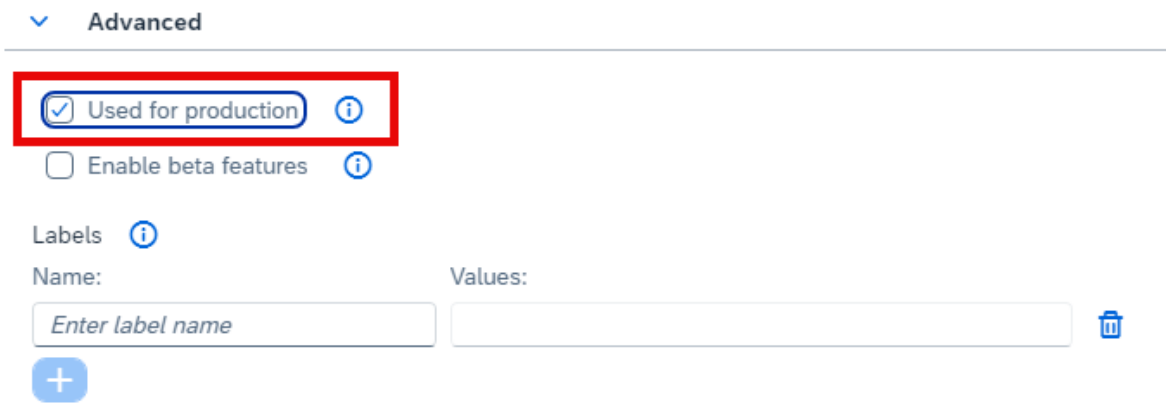
2. In the *Create Subaccount* dialog, enter a name for your subaccount and select the region.

   The parent defaults to the name of your global account.

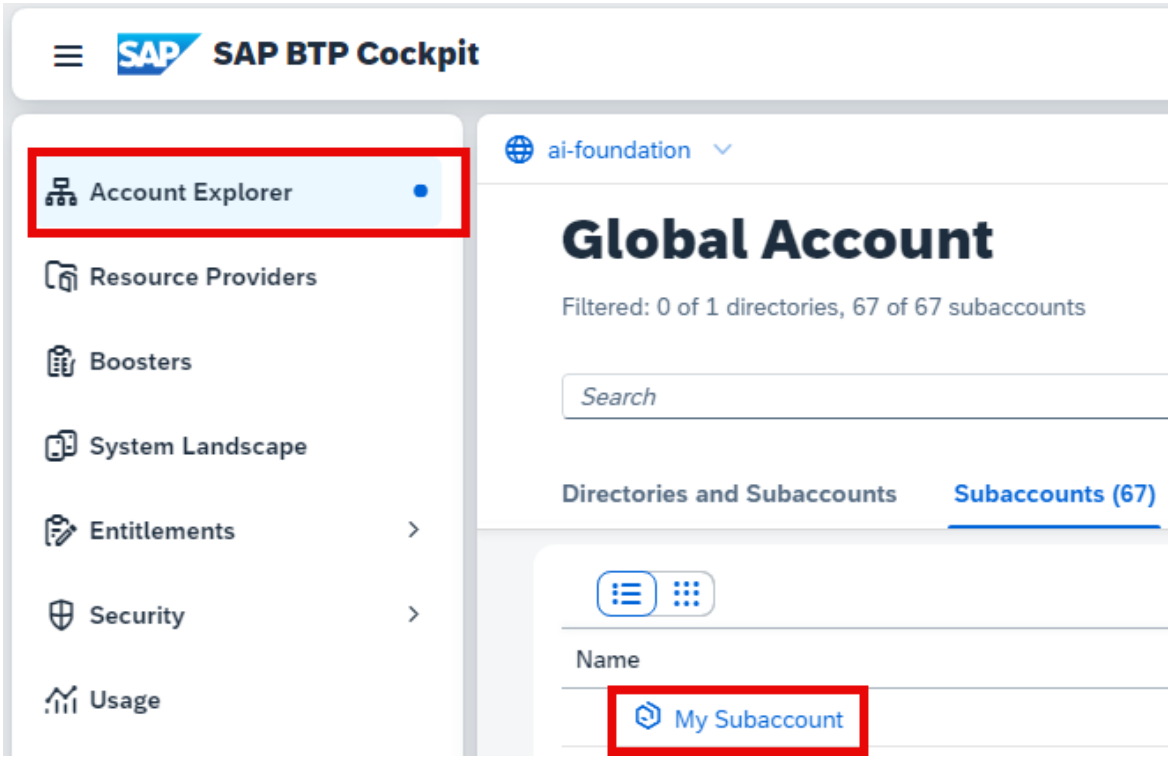


3. **Optional:** If your subaccount is used for production purposes, under *Advanced* select the *Used for production* checkbox.

   This setting does not change the configuration of your subaccount. It is intended to help you manage the production subaccounts in your global account. For example, your cloud operator can refer to it when handling incidents related to mission-critical accounts.



4. Click *Create*.
5. Return to the *Account Explorer* to view your subaccount.



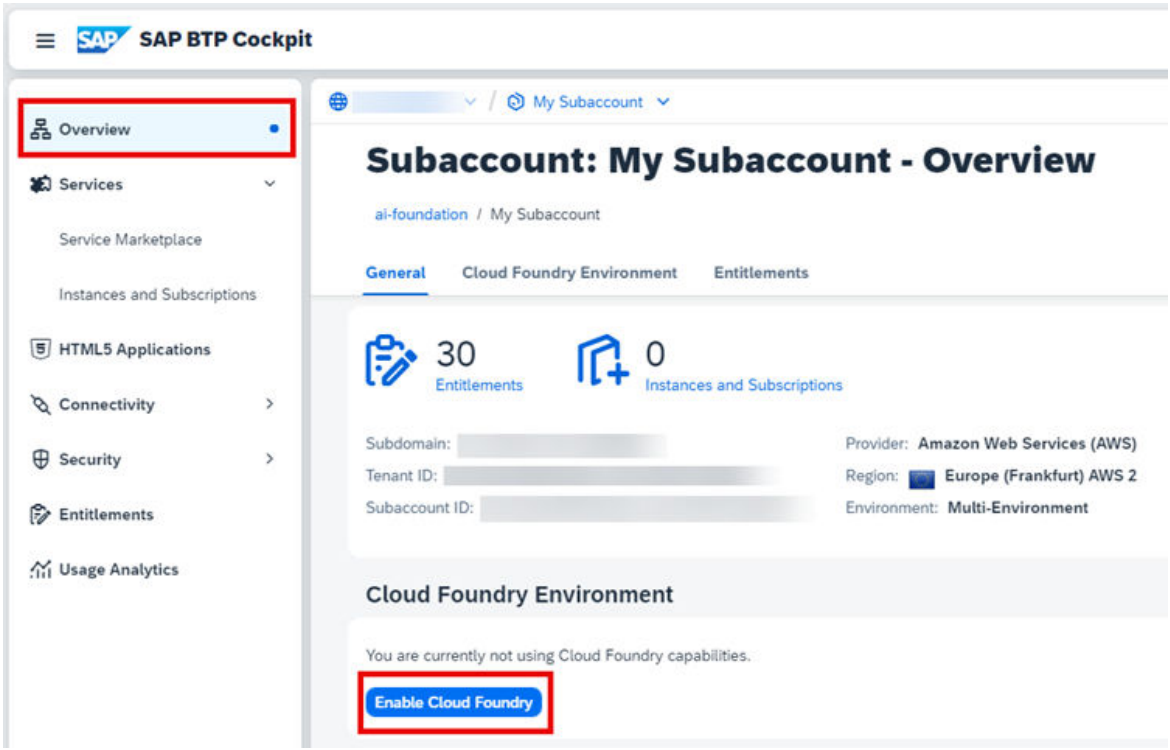
Related Information

SAP BTP Cockpit

4.1.2 Enable Cloud Foundry

Procedure

1. Click your subaccount and on the *Overview* page, choose *Enable Cloud Foundry*.

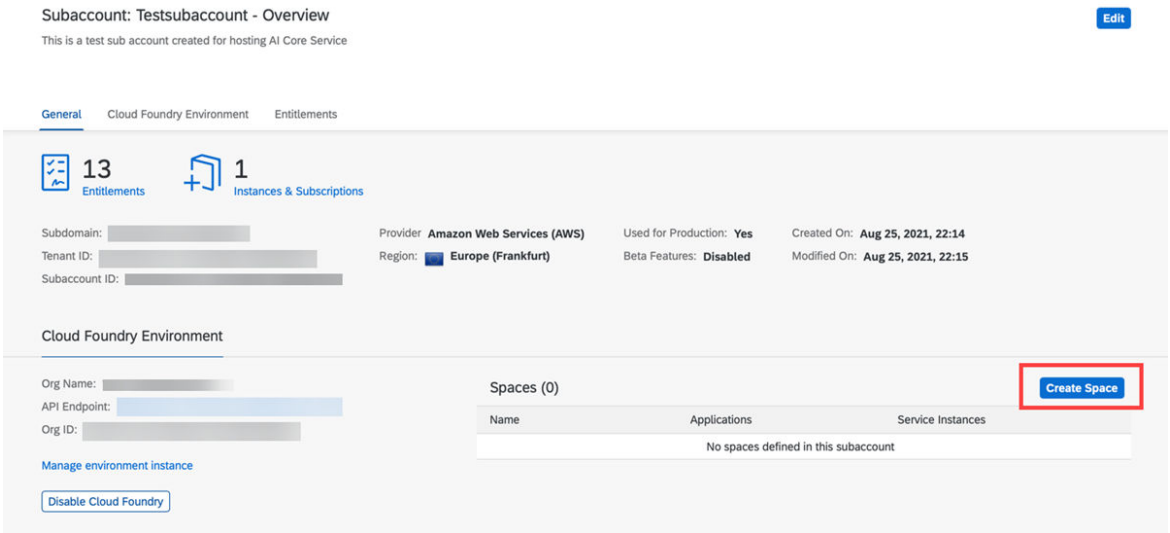


2. Enter the basic information for your Cloud Foundry environment instance and click *Create*.

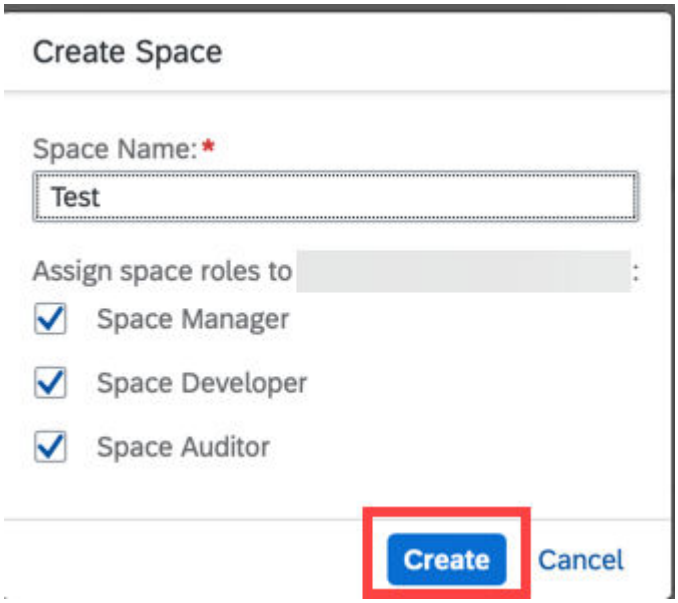
4.1.3 Create a Space

Procedure

1. On the overview page for your subaccount, choose *Create Space*.



2. Enter a name for your space, assign the required roles, and click *Create*.



Related Information

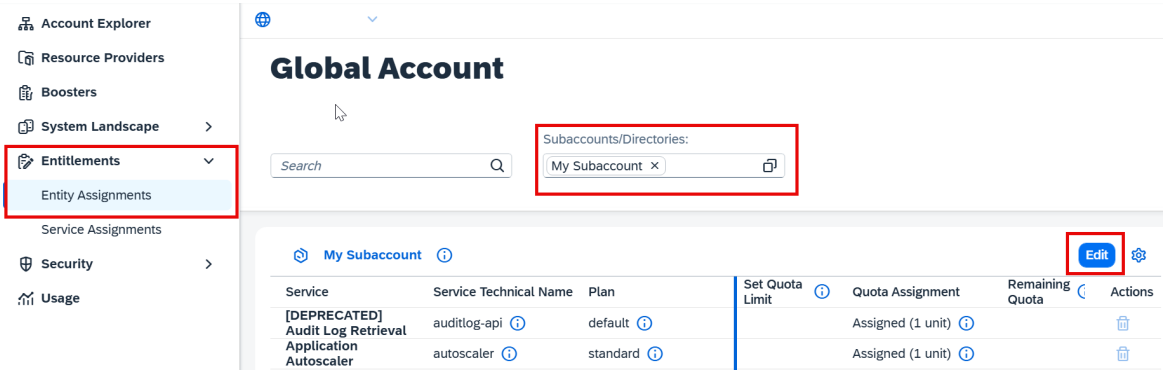
About Roles in the Cloud Foundry Environment

4.1.4 Add a Service Plan

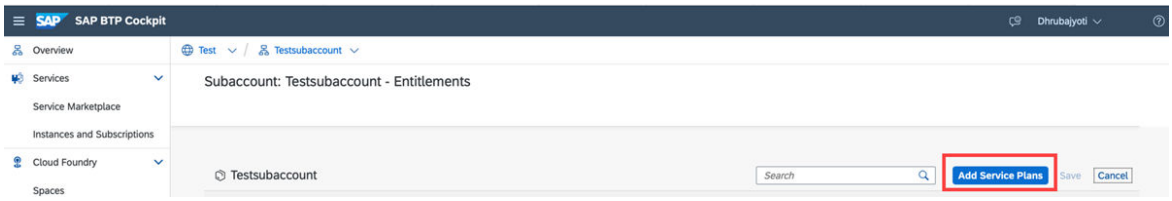
Configure the required entitlements to make SAP AI Core accessible in your subaccount.

Procedure

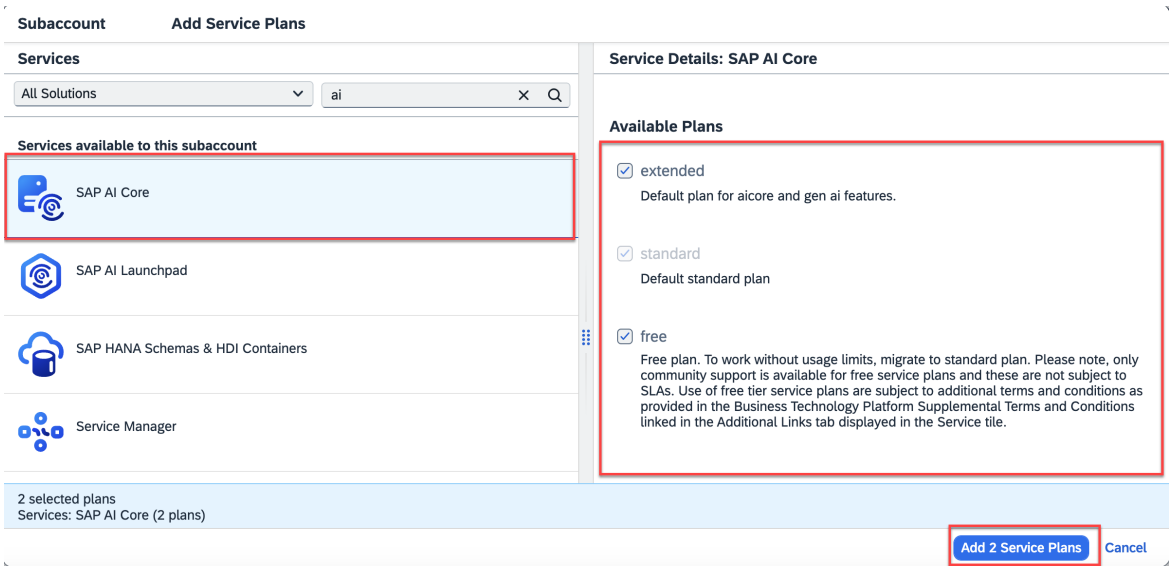
1. In SAP BTP cockpit, navigate to your global account and choose *Entitlements* and then *Entity Assignments*.
2. Use the input help to select your subaccount and click *Edit*.
3. Choose *Edit*.



4. Choose *Add Service Plans*.

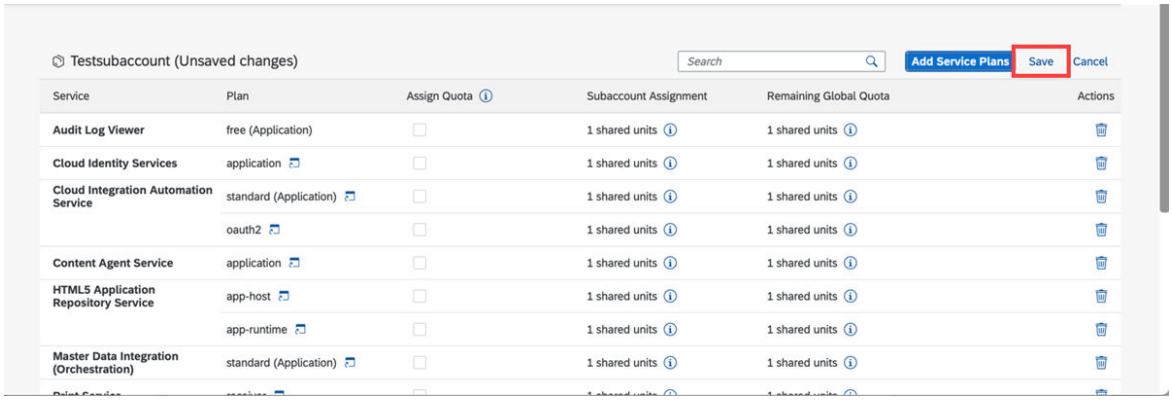


5. Select SAP AI Core and pick your chosen service plan or plans.



> ### Note
>
> To use generative AI capabilities, choose the extended plan. For more information, see Service Plans [page 134].

6. Save your changes.



4.1.4.1 Service Plans

The SAP AI Core service plan you choose determines pricing, conditions of use, resources, available services, and hosts.

Your choice depends on your use case:

- Use the *Standard* plan for production workloads without generative AI.
- Use the *Extended* plan for production workloads with generative AI hub access.

> → Tip
>
> If you want a guided, hands-on environment, you can explore the generative AI hub using the 30-day free trial. For more information, see Try now: 30-Day Basic Trial.

Comparison of Service Plans

Feature / Plan	Standard (Production)	Extended (Production + Generative AI)
Account type	Enterprise account	Enterprise account
Support	Full SAP support with SLA	Full SAP support with SLA
Generative AI hub	✕ Not included	✓ Included
Pricing	Pay per resource + baseline charge	Pay per resource + model/ token usage
Instances per subaccount	1+ instances	1+ instances
Executions/Deployments	Multiple	Multiple
Resource groups	Multiple (quota applies)	Multiple (quota applies)
Resource plan	Choice of resource plans	Choice of resource plans
Upgrade/downgrade rules	You can upgrade from the standard plan to the extended plan.	You cannot downgrade from the extended plan to the standard plan.

> ⓘ Note
>
> For supported regions, see SAP Discovery Center.

Quotas

Deployment Quotas

Each tenant is assigned a default quota that limits the number of deployments and replicas per deployment. If you reach this quota, your deployment will not be created and you will be notified accordingly. You can free up your quota by deleting existing deployments.

Alternatively, you can request an increase to your quota by creating a ticket on component **`CA-ML-AIC`**. Enter the description **`Request to Increase Quota`** and include details about the size of your increase, whether you want to include deployments, replicas, or both, and your subaccount ID.

Resource Group Quotas

> ⚠ Restriction
>
> The maximum number of resource groups is limited at tenant level to 50. If you reach this limit, you receive an error message. To free up space, delete some resource groups. Alternatively, raise a ticket to increase your quota.
>
> For more information, see Delete a Resource Group.

Tenant-Wide Generic Secrets Quotas

Each tenant can have a maximum of five tenant-wide secrets. If you reach this limit, you receive an error message. To free up space, delete tenant-wide secrets as described at Delete a Generic Secret. Alternatively, submit a ticket to request an increase in your quota.

Related Information

SAP Discovery Center
SAP BTP Service Description Guide
Choose an Instance

4.1.4.1.1 Update a Service Plan

Update your SAP AI Core service instance from the standard plan to the extended plan while keeping your data and models.

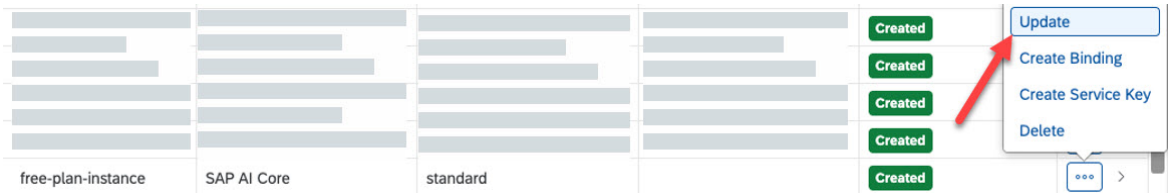
Context

During the update, all metadata and transaction data, including trained models, is retained.

> ⚠ Restriction
>
> You cannot downgrade from the extended plan to the standard plan.
>
> If a standard or extended instance is deleted, you cannot create a new standard plan instance.

Procedure

1. Open your global account in the SAP BTP cockpit.
2. Go to your subaccount.
3. In the navigation area, choose *Instances and Subscriptions*.
4. Search for SAP AI Core.
5. At the end of the subscription row, select the ellipsis (...) and choose *Update*.



6. In the wizard that opens, select *default* and click *Update Subscription*.

Results

All data is migrated automatically to your new plan.

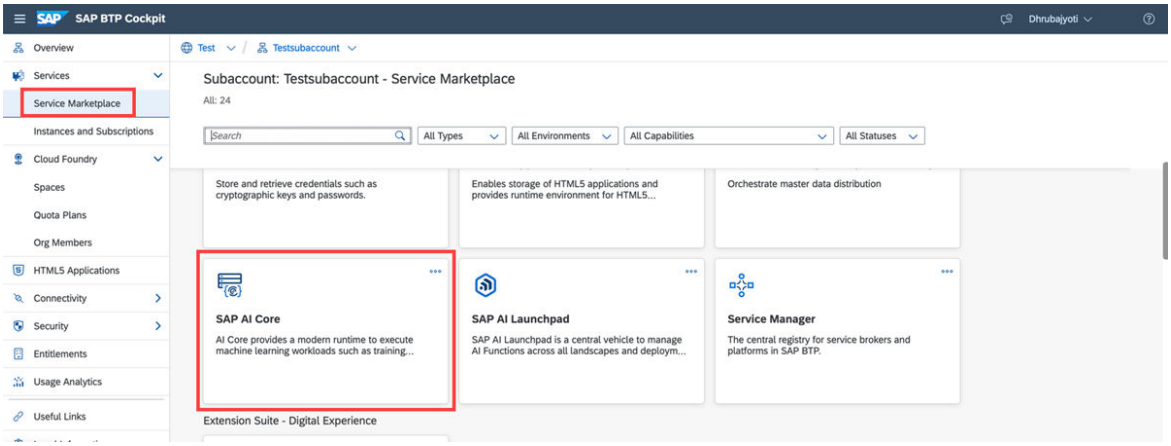
User permissions remain unchanged.

4.1.5 Create a Service Instance

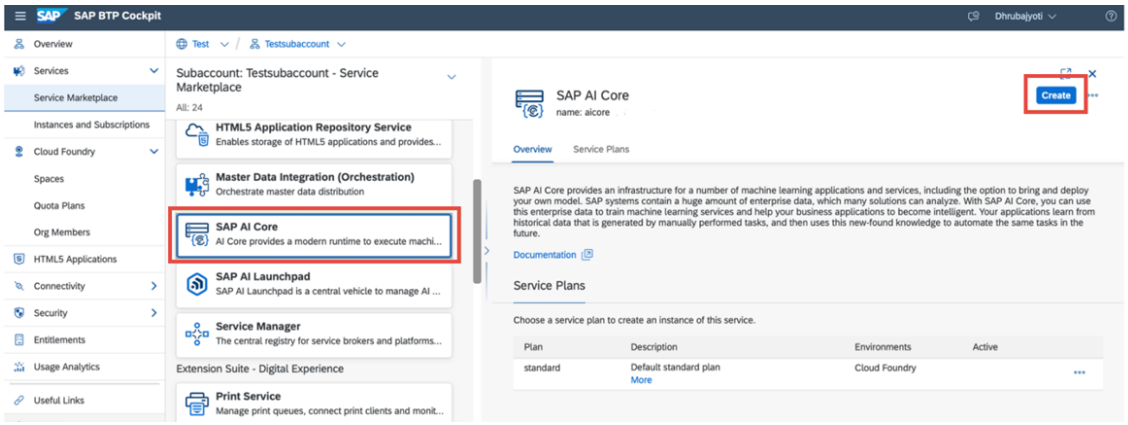
Procedure

1. In SAP BTP cockpit, navigate to your subaccount within your global account and choose *Service Marketplace*.

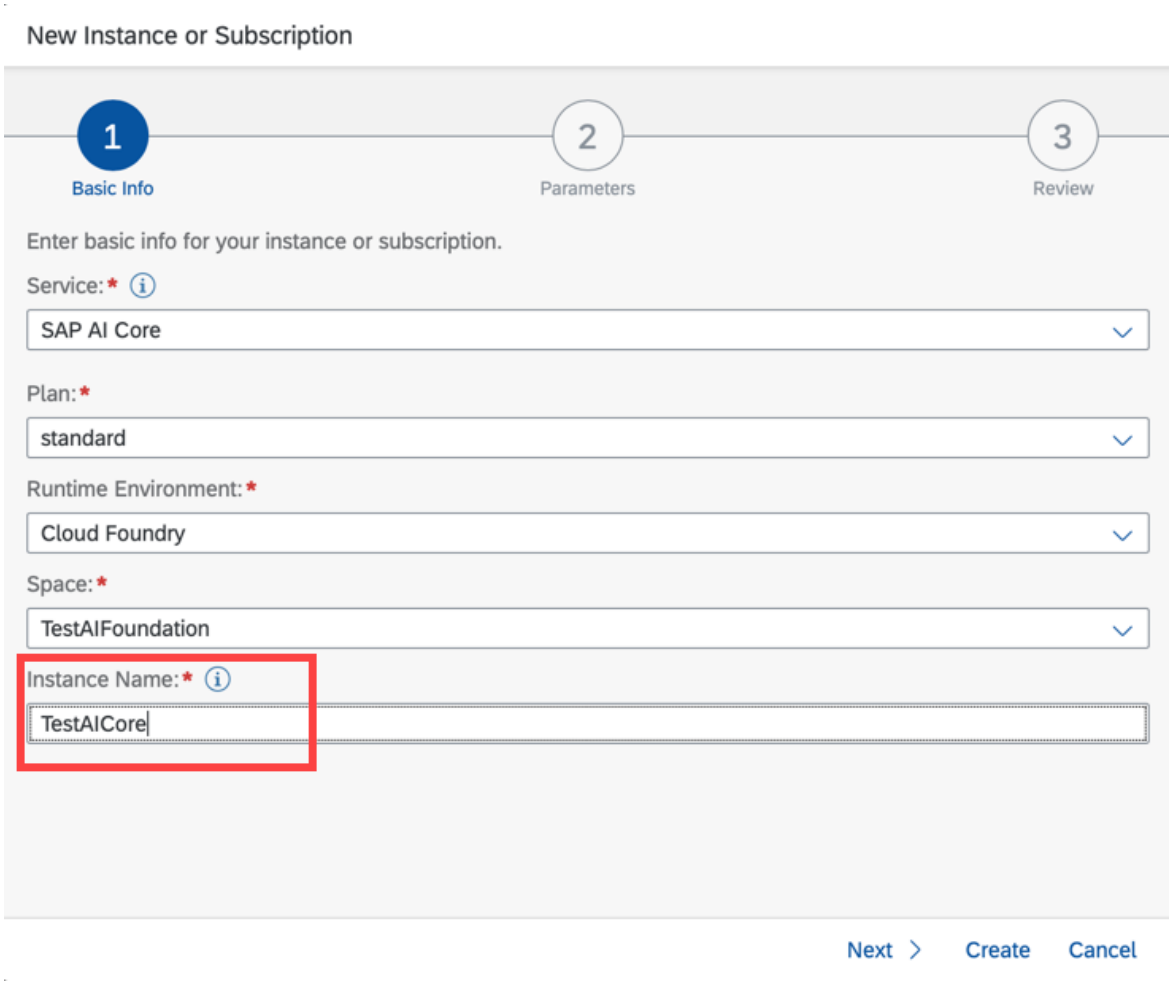
   You will see a tile for SAP AI Core.



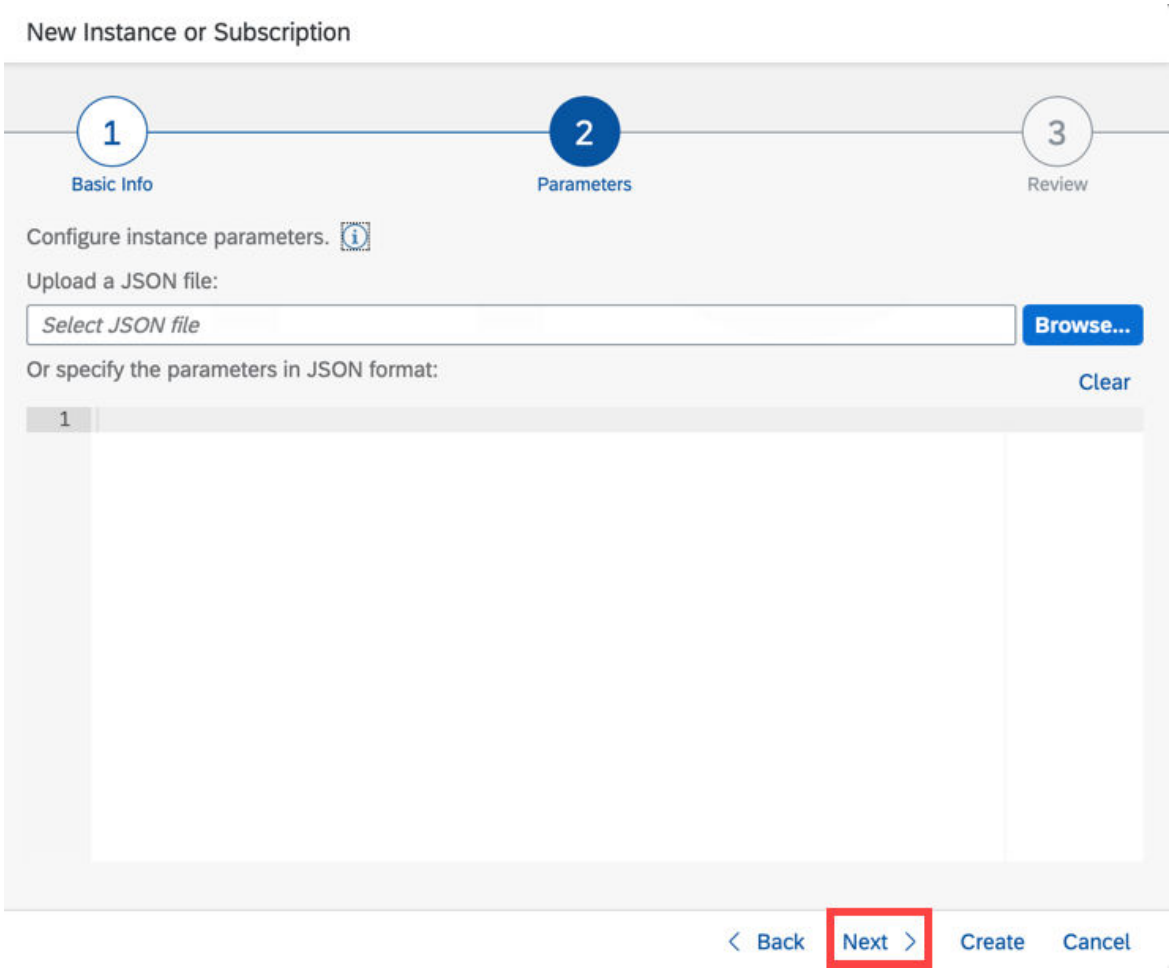
2. Open the tile and click *Create*.



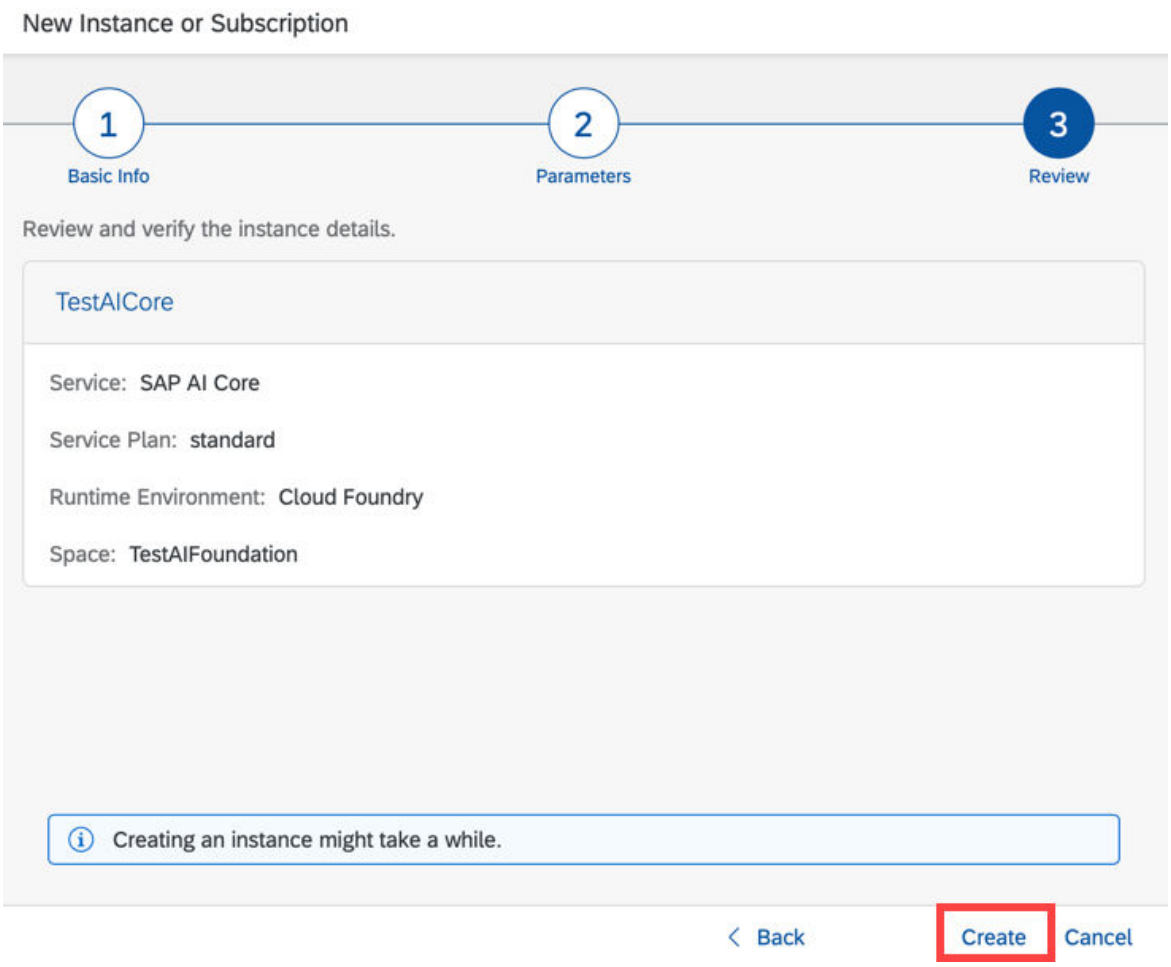
3. Enter a name for your service instance and choose *Next* (all other details will be filled by default).



4. At present, the JSON file upload feature is not supported. Choose *Next* to proceed.

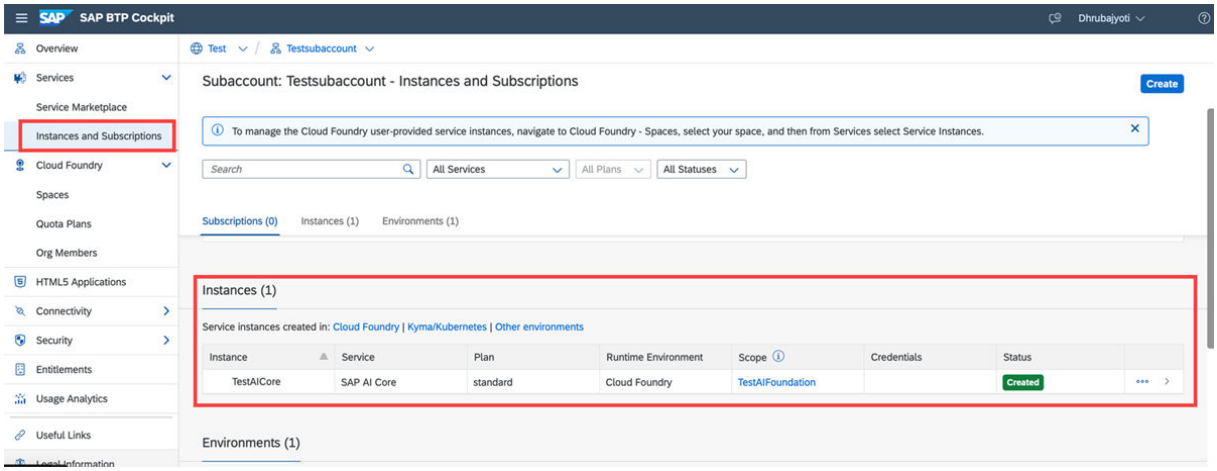


5. Check the data and choose *Create*.



Results

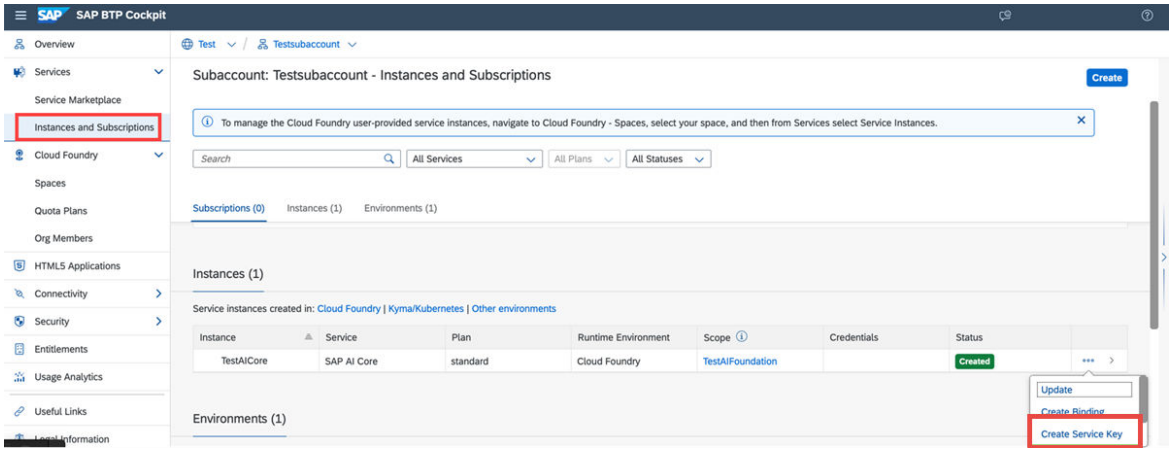
When your service instance is created, you can view it on the *Instances and Subscriptions* page of your subaccount.



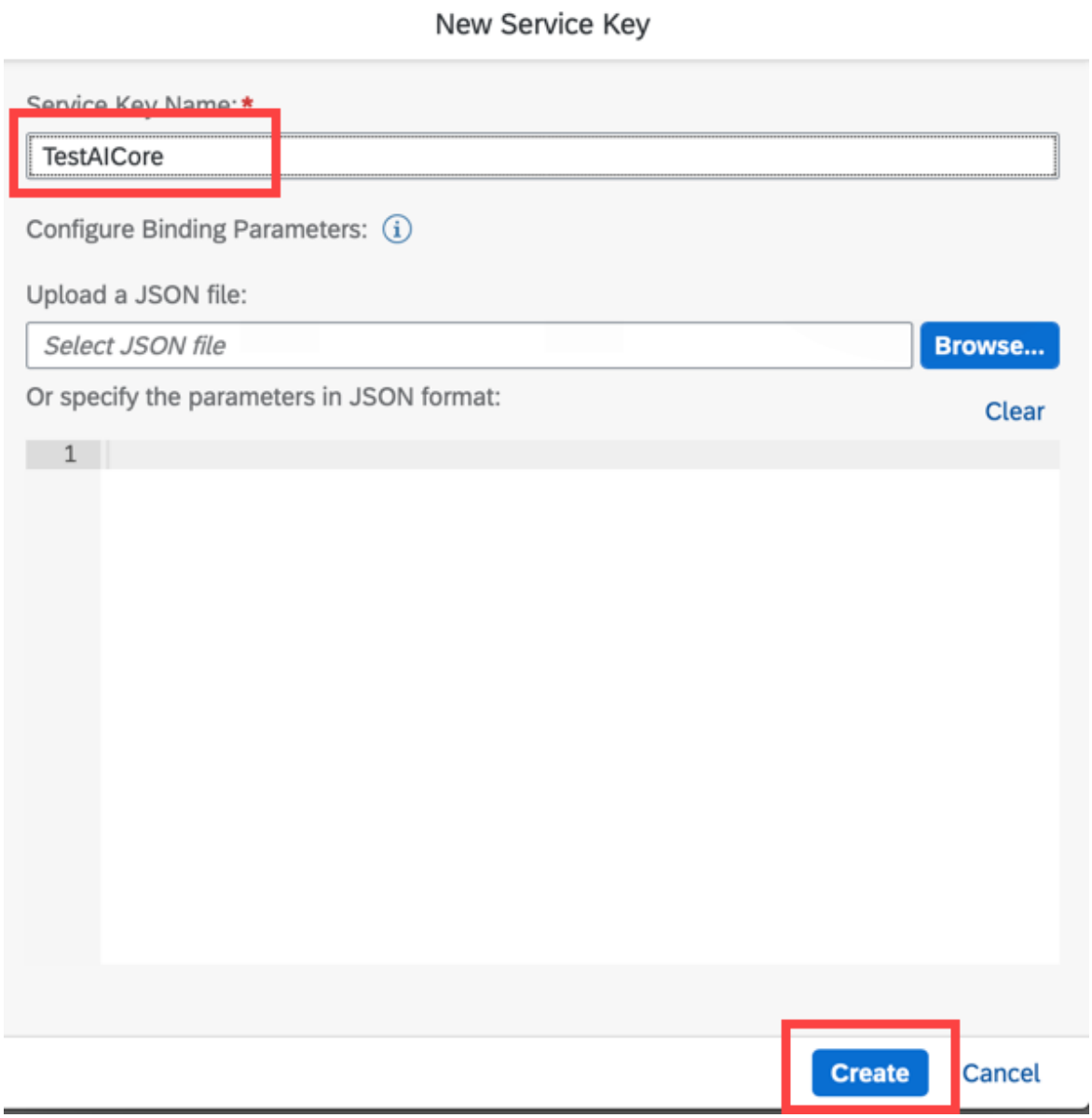
4.1.6 Create a Service Key

Procedure

1. On the *Instances and Subscriptions* page, find your new instance and choose *Create Service Key* from the dropdown.



2. Enter a name for your service key and click *Create*.



> → Remember
>
> You are responsible for the rotation of your access credentials and certificates of SAP AI Core within BTP according to regional policy.

3. **Optional:** To use an x.509 certificate instead of client secret credentials, specify the credentials by adjusting and uploading the following JSON code:

```
{
  "xsuaa": {
    "credential-type": "x509",
    "x509": {
      "key-length": 2048,
      "validity": 7,
      "validity-type": "DAYS"
    }
  }
```

```
}
```

- `key-length`: The byte length of the generated private key. Default: 2048.
- `validity`: The validity time unit. DAYS, MONTHS and YEARS are supported. Default: DAYS.
- `validity-type`: The number of time units. Default: 7.

The default combination is a key of length 2048, valid for 7 DAYS.

4. Download your service key to save it.

Results

You now have your service key, which provides URLs and credentials for accessing the SAP AI Core instance through SAP AI Launchpad, SAP AI Core toolkit, a Third-Party API Platform, or curl.

> → Remember
>
> You are responsible for the rotation of your access credentials and certificates of SAP AI Core within BTP according to regional policy.



If you have generated a client secret, your key will include:

- `clientid`, `clientsecret`, and `url` can be used to generate your authentication token.
- `identityzone` and `identityzoneid` represent your tenant ID.
- `appname` provides the service instance details if service instance isolation is implemented.
- `serviceurls` allow you to interact with SAP AI Core once your authentication token has been generated.

- `AI_API_URL`: Unified AI API to handle ML artifacts (such as training, data, models, and deployments) across multiple hyperscalers.

If you have generated an x.509 certificate, your key will include:

- `certificate`
- `certurl` can be used to generate your authentication token.
- `key` your RSA private key.
- `identityzone` and `identityzoneid` represent your tenant ID.
- `appname` provides the service instance details if service instance isolation is implemented.
- `serviceurls` allow you to interact with SAP AI Core once your authentication token has been generated.
  - `AI_API_URL`: Unified AI API to handle ML artifacts (such as training, data, models, and deployments) across multiple hyperscalers.

4.1.7 Use a Service Key

After you have created your service key, it can be used by local clients, apps in other spaces, or entities outside your deployment to access SAP AI Core through one of the available interfaces.

Using a Third-Party API Platform

Prerequisites

- You have downloaded and installed the API platform of your choice.
- You have familiarized yourself with the documentation and interface of the platform.

Procedure

1. Download the JSON collection from https://api.sap.com/api/AI_CORE_API/overview.
2. Import the JSON file to the API platform.
3. After the import is complete, highlight the collection and select the *Authorization* tab.
4. Navigate to *Configure New Token*, enter the credentials from your service key, and save your changes.

> **Note**
>
> - The *Token Name* field is your choice of descriptive identifier.
> - The *Access Token URL* is labeled *url* in your service key. Add **`/oauth/token`** to the end of the URL.
> - The *Grant Type* should be **`Client Credentials`**.
>
> If you see an alert relating to the characters in your credentials, ignore it.

> ### Note
>
> If you have generated a x.509 certificate instead of client secret credentials, you'll need to use your `certificate`, `key` and `certUrl` to create your token.
>
> For example: `curl --cert <cert.pem> --key <key.pem> -XPOST <certUrl>/oauth/token -d 'grant_type=client_credentials&client_id=<client id>'`

5. Select the *Variables* tab, and set your `baseUrl` from your credentials.

   The `baseUrl` is labeled *AI_API_URL* in your service key.
6. Choose *Save*.
7. In the *Authorization* tab, choose *Get New Access Token*. and wait for the authentication process.
   When the authentication process is complete, select *Use Token* to finish. Check that the token is stored in your environment variables. If it is not stored automatically, you can copy and paste it to the *Token* field manually.

Next Steps

To train and deploy your own AI models, follow the procedure in Administration.

To use generative AI models provided in the generative AI hub, see Generative AI Hub.

Using Curl

Prerequisites

curl is likely to be installed on your operating system by default. To check, open a command prompt and enter **`curl -V`**. If curl isn't installed, download and install it from https://curl.se/ .

> ### Note
>
> On macOS, you may need to install `jq` so that you can follow the curl commands.
>
> 1. Install brew from https://brew.sh/ .
> 2. In a Terminal session, run `brew install jq` to install `jq` in your shell environment.

Procedure

1. Set up your environment as follows:

   For Linux:

```
# XSUAA details
# URLs should be without trailing slash '/'
```

```
export CLIENTID=<clientid> from service key
export CLIENTSECRET=<clientsecret> from service key
export XSUAA_URL=<url> from service key
export AI_API_URL=<AI_API_URL> from service key
```

For Windows PowerShell:

```
$env:CLIENTID = <clientid> from service key
$env:CLIENTSECRET = <clientsecret> from service key
$env:XSUAA_URL = <url> from service key
$env:AI_API_URL = <AI_API_URL> from service key
```

> ### Note
>
> The `export` command sets the values of your keys to your environment variable, meaning that they will be retained after you close your terminal session. It is possible to set the environment variables without the `export`, for the current session only.

2. Get the XSUAA OAuth Token using `clientid` and `clientsecret` from the service key to call the APIs.

   The XSUAA OAuth token is required for authentication when making AI API calls.

   For Linux:

   ```
   SECRET=`echo -n "$CLIENTID:$CLIENTSECRET" | base64 -i - `
   TOKEN=`curl --request POST \
     --url "$XSUAA_URL/oauth/token" \
     --header "Content-Type: application/x-www-form-urlencoded" \
     --data "grant_type=client_credentials" \
     --data "client_id=$CLIENTID" \
     --data "client_secret=$CLIENTSECRET"`
   ```

   For Windows PowerShell:

   ```
   $SECRET = $env:CLIENTID + ":" + $env:CLIENTSECRET
   $base64SECRET =
   [Convert]::ToBase64String([System.Text.Encoding]::UTF8.GetBytes($SECRET))
   $TOKENRESPONSE = Invoke-WebRequest -Method Post "$env:XSUAA_URL/oauth/token"
   -Headers @{ "Authorization" = "Basic $base64SECRET"; "Content-Type" =
   "application/x-www-form-urlencoded" } -Body "grant_type=client_credentials"
   $TOKENJSON = $TOKENRESPONSE.Content | ConvertFrom-Json
   $TOKEN = $TOKENJSON.access_token
   ```

   > ### Note
   >
   > The token is valid for a limited time. Once it has expired, create a new token, using the same code snippet.

   > ### Note
   >
   > If you have generated a x.509 certificate instead of client secret credentials, you'll need to use your `certificate`, `key` and `certUrl` to create your token.
   >
   > For example: `curl --cert <cert.pem> --key <key.pem> -XPOST <certUrl>/oauth/token -d 'grant_type=client_credentials&client_id=<client id>'`

3. Verify that the token has been fetched properly:

   ```
   echo $TOKEN
   ```

You should see a long string of alphanumeric characters:

```
eyJhbGciOiJSUzI1NiIsImprdSI6Imh0dHBzOi8vYWktYWxwaGEtdmFsaWRhdGlvbi0yLmF1dGhlbn
RpY2F0aW9uLnNhcC5oYW5hLm9uZGVtYW5kLmNvbS90b2tlbl9rZXlzIiwia2lkIjoiZGVmYXVsdC1q
d3Qta2V5LTMyODMxMjg2NCIsInR5cCI6IkpXVCJ94ZGU5YjAxNmQ0MDk5YjlmM...
.............................................
...ALdfbMsHoYTtF6fNFbf3ZQ
```

Next Steps

To train and deploy your own AI models, follow the procedure in Administration.

To use generative AI models provided in the generative AI hub, see Generative AI Hub.

4.2 Enabling the Service in the Kyma Environment

Enable SAP AI Core using the standard procedures for the SAP BTP Kyma environment.

Procedure

1. Create a service instance in the Kyma environment.
2. You can then bind the service instance to your application, or you can create a service key to communicate directly with the service instance. See Using SAP BTP Services in the Kyma Environment.

Task overview: Initial Setup [page 127]

Related Information

Enabling the Service in Cloud Foundry [page 127]
Using SAP BTP Services in the Kyma Environment

5 Tutorials

All available missions for SAP AI Core.

AI Use Case	Tutorial Group/Mission	Description
Generative AI	Generative AI with SAP AI Core - Setup	Set up your SAP Business Technology Platform environment to explore SAP AI Core.
Generative AI	Generative AI with SAP AI Core - Orchestration	Get started with generative AI workflows with different vendors of Large language models in SAP AI Core and learn the fundamentals of promoting and embedding with generative AI SDK in SAP AI Core.
Generative AI	Generative AI with SAP AI Core - Foundation Models	Explore foundation large language models that are part of SAP AI Core in a number of use cases.
Getting Started Custom AI	Custom AI with SAP AI Core	Get started with SAP AI Core, learn the fundamentals, create your first custom AI workflow, and move your machine learning code to a production cloud.

6 APIs and API Extensions

Explore APIs and API extensions that you can use with SAP AI Core.

APIs

Resource	Description	More Information
AI Core API	Explore the entities, methods, and endpoints provided by the API for SAP AI Core.	AI Core API
AI API	Explore the entities, methods, and endpoints provided by the runtime-agnostic AI API.	AI API swagger specification
Prompt Registry	Simplify the lifecycle management of prompt templates of your business AI scenarios.	Prompt Registry API
Orchestration	Enhance content generation with additional capabilities for business AI scenarios.	Orchestration API

API Extensions

Resource	Description	More Information
Dataset Management Extension	Manage your dataset files.	AI API Dataset Extension swagger specification.
Resource Groups Extension	Access usage information for a tenant using the AI API.	AI API Admin API swagger specification
Analytics Extension	Add analytics capabilities to the AI API.	AI API Analytics Extension swagger specification
Metrics Extension	Query which capabilities of the metrics endpoint are supported.	AI API Metrics API swagger specification

7 Libraries and SDKs

Explore additional SDKs and libraries that you can use with SAP AI Core.

SDKs Available with SAP AI Core

Resource	Description	More Information
SAP Cloud SDK for AI : JavaScript 	SAP Cloud SDK for AI is the official SDK for SAP AI Core, generative AI hub, and orchestration. You can use SAP Cloud SDK for AI to: • Integrate chat completion into your business applications with SAP Cloud SDK for AI. • Leverage the generative AI hub capabilities of SAP AI Core such as templating, grounding, data masking, content filtering. For more information, see SAP AI Core. • Set up your SAP AI Core instance with SAP Cloud SDK for AI.	• GitHub Repository • Documentation • NPM
SAP Cloud SDK for AI : Java 	SAP Cloud SDK for AI is the official SDK for SAP AI Core, generative AI hub, and orchestration. You can use SAP Cloud SDK for AI to: • Integrate chat completion into your business applications with SAP Cloud SDK for AI. • Leverage the generative AI hub capabilities of SAP AI Core such as templating, grounding, data masking, content filtering. For more information, see SAP AI Core. • Set up your SAP AI Core instance with SAP Cloud SDK for AI.	• GitHub Repository • Documentation • Maven

Resource	Description	More Information
SAP Cloud SDK for AI : Python 	SAP Cloud SDK for AI is the official SDK for SAP AI Core, generative AI hub, and orchestration. The SDK is composed of three python distributions: You can use sap-ai-sdk-base to access the AI API via Python methods and data structures. You can use sap-ai-sdk-core to interact with SAP AI Core for administration and public lifecycle management. You can use sap-ai-sdk-gen to: • Integrate native SDK libraries and LangChain for accessing models on generative AI hub in SAP AI Core. • Leverage the orchestration service of generative AI hub with capabilities such as templating, grounding, data masking, and content filtering. For more information, see Generative AI Hub.	• Pypi Base • Pypi Core • Pypi Gen • Documentation
LiteLLM	The LLMs in the generative AI hub can be accessed from LiteLLM. LiteLLM is an open-source library that supports over 100 LLMs from various providers. It lets you connect LLM and agentic frameworks to the generative AI hub.	• LiteLLM - Getting Started • Agentic code samples

8 Content Packages

Explore additional Content Packages for use with SAP AI Core.

Content Packages Available with SAP AI Core

Resource	Description	More Information
Content package for large language models for SAP AI Core	The content package for large language models for SAP AI Core simplifies the deployment of large language models with integrated and automated workflows.	Content package for large language models for SAP AI Core
Data robot package	The content package for DataRobot integration for SAP AI Core.	Data robot package for SAP AI Core
Computer vision package	The content package for image use cases in SAP AI Core adds image classification and feature extraction and is used with the SAP AI SDK Core.	Computer vision package for SAP AI Core

9 Advanced Features

Explore advanced features, within SAP AI Core.

AI Content as a Service [page 153]
With SAP AI Core, you can publish AI content such as workflows, serving templates, or Docker images as a managed service on the SAP BTP *Service Marketplace*. This allows other tenants to consume your content through standard APIs.

9.1 AI Content as a Service

With SAP AI Core, you can publish AI content such as workflows, serving templates, or Docker images as a managed service on the SAP BTP *Service Marketplace*. This allows other tenants to consume your content through standard APIs.

A **service provider** is the main SAP AI Core tenant that publishes AI content as a service. For example, the service provider provides a workflow template for other users.

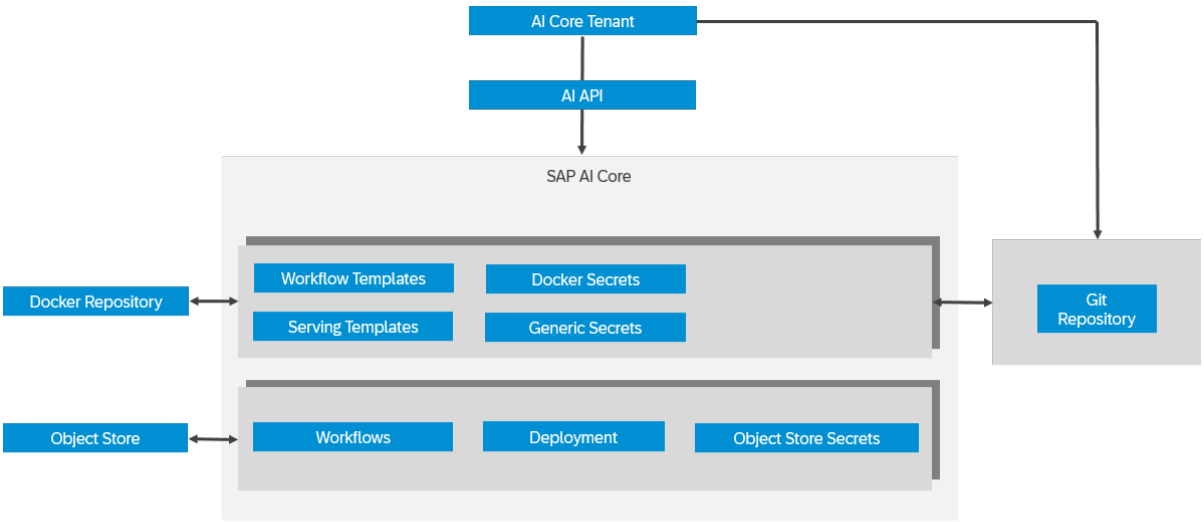
> **Note**
>
> A service provider tenant can provision only one service.

A **service consumer** is the tenant that consumes AI content published by a provider. Consumers create service instances, generate service keys, and use the provided service URL to access the content. They can also start executions or deployments.

Service Provider Flow

As a service provider, you publish your AI content to the SAP BTP *Service Marketplace* as follows:

1. Create consumer-ready AI content (for example, a workflow, serving template, or Docker image).
2. Create a generic secret for broker registration. For more information, see Create a Generic Secret.
3. Provide a service custom resource YAML in a registered git repository.
4. Fetch the service broker information by calling the endpoint: `{{apiurl}}/v2/admin/services`.
5. Register the service broker in the SAP BTP *Service Marketplace* and SAP Cloud Management service.

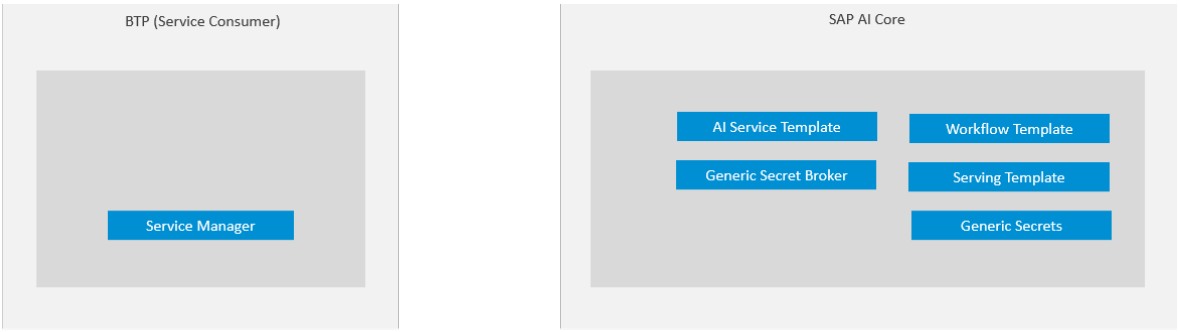


Your AI content is now available in the SAP BTP *Service Marketplace* for consumers to use. The service broker manages onboarding and offboarding of consumers automatically.

Service Consumer Flow

As a service consumer, you create and use a service instance based on the provider's content.

1. In the SAP BTP *Service Marketplace*, create a service instance.
   SAP AI Core creates a resource group for you with:
   `<resourceGroupId> == serviceInstanceId`.
2. Create a service key to authenticate against the service.
3. Use the service URL to start executions or deployments with the AI API (`serviceUrl`).



Parent topic: Advanced Features [page 153]

9.1.1 Service Custom Resource

The service provider main tenant needs to prepare the service custom resource. The custom resource contains service details, reference to broker credentials or secrets, and capabilities configured for service consumers.

An example service custom resource is provided in the following code block:

```
apiVersion: ai.sap.com/v1alpha1
kind: Service
metadata:
    name: sample-service
spec:
    brokerSecret:
        name: broker-credentials
        usernameKeyRef: username
        passwordKeyRef: password
    description: Service used for demos
    capabilities:
        basic:
            staticDeployments: true
            userDeployments: true
            createExecutions: true
            userPromptTemplates: true
                        secrets: true
        logs:
            executions: true
            deployments: true
    serviceCatalog:
    - extendCredentials:
        shared:
            serviceUrls:
                AI_API_URL: https://api.ai.internalprod.eu-
central-1.aws.ml.hana.ondemand.com
      extendCatalog:
        name: sample-service
        id: sample-service-broker-id
        description: sample service
        bindable: true
        plans:
            - id: sample-service-standard
            description: Standard plan for sample service
            name: standard
            free: false
            metadata:
                supportedPlatforms:
                - cloudfoundry
                - kubernetes
                - sapbtp
```

This can be used as a guide, with values amended as required. To create YAML descriptors, use any text editor with a YAML plugin.

For details about parameters, refer to the following table:

Service Parameter Details

Type	Parameter		Description
metadata	name		Name for the service
brokerSecret	name		Secret's name containing credentials to register Service Broker. **Note** It is mandatory to have this secret registered as a generic secret.
	usernameKeyRef		Key reference for username from registered Secret.
	passwordKeyRef		Key reference for password from registered Secret.
capabilities	basic	staticDeployments	Consumers can use existing deployments, but not allowed to Create, Update or Delete (default: true).
		userDeployments	Consumers can create deployments (default: true).
		createExecutions	Consumers can create executions (default: true).
		userPromptTemplates	Consumers can create prompt templates (default: true).
		secrets	Consumers can access to create and read generic secret (default: false)
	logs	executions	Consumers can access execution logs.
		deployments	Consumers can access deployment logs.
		by default	Consumer has access to: • Create, read artifacts & configurations • Download, upload and delete datasets • Create, read, update & delete object store secrets
serviceCatalog	extendCredentials		Used to mention to be consumed Service URL. It will reflect in the service-key.

Type	Parameter	Description
	extendCatalog	Used to extend service catalog.
	enableSharedResourceGroup	Flag used to enable or disable share resource group. For more information, see Shared Resource Group [page 160].

> ⚠ Restriction
>
> Updating of Service Custom Resource is not Supported
>
> Once a service custom resource has synced successfully, updates of any parameters in templates will not take any effect.

9.1.2 Getting Started as a Service Provider

To onboard a service, complete the following:

1. Create a brokerSecret to be used as credentials when registering the service broker.
    1. Use a new POST request to URL `{{apiurl}}/v2/admin/secrets`
    2. Provide credentials (base64 encoded) in the request body in JSON format:

```
{
"name": "broker-credentials",
"data": {
        "username": "bXktc2VjcmV0LWNyZWRlbnRpYWw=",
        "password": "bXktc2VjcmV0LW90aGVyLWNyZWRlbnRpYWw="
        }
}
```

    3. Specify the scope of the request in the header: AI-Tenant-Scope: true
    4. Send the request.
2. Modify the brokerSecret specification section using these details:

```
apiVersion: ai.sap.com/v1alpha1
kind: Service
metadata:
    name: sample-service
spec:
    brokerSecret:
        name: broker-credentials
        usernameKeyRef: username
        passwordKeyRef: password
    description: Service used for demos
    capabilities:
        basic:
            staticDeployments: true
            userDeployments: true
            createExecutions: true
        logs:
            executions: true
            deployments: true
```

```
    serviceCatalog:
        - extendCredentials:
            shared:
                serviceUrls:
                    AI_SVC_URL: https://api.ai.internalprod.eu-
central-1.aws.ml.hana.ondemand.com
        extendCatalog:
            name: sample-service
            id: sample-service-broker-id
            description: sample service
            bindable: true
            plans:
                - id: sample-service-standard
                description: Standard plan for sample Service
                name: standard
                free: false
                metadata:
                    supportedPlatforms:
                        - cloudfoundry
                        - kubernetes
                        - sapbtp
```

> ### Note
>
> The username and password are key names from the broker credentials in the previous step.

3. Update the spec.serviceCatalog[].extendCredentials with the service URL you want to provide to the consumer, which will be part of the service key. Provide catalog details under spec.serviceCatalog[].extendCatalog.
4. Push your service custom resource to your registered GitHub repository, and wait for the sync to be successful.
5. Once it has synced, Check the service details by sending a GET request to URL `{{apiurl}}/v2/admin/services`.

```
{
    "count": 1
    "resources": [
        {
        "name": "sample-service",
        "description": "Service used for demos",
        "status": "PROVISIONED",
        "url": "https://aif-xyzabc.servicebroker.internalprod.eu-
central.aws.ml.hana.ondemand.com"
        }
    ]
}
```

Take note of the service broker URL.

6. Register the service broker using smctl as subaccount-scoped.
    1. Test the registration of the service broker first as subaccount-scoped, before you register it globally. Subaccount-scoped means that your service is automatically visible in the catalog of environments where it's registered.
    Once smctl is installed, login as shown:

```
# env variables
SERVICE_MANAGER_URL=<sm url e.g. https://service-
manager.cfapps.sap.hana.ondemand.com/>
SVC_SUBACCOUNT_USER=<user-with-servicemanager-
role>SVC_SUBACCOUNT_PWD=<password + 2FA>
SERVICE_BROKER_URL=https://aif-xyzabc.servicebroker.internalprod.eu-
central.aws.ml.hana.ondemand.com
```

```
SVC_SUBACCOUNT_SUBDOMAIN=<subaccount e.g. subaccountxyz>
SERVICE_BROKER_USER=<broker username provided in secret>
SERVICE_BROKER_PWD=<broker password provided in secret>
# smctl login
smctl login -a $SERVICE_MANAGER_URL \
    --param subdomain=$SVC_SUBACCOUNT_SUBDOMAIN \
    -u=$SVC_SUBACCOUNT_USER \
    -p=$SVC_SUBACCOUNT_PWD
```

2. Register your service-broker by providing the broker-name, URL, and credentials.

```
# register service-broker
smctl register-broker sample-service $SERVICE_BROKER_URL -b
$SERVICE_BROKER_USER:$SERVICE_BROKER_PWD
# service-broker registration should complete successfully
```

With the service-broker registered successfully, the service is available in the *Service Marketplace*.

```
# assuming you are logged in to Cloud
                                Foundry, provided correct subaccount, get
service plan information
cf marketplace -s sample-service
```

Consumers can now create a service instance and service-key. On creation of service instance, SAP AI Core will create a corresponding resource-group with id = instance id, and the service is now ready for use.

9.1.3 Metering

Describes how SubaccountID and ServiceInstanceID are available as environment variables in the workflow runtime for metering.

For metering purposes, the workflow runtime automatically injects the following environment variables into each workflow pod:

- `AICORE_CAS_SUBACCOUNT_ID`: The Subaccount ID of the CaaS consumer.
- `AICORE_CAS_SERVICE_INSTANCE_ID`: The Service Instance ID of the CaaS consumer.

These environment variables are available to all processes running inside the workflow pod. You can use them to track usage, or integrate with metering system.

In your workflow container, you can access these variables using the following bash script:

```
echo "Subaccount ID: $AICORE_SUBACCOUNT_ID"
echo "Service Instance ID: $AICORE_SERVICE_INSTANCE_ID"
```

9.1.4 Offboarding

To prevent accidental deletion of the service, service providers must provide a deletion strategy as follows:

```
metadata:
    name: sample-service
    annotations:
        ai.sap.com/serviceDeletionStrategy: "delete"
```

With the serviceDeletionStrategy annotation, service providers can delete service custom resources from the git repository and proceed for offboarding. For successful service offboarding, all the consumer service instances should be deleted.

9.1.5 Shared Resource Group

Context

When a consumer creates a service instance of the custom resource, a resource group is automatically created with the ID of the service instance ID. A consumer's access to resources is restricted to this resource group.

A shared resource group can be used i a service provider wants to manage a central deployment and make it available to all of the service consumers

> ⚠ Restriction
>
> Shared resource groups only allow deployment creation for models available in the `foundation-models` scenario, in generative AI hub.

Procedure

1. To enable shared resource group, set `enableSharedResourceGroup` to `true` in the service custom resource.

   > {·} Sample Code
   >
   ```
   ...
       enableSharedResourceGroup: true
       serviceCatalog:
       ...
   ```

   Tenants cannot create, modify or delete a shared resource group using the API. Onboarding for shared resource groups is only possible through the service custom resource..

2. Check the status of your shared resource group by sending a GET request to the endpoint `{{apiurl}}/v2/admin/services/<service-name>`.

   > {·} Sample Code
   >
   ```
   curl --location '$AI_API_URL/v2/admin/services/aisvc-spam-detection' \
   --header 'Authorization: Bearer $TOKEN'
   # API Response:
   {
       "brokerSecret": {
           "name": "broker-credentials",
   ```

```
        "passwordKeyRef": "password",
        "usernameKeyRef": "username"
    },
    "capabilities": {
        "basic": {
            "createExecutions": true,
            "staticDeployments": true,
            "userDeployments": true
        },
        "logs": {
            "deployments": true,
            "executions": true
        }
    },
    "description": "Service exposing AI Content for Spam Detection use-
case",
    "name": "aisvc-spam-detection",
    "serviceCatalog": [
        {
            "extendCatalog": {
                "bindable": true,
                "description": "Service exposing AI Content for Spam
Detection use-case",
                "id": "aisvc-spam-detection-0",
                "name": "aisvc-spam-detection",
                "plans": [
                    {
                        "description": "Standard plan for Spam Detection
Service",
                        "free": false,
                        "id": "standard",
                        "metadata": {
                            "supportedPlatforms": [
                                "cloudfoundry",
                                "kubernetes",
                                "sapbtp"
                            ]
                        },
                        "name": "standard"
                    }
                ]
            },
            "extendCredentials": {
                "shared": {
                    "serviceUrls": {
                        "AI_API_URL": "https://api.ai.internalprod.eu-
central-1.aws.ml.hana.ondemand.com"
                    }
                }
            }
        }
    ],
    "sharedResourceGroupStatus": {
        "id": "shared",
        "isEnabled": true,
        "state": "Provisioned"
    },
    "status": "PROVISIONED",
    "statusMessage": "",
    "url": "https://aisvc-425c28c5-aisvc-spam-
detection.servicebroker.internalprod.eu-central-1.aws.ml.hana.ondemand.com"
}
```

Results

If the `sharedResourceGroupStatus` is `Provisioned`, your shared resource group is provisioned successfully.

Next Steps

To add resources to your shared resource group. include the header `AI-Resource-Group: shared`. Consumers accessing service must also use the header `AI-Resource-Group: shared` to access the shared resource group.

10 Security

Here, we'll explain some of the security aspects of SAP AI Core.

10.1 Security Features of Data, Data Flow, and Processes

The table below shows an overview of the data flow for SAP AI Core.

Overview of Data and Security Measures

Step	Description	Security Measure
1	Transmission control/communication security	Encrypted (HTTPS) communication. Data-in-transit is encrypted using state-of-the-art TLS settings.
2	Application data residing in persistence layer	Encrypted at rest with state-of-the-art encryption keys generated and maintained by SAP.
3	Application data residing in persistence layer	Backup and restore capabilities are implemented and tested regularly. Backups are stored in remote locations and backups are encrypted at rest with state-of-the-art encryption keys generated and maintained by SAP.
4	Access control and separation by purpose	Roles and scopes are available for implementing access control. Customer admin can use standard BTP security administration capabilities to assign roles to users to ensure "least privilege" and "segregation of duties".

10.2 Encryption in Transit

Communication with the service, including data upload and download, is encrypted using the transport layer security (TLS) protocol. SAP services support only the latest protocol versions (that is, TLS v1.2 and later) and strong cipher suites. Your systems must use the supported protocol versions and cipher suites to set up secure communication with the services. They must also validate the certificates against the services' domain names to avoid man-in-the-middle attacks.

10.3 Encryption of Data

SAP AI Core encrypts customer data using tenant-specific encryption keys and integrates with SAP Data Custodian Key Management Service (KMS) to support customer-managed root encryption keys.

SAP AI Core integrates with SAP Data Custodian Key Management Service (KMS). You can use KMS to supply your own root encryption keys and manage customer-managed key (CMK) keychains.

For more information, see SAP Data Custodian Key Management Service.

Data created in SAP AI Core is automatically encrypted using tenant-specific data encryption keys (L4). The L4 keys are encrypted by key encryption keys (L2 and L3), which are protected by the customer-supplied root keys (L1).

> **Note**
>
> Disabling a root key can affect SAP AI Core operations:
>
> - Data encrypted with the key becomes inaccessible.
> - Data currently being processed cannot be persisted.
> - Connected systems may experience service disruptions.

> **Note**
>
> If no root key is provided through Key Management Service, SAP AI Core continues to encrypt data using tenant-specific data encryption keys.

Encrypted Data in SAP AI Core

The following data is encrypted using tenant-specific client-side (L4) encryption keys:

- Data Destination: configuration
- Tabular Artifact: path, CSN document, CSN document reference path
- Workflow templates
- Serving templates

10.4 Authentication and Administration

SAP AI Core uses the authentication mechanisms provided by SAP Authorization and Trust Management service (XSUAA). The credentials to authenticate against the XSUAA to access SAP AI Core are provided as part of the service key for the SAP AI Core service.

For information about SAP Authorization and Trust Management service (XSUAA) in SAP BTP, see What Is the SAP Authorization and Trust Management Service?

10.5 Docker Images

SAP AI Core supports tenant-specific Docker registries (registered via the administration APIs). Additional tenant workloads, such as for execution and deployments, can be created by referencing the Docker images from this Docker registry.

Docker images are cached on virtual machines. These cached Docker images cannot be accessed by other tenants and will not be accessed by SAP.

Cached Docker images are not deleted immediately upon tenant offboarding but are cleaned up as part of operational events such as cluster scaling-down behavior, maintenance, and upgrade of virtual machines.

With every service that you consume, there is a shared security responsibility between you and SAP. Because the creation of a Docker image is the responsibility of the tenant, we strongly recommend that you do not embed or hard-code personal data, sensitive data, or machine learning models inside your Docker images.

For security reasons, Docker containers in SAP AI Core are run as non-root only. For more information, see Workflow Templates.

10.6 AI Content Security

AI content covers workflow templates and serving templates, as well as the Docker images used in them. Docker images contain the machine-learning algorithms or code, along with the machine-learning libraries, frameworks, and other dependent packages. Ensure that you follow standard practices for developing secure software when working with AI content.

> **Note**
>
> Users of AI core are responsible of the content of their docker images and assume the risk of running compromised containers in the platform.

Security Practices

Practice	Description
Threat Modeling	Hold threat modeling workshops to identify and assess security risks or threats in the AI content.
Static Code Scans	Use static code scan (SAST) tools to scan and analyze the code for vulnerabilities.
Open-Source Vulnerability Scan	Scan any open-source components used by the product for vulnerabilities, and patch any vulnerable OSS components found.
Open-Source Strategy	Develop an update strategy that defines when the open-source components used in a product or service are updated to the latest secure version.
Code Reviews	Perform peer code reviews of each code change. The reviewer should take a closer look at the code from a security perspective.
Malware Scanning	Scan for malware in the data uploaded for AI content.

Practice	Description
Secure Code Protection	Support security assurance starting with source code through deployed service. For example, use Docker image digest and image sign verifications.
Docker Base Image Security	Use a secure, light base image for building the Docker images for the AI content. Ensure you use the latest available base image and remove unused components from the Docker image.

10.7 Kubernetes Security

We recommend that you enable the relevant and applicable Kubernetes security features on your workflow templates and serving templates. Ensure that you enable the appropriate Kubernetes features for your workloads.

Restricted Pod Settings

To ensure proper security configurations for containers, certain security-related pod fields must adhere to specific values. There is a large number of pod fields, which change over time. Here, we list the most important fields and their required settings. You do not have to explicitly set these values on your own, however if you set these fields incorrectly, the pod will be rejected.

Setting for `PodSecurityContext`

Capability	Setting
`Add`	Must not be set
`Drop`	Must contain exactly one element: `['all']`
`RunAsUser`	Must be set to `65534`
`RunAsGroup`	Must be set to `65534`
`RunAsNonRoot`	Must be set to `true`. The pod must run as a non-root user
`AllowPrivilegeEscalation`	Must be set to `false`

These settings are enforced internally. If your pod is rejected, it is likely due to one or more of these fields being set incorrectly. Please review your configuration and ensure compliance with the above requirements.

Related Information

Security Best Practices for Kubernetes Deployment

10.8 Configuration Data and Secrets

Workloads can access network resources other than object stores by using credentials at runtime. The ways of including this information at runtime have different standards when it comes to confidentiality.

- For **sensitive** information:
  SAP AI Core allows you to include secrets in the form of generic secrets. Generic secrets are created and managed by the REST APIs in SAP AI Core and consumed securely in a workload.
- For **mTLS authentication** scenarios:
  SAP AI Core provides mTLS certificate secrets. These secrets use managed certificates issued by the SAP BTP Certificate Service. Unlike generic secrets, you do not supply the credential payload, SAP AI Core generates the certificate and private key for you. For more information, see Manage mTLS Certificate Secrets.
- For **non-sensitive** information:
  You can include non-sensitive parameters using configurations or labels.

  > ### Note
  >
  > These parameters may be returned in clear text (for example, in GET requests).

10.9 Output Encoding

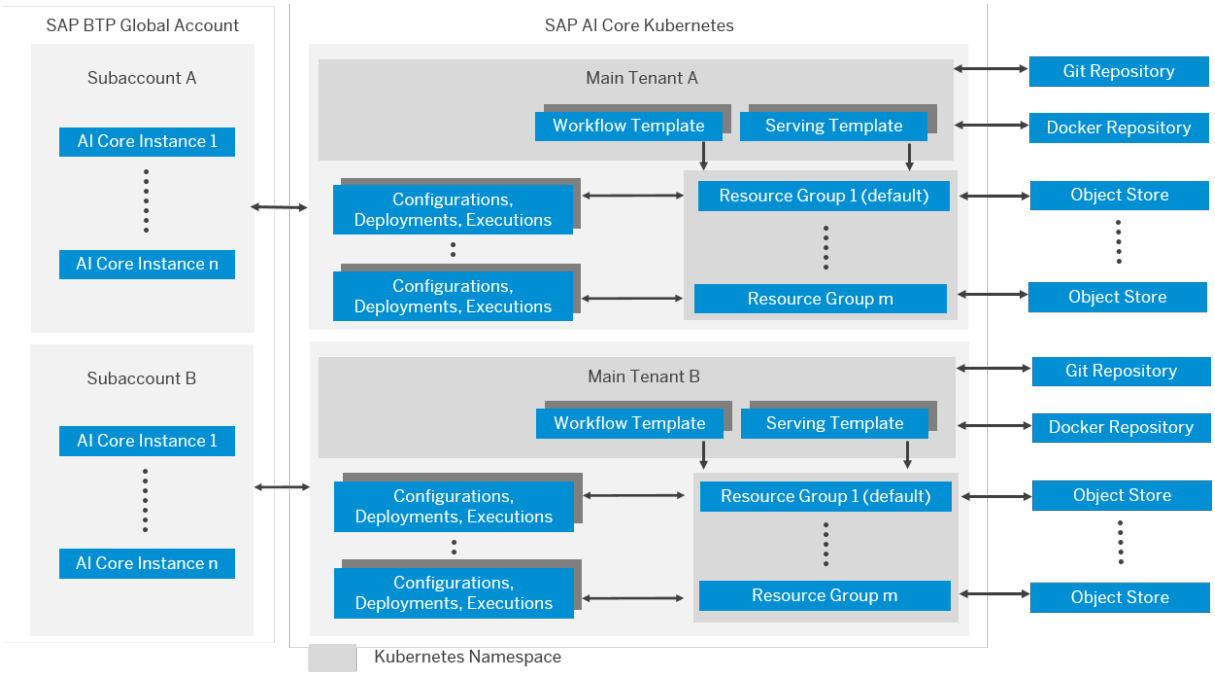
To avoid breaking the business functionality, SAP AI Core does not sanitize any user input. Consumers or applications that consume the AI API are expected to perform necessary output encoding based on the usage context to prevent XSS attacks.

Related Information

Cross Site Scripting (XSS) – OWASP Site
Cross Site Scripting Prevention Cheat Sheet – OWASP Site

10.10 Multitenancy

SAP AI Core is a tenant-aware BTP reuse service, supporting main tenants and resource groups. Resources are defined for tenants or resource groups as outlined below:



Each main tenant and resource group is mapped to a namespace. The main tenant namespace only contains templates for workflows and model serving. The instances of these objects are created in the respective resource group namespaces and reference the corresponding templates in the main tenant namespace. Each main tenant has a default resource group, which can be used for workloads from the main tenant.

Tenant-Level Resources

Tenant-level resources include executables such as:

- Workflow templates
- Serving templates
- Docker registries that contain Docker images
- User authentication and authorization (UAA)

User authentication and authorization are based on the SAP AI Core tenant (access token obtained using the service key for SAP AI Core). At runtime or when managing the lifecycle via AI API, the SAP AI Core tenant must set the appropriate resource group in the request header.

Resource-Group-Level Resources

Executables at the tenant level are shared across all resource groups. In contrast, runtime entities such as executions, deployments, configurations, and artifacts belong to a specific resource group and cannot be shared across resource groups. Similarly, generic secrets created within a resource group can be used only for workloads within that group.

You can register an object store at the resource-group level by setting the resource group header. We recommend that you do not use the same object store bucket with the same IAM user for multiple resource groups.

Tenant Isolation of Workloads

Workloads run in a sandbox environment and cannot access workflows of other tenants or resource groups. Only TCP is supported for inbound or outbound traffic from a workload. Opening workloads to open Sockets on UDP ports is strongly discouraged. They are not usable, but may pose a theoretical security problem for the workload.

10.11 Auditing and Logging Information

Here you can find a list of the security events that are logged by SAP AI Core.

Security Events Written in Audit Logs

What Events Are Logged	How to Identify Related Log Events
Creation of object store secret	Message containing a time stamp, tenant IDs, and attributes containing the API call and `"new": "Succeeded"`.
Deletion of object store secret	Message containing a time stamp, tenant IDs, and attributes containing the API call and `"new": "Succeeded"`.
Successful retrieval of object store secret	Successfully read object store connection details for resource group `<resource group name>` and object store `<object store name>`
Provisioning of resource group	Message containing a time stamp, tenant IDs, and attributes containing the API call and `"new": "Succeeded"`.
Deprovisioning of resource group	Message containing a time stamp, tenant IDs, and attributes containing the API call and `"new": "Succeeded"`.
Creation of docker registry secret	Message containing a time stamp, tenant IDs, and attributes containing the API call and `"new": "Succeeded"`.

What Events Are Logged	How to Identify Related Log Events
Deletion of docker registry secret	Message containing a time stamp, tenant IDs, and attributes containing the API call and `"new": "Succeeded"`.
Creation of deployments	Message containing a time stamp, tenant IDs, and attributes containing the API call and `"new": "Succeeded"`.
Deletion of deployments	Message containing a time stamp, tenant IDs, and attributes containing the API call and `"new": "Succeeded"`.
Creation of executions	Message containing a time stamp, tenant IDs, and attributes containing the API call and `"new": "Succeeded"`.
Deletion of executions	Message containing a time stamp, tenant IDs, and attributes containing the API call and `"new": "Succeeded"`.
Creation of ArgoCD application	Message containing a time stamp, tenant IDs, and attributes containing the API call and `"new": "Succeeded"`.
Deletion of ArgoCD application	Message containing a time stamp, tenant IDs, and attributes containing the API call and `"new": "Succeeded"`.
Creation of repositories	Message containing a time stamp, tenant IDs, and attributes containing the API call and `"new": "Succeeded"`.
Deletion of repositories	Message containing a time stamp, tenant IDs, and attributes containing the API call and `"new": "Succeeded"`.
Provisioning of tenant	Message containing a time stamp, tenant IDs, and attributes containing `"name": "state",`, `"new": "PENDING"` and `"name": "cfAccountState`, `"new": "ACTIVE"` and success value `true`
Retrieval of tenant	Message containing `Attribute <attribute detail> was read`.
Deprovisioning of tenant	Message containing a time stamp, tenant IDs, and attributes containing `"name": "state",`, `"new": "PENDING"` and `"name": "cfAccountState`, `"new": "DELETED"` and success value `true`.
List Get Watch	Message containing a time stamp, tenant IDs, source IPs, the request URI, and the level.
Create Update Patch	Message containing a time stamp, tenant IDs, source IPs, the request URI, the level, and the request and response objects.

What Events Are Logged	How to Identify Related Log Events
Delete	Message containing a time stamp, tenant IDs, source IPs, the request URI, the level, the and response object.
Token expired	`Jwt is expired`
No authorization header	`RBAC: access denied`
Invalid token	`Jwt issuer is not configured`
Wrong `tenantId`	`Jwt verification fails`

The following information is described in the table columns:

- *Event grouping* - Events that are logged with a similar format or are related to the same entities.
- *What events are logged* - Description of the security or data protection and privacy related event that is logged.
- *How to identify related log events* - Search criteria or key words, that are specific for a log event that is created along with the logged event.
- *Additional information* - Any related information that can be helpful.

Related Information

Audit Logging in the Cloud Foundry Environment
Audit Logging in the Neo Environment

10.12 Data Protection and Privacy

For general information about data protection and privacy on SAP Business Technology Platform, see Data Protection and Privacy.

Data protection is associated with numerous legal requirements and privacy concerns. In addition to compliance with general data protection and privacy acts, it is necessary to consider compliance with industry-specific legislation in different countries/regions. This section describes the specific features and functions that SAP AI Core provides to support compliance with the relevant legal requirements and data privacy.

This guide does not give advice on whether these features and functions are the best method to support company, industry, regional, or country/region-specific requirements. Furthermore, this guide does not give advice or recommendations about additional features that would be required in a particular environment. Decisions related to data protection must be made on a case-by-case basis and under consideration of the given system landscape and the applicable legal requirements.

> ### Note
>
> SAP does not provide legal advice in any form. SAP software supports data protection compliance by providing security features and specific data protection-relevant functions, such as simplified blocking and

deletion of personal data. In many cases, compliance with applicable data protection and privacy laws will not be covered by a product feature. Definitions and other terms used in this document are not taken from a particular legal source.

The extent to which data protection is ensured depends on secure system operation. Network security, security note implementation, adequate logging of system changes, and appropriate usage of the system are the basic technical requirements for compliance with data privacy legislation and other legislation.

For a glossary of Data Protection and Privacy terms in SAP BTP, see the SAP BTP Glossary for Data Protection and Privacy.

10.12.1 Data Storage and Processing

SAP AI Core provides functionality that allows you to process data, such as configuration files or Machine Learning (ML) Training or ML Serving.

SAP AI Core acts as the data processor and is not aware of the type of data or category of data. SAP AI Core customers, as Data Controllers, are responsible for fulfilling data protection and privacy (DPP) responsibilities for data storage and processing requirements.

10.12.2 Change Logging and Read-Access Logging

SAP AI Core acts as the data processor and is not aware of the type of data or category of data. SAP AI Core customers, as Data Controllers, are responsible for fulfilling data protection and privacy (DPP) responsibilities for data storage and processing requirements.

For any applications or services you develop using SAP AI Core, you must ensure that they include relevant logging functions, and ensure compliance with the data privacy laws by making sure that the data is properly logged.

10.12.3 Consent

SAP AI Core acts as the data processor and is not aware of the type of data or category of data. SAP AI Core customers, as Data Controllers, are responsible for fulfilling asking for consent from data subjects before collecting any personal data.

10.12.4 Deletion

SAP AI Core supports **Bring your object store**, whereby customers register the object store secret where artifact relevant files (such as training data or machine learning models or other types) are stored. Such data is used by the ML workloads during processing, such as ML Training or ML Serving.

The notion of artifacts is limited to capture the metadata of the data. Data is physically stored in the object store and SAP AI Core is not responsible for deletion of files. When artifacts are deleted using AI API, corresponding metadata are deleted from SAP AI Core service and the actual files are not deleted from the object store. For more information about the AI API, see AI API Overview [page 120].

Upon offboarding, SAP AI Core will clean up the cached data within AI Core used for processing purposes.

SAP AI Core customers, as Data Controllers, are responsible for deletion of data from the registered object store.

10.12.5 Security and Customer Data Protection

SAP product standard security and the data protection and privacy (DPP) requirements set high standards and obligations when it comes to securing and protecting customer data that is entrusted to SAP.

Customer data protection is handled in three ways:

- Customer data is imported, output, and processed by the services for no purpose beyond that to which the customer has subscribed.
- Customer data is protected from malicious access by security technologies that include authentication and authorization.
- Customer data is protected from accidental exposure to SAP administrators or support persons by security policies, access controls, and monitoring.

11 Accessibility Features in SAP AI Core

To optimize your experience of SAP AI Core, SAP AI Core provides features and settings that help you use the software efficiently.

> **Note**
>
> SAP AI Core uses SAP AI Launchpad for its interface, which is based on SAPUI5. For this reason, accessibility features for SAPUI5 also apply. See the accessibility documentation for SAPUI5 on SAP Help Portal at Accessibility for End Users.

For more information on-screen reader support and keyboard shortcuts, see Screen-Reader Support for SAPUI5 Controls and Keyboard Handling for SAPUI5 Elements.

12 Monitoring and Troubleshooting

Explore solutions to potential issues, and find out how to get support.

Getting Support

If you encounter an issue with this service, we recommend that you follow the procedure below.

Check Platform Status

Check the availability of the platform at SAP Trust Center.

For more information about platform availability, updates and notifications, see Platform Updates and Notifications in the Cloud Foundry Environment.

Contact SAP Support

You can report an incident or error through the SAP Support Portal.

Please use the following component(s) for your incident:

Component Name	Component Description
`CA-ML-AIC`	SAP AI Core

We recommend that you include the following information when you submit the incident:

- Region information
- Subaccount technical name
- URL of the page where the incident or error occurs
- Steps or clicks used to replicate the error
- Screenshots, videos, or the code entered

12.1 Troubleshooting

For troubleshooting information, see the following sections:

Repository [page 176]

Configuration [page 177]

Artifacts [page 179]

Application [page 181]

Execution [page 186]

Docker [page 188]

Deployment [page 189]

Miscellaneous [page 190]

12.1.1 Repository

Repository `ra-aicore-test` not found for tenant

You get the result:

```
{
    "error":{
        "code": "500",
        "details": [
            {
                "code": null,
                "message": " "Repository ra-aicore-test not found for tenant
b82a8318"
            }
        ],
        "message": "Repository ra-aicore-test not found for tenant b82a8318"
        "request_id": null,
        "target": "/api/v4alpha/repositories"
    }
}
```

and:

```
AIAPIServerException: Failed to post /admin/repositories: Repository ra-aicore-
test not found for tenant 68
```

Follow the solution:

Use a different name for the value of the `name` parameter. The exception is raised by reuse of the name `aicore-test`.

```
response = ai_api_client.rest_clinet.post(
    path="/admin/repositories",
    body={
        "name": "aicore-test-1",
        "url": "https://github.com/John/aicore-test",
    }
)
print(response)
{'message': 'Repository has been on-boarded.'
```

Parent topic: Troubleshooting [page 175]

Related Information

Configuration [page 177]
Artifacts [page 179]
Application [page 181]
Execution [page 186]
Docker [page 188]
Deployment [page 189]
Miscellaneous [page 190]

12.1.2 Configuration

Could not create configuration, because executable <x> for scenario <y> is not found

When you try to create a configuration, you are told that an executable cannot be found for your scenario.

Check the following:

1. Check that you are using the `name` value from your workflow for the executable ID in the configuration.

```
apiVersion: argoproj.io/v1alpha1
kind: WorkflowTemplate
metadata:
        name: text-clf-train-tutorial
        annotations:
            scenarios.ai.sap.com/description: "SAP developers tutorial
scenario"
...
```

> ⓘ Note
>
> Do not use the value from `executables.ai.sap.com/id` as an executable ID.

2. Check that you are using the value of `executables.ai.sap.com/id` from your workflows as your scenario ID.

```
...
        artifacts.ai.sap.com/text-data.kind: "dataset"
        artifacts.ai.sap.com/text-model-tutorial.kind: "model"
    labels:
        scenarios.ai.sap.com/id: "text-clf-tutorial"
        ai.sap.com/version: "2.1.0"
    spec:
...
```

Log message: using minio client

Check the following:

1. Check that you are using the `name` value from your workflow for the executable ID in the configuration.

```
apiVersion: argoproj.io/v1alpha1
kind: WorkflowTemplate
metadata:
        name: text-clf-train-tutorial
        annotations:
            scenarios.ai.sap.com/description: "SAP developers tutorial
scenario"
...
```

> ### Note
>
> Do not use the value from `executables.ai.sap.com/id` as an executable ID.

2. Check that you are using the value of `executables.ai.sap.com/id` from your workflows as your scenario ID.

```
...
        artifacts.ai.sap.com/text-data.kind: "dataset"
        artifacts.ai.sap.com/text-model-tutorial.kind: "model"
    labels:
        scenarios.ai.sap.com/id: "text-clf-tutorial"
        ai.sap.com/version: "2.1.0"
    spec:
...
```

Parent topic: Troubleshooting [page 175]

Related Information

Repository [page 176]
Artifacts [page 179]
Application [page 181]
Execution [page 186]
Docker [page 188]
Deployment [page 189]
Miscellaneous [page 190]

12.1.3 Artifacts

No output artifact has been generated

Complete the following:

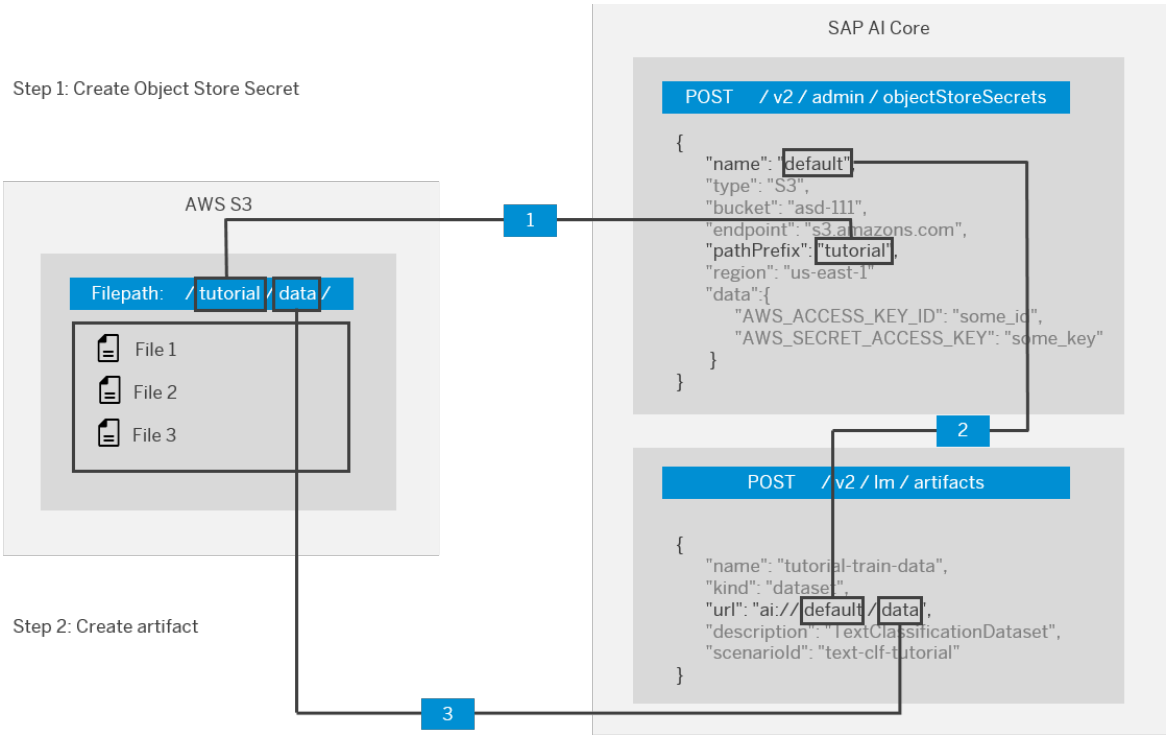
Define the `globalName` parameter for the output artifact in your workflow:

```
...
    executables.ai.sap.com/description: "Text classification Scikit training
executable"
    executables.ai.sap.com/name: "text-clf-train-tutorial-exec"
    artifacts.ai.sap.com/text-data.kind: "dataset"
    artifacts.ai.sap.com/text-model-tutorial.kind: "mind"
labels:
    scenarios.ai.sap.com/id: "text-clf-tutorial"
    ai.sap.com/version: "2.1.0"
...
...
        outputs:
            artifacts:
                -name: text-model-tutorial
                path: /app/model
                global name: text-model-tutorial
                archive:
                    none: {}
        container:
...
```

Failed to load artifacts: The specified key does not exist

Complete the following:

1. Ensure you have created an object store secret using the naming convention `<name>` and the `pathPrefix` from your AWS S3 path. Refer to the following diagram:



2. When creating the artifact, don't add the trailing forward slash (/) in URL parameter:
   - Incorrect usage: `"url": "ai://yourObjectStoreSecretName/folder/subfolder/"`
   - Correct usage: `"url": "ai://yourObjectStoreSecretName/folder/subfolder"`

Parent topic: Troubleshooting [page 175]

Related Information

Repository [page 176]
Configuration [page 177]
Application [page 181]
Execution [page 186]
Docker [page 188]
Deployment [page 189]
Miscellaneous [page 190]

12.1.4 Application

Templates are not synced via applications

After you have deleted or created an application, templates are not synced.

Follow the solution:

1. Delete the synced application using the endpoint:
   `DELETE {{apiurl}}/v2/admin/applications/{{appName}}`
2. Offboard the associated GitHub repository using the endpoint:
   `DELETE {{apiurl}}/v2/admin/repositories/{{repositoryName}}`
3. Onboard the associated GitHub repository using a personal access token instead of your GitHub password. For more information, see Creating a personal access token .
   `POST {{apiurl}}/v2/admin/repositories`

   Sample Code

   Body:

   ```
   {
   "name": "aicore-test",
    "url": "https://github.com/john/aicore-test",
    "username": "john",
    "password": "yourGitHubPersonalAccessToken"
   }
   ```
4. Create the application using the endpoint:
   `POST {{apiurl}}/v2/admin/applications`
5. Check the ArgoCD applications to determine if the repository has been synchronized correctly for the tenant. For example, check that there are no duplicated workflow names. The value of the `name` parameter is considered as an executable ID.

   ```
   name: text-clf-train-tutorial
           annotations:
                ...apiVersion: argoproj.io/v1alpha1
   kind: WorkflowTemplate
   metadata:
   ```
6. Check that you're calling SAP AI Core using the expected tenant.
7. Check if the workflow templates contain the correct scenario label.
8. Get your application sync status using the endpoint:
   `GET {{apiurl}}/v2/admin/applications/{{appName}}/status`
   The status will any return errors in your templates. When your templates are updated, the application will resync automatically after approximately three minutes.

Executables do not appear when you retrieve them from a scenario

Follow the solution:

1. Delete the synced application using the endpoint:
   `DELETE {{apiurl}}/v2/admin/repositories/{{appName}}`
2. Offboard the associated GitHub repository using the endpoint:
   `DELETE {{apiurl}}/v2/admin/repositories/{{repositoryName}}`
3. Onboard the associated GitHub repository using a personal access token instead of your GitHub password. For more information, see Creating a personal access token .
   `POST {{apiurl}}/v2/admin/repositories`

   Sample Code

   Body:

   ```
   apiVersion: argoproj.io/v1alpha1{
   "name": "aicore-test",
    "url": "https://github.com/john/aicore-test",
    "username": "john",
    "password": "yourGitHubPersonalAccessToken"
   }
   ```
4. Create the application using the endpoint:
   `POST {{apiurl}}/v2/admin/applications`
5. Check the ArgoCD applications to determine if the repository has been synchronized correctly for the tenant. For example, check that there are no duplicated workflow names. The value of the `name` parameter is considered as an executable ID.

   ```
   apiVersion: argoproj.io/v1alpha1
   kind: WorkflowTemplate
   metadata:
       name: text-clf-train-tutorial
           annotations:
               ...
   ```
6. Check if the user is calling SAP AI Core using the expected tenant.
7. Check that the scenario label is correct in the workflow templates.
8. Get your application sync status using the endpoint:
   `GET {{apiurl}}/v2/admin/applications/{{appName}}/status`
   The status will any return errors in your templates. When your templates are updated, the application will resync automatically after approximately three minutes.

Application status returns `healthy`, but most other properties are `unknown`

Follow the solution:

1. Delete the synced application using the endpoint:
   `DELETE {{apiurl}}/v2/admin/applications/{{appName}}`
2. Offboard the associated GitHub repository using the endpoint:
   `DELETE`
   `{{apiurl}}/v2/admin/repositories/{{repositoryName}}`

3. Onboard the associated GitHub repository using a personal access token instead of your GitHub password. For more information, see Creating a personal access token .
   `POST {{apiurl}}/v2/admin/repositories`

   Sample Code

   Body:

   ```
   {
   "name": "aicore-test",
    "url": "https://github.com/john/aicore-test",
    "username": "john",
    "password": "yourGitHubPersonalAccessTokenHere"
   }
   ```

4. Create the application using the endpoint:
   `POST {{apiurl}}/v2/admin/applications`
5. Check the ArgoCD applications to determine if the repository has been synchronized correctly for the tenant. For example, check that there are no duplicated workflow names. The value of the `name` parameter is considered as an executable ID.

   ```
   apiVersion: argoproj.io/v1alpha1
   kind: WorkflowTemplate
   metadata:
       name: text-clf-train-tutorial
           annotations:
               ...
   ```

6. Check if the user is calling SAP AI Core using the expected tenant.
7. Check if the workflow templates contain the correct scenario label.
8. Get your application sync status using the endpoint:
   `GET {{apiurl}}/v2/admin/applications/{{appName}}/status`
   The status will any return errors in your templates. When your templates are updated, the application will resync automatically after approximately three minutes.

Application status message: `rpc error: code = Unknown desc = my-path: app path does not exist`

The specified path in your application doesn't exist in your repository.

Follow the solution:

Delete your application and create a new one using the correct path.

Application status message: `application repo <your git repository> is not permitted in project 'xyz'`

The repository URL cannot be found in your onboarded repositories.

Check the following:

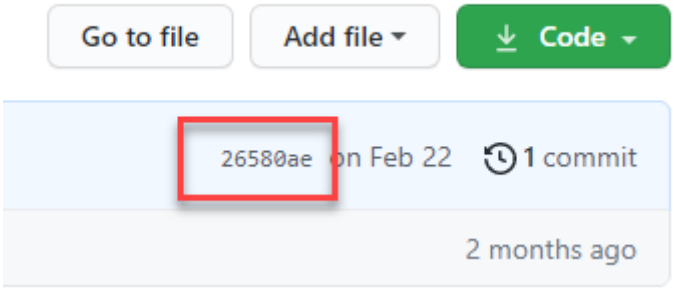
Make sure that the repository specified in your application has been successfully onboarded by using **GET** `{{apiurl}}/v2/admin/applications` and checking that the repository url has `"status": "COMPLETED"`

Application status message: `rpc error: code = Unknown desc = Unable to resolve 'not-existing-branch' to a commit SHA`

The revision you specified in your application doesn't exist in your repository.

Follow the solution:

Delete your application and create a new one using the correct revision. The revision number is found in GitHub here:



Alternatively, enter **HEAD** to refer to the latest commit.

Application status message: `rpc error: code = FailedPrecondition desc = Failed to unmarshal \"worklflow.yaml\": failed to unmarshal manifest: error converting YAML to JSON: yaml: line 7: mapping values are not allowed in this context`

You have a syntax error in your workflow template.

Follow the solution:

Use Argo Lint to identify syntax errors in your workflow template. To set up the Argo Lint IDE, refer to Argo Lint IDE setup

Application status message: `spec.source.repoURL and spec.source.path either spec.source.chart are required`

You specified an empty path in your application.

```
{
    "healthStatus": "Unknown",
```

```
    "message": "spec.source.repoURL and spec.source.path either
spec.source.chart are required",
    "reconciledAt": "Unknown",
    "source": {
        "path": "Unknown",
        "repoURL": "Unknown",
        "revision": "Unknown"
    },
    "syncFinishedAt": "Unknown",
    "syncRessourcesStatus": [],
    "syncStartedAt": "Unknown",
    "syncStatus": "Unknown"
}
```

Follow the solution:

Delete your application and create a new one, specifying a path. Check its status by using the endpoint: `{{apiurl}}/v2/admin/applications/{{appName}}/status`

Sync an Application Manually

Applications sync with your GitHub repository automatically at intervals of ~3 minutes. Use the endpoint below to manually request a sync:`{{apiurl}}/admin/applications/{{appName}}/refresh`

Parent topic: Troubleshooting [page 175]

Related Information

Repository [page 176]
Configuration [page 177]
Artifacts [page 179]
Execution [page 186]
Docker [page 188]
Deployment [page 189]
Miscellaneous [page 190]

12.1.5 Execution

Execution status is DEAD or PENDING for a long time

Check the following:

1. Check the execution logs.
2. Check that the parameter name and execution name that you provided match those in your template. Note that the names are case-sensitive.

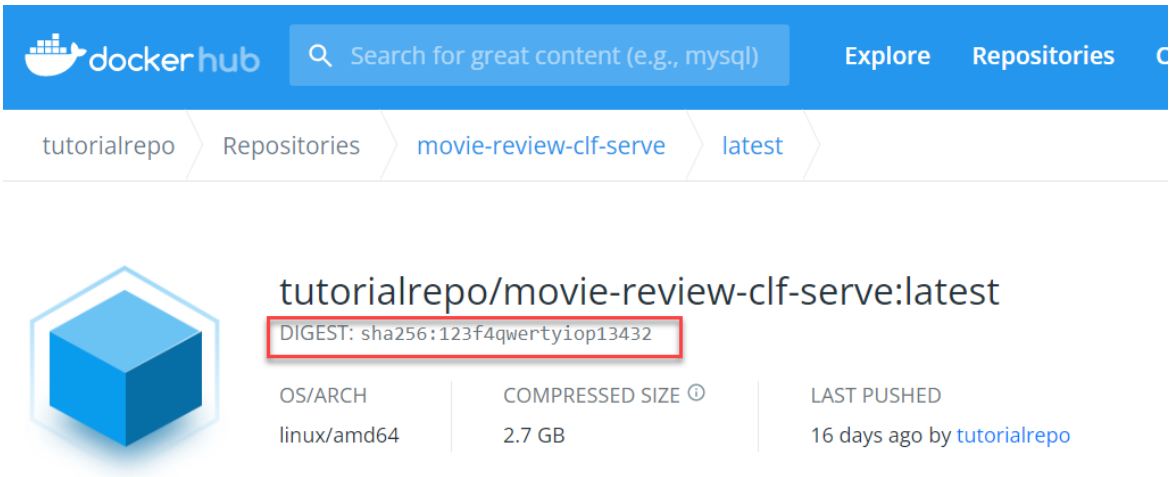
Images from private docker.io cannot be pulled for execution

Follow the solution:

1. Specify the domain name of your Docker Hub. For example:

```
...
                none: {}
        container:
            image: "docker.io/tutorialrepo/movie-review-clf-
serve@sha256:123f4qwertyiop13432"
            imagePullPolicy: Always
...
```

If you do not already know your domain name, refer to the image:



2. Specify your Docker Image digest instead of the image version in your templates:
image tutorialrepo/movie-review-clf-serve@sha256:123f4qwertyiop13432

Output artifacts in the execution list are empty and have status UNKNOWN for a long time

Check the following:

1. Check that an object secret with the name `default` exists in the current resource group. For more information, see Register an Object Store Secret
2. Check that you have created the Docker Registry secrets required to pull your Docker Images. Logs will be available only after the execution has started.

`GET` execution has the status UNKNOWN for a long time

Check the following:

1. Check if an object secret with the name `default` exists in the current resource group. For more information, see Register an Object Store Secret
2. Check that you have created the Docker Registry secrets required to pull your Docker Images. Logs will be available only after the execution has started.

Execution status changes to DEAD without any log

Check the following:

1. Check that your templates meet the Argo specifications and that they can be executed by SAP AI Core.
2. To identify errors automatically in your templates, use the Argo linter.
3. Check that you have created Docker folders, to store artifacts that will be created during an execution.

Parent topic: Troubleshooting [page 175]

Related Information

Repository [page 176]
Configuration [page 177]
Artifacts [page 179]
Application [page 181]
Docker [page 188]
Deployment [page 189]
Miscellaneous [page 190]

12.1.6 Docker

Error pulling container main logs. Back-off pulling image

You push a template to SAP AI Core and you get the error `"Error pulling container main logs". "Back-off pulling image"`.

Complete the following steps:

1. Make sure that your Docker Registry is public-facing and not protected behind the firewall of your organization.
   If your Docker Image isn't public, verify that you have created a Docker Registry secret in SAP AI Core.
2. Check that you can pull your Docker Image on your local computer using the credentials from the previous step.
3. Specify the same Docker Registry secret in your executable.
4. Specify the path to your Docker Image in your executable in following format:
   `<DOCKER_REGISTRY>/<REPO_NAME>/<DOCKER_IMAGE>:<TAGNAME>`

   > Example
   >
   > `docker.io/tutorialrepo/text-clf-train:0.0.1`

Parent topic: Troubleshooting [page 175]

Related Information

Repository [page 176]
Configuration [page 177]
Artifacts [page 179]
Application [page 181]
Execution [page 186]
Deployment [page 189]
Miscellaneous [page 190]

12.1.7 Deployment

You want to force a deployment with status UNKNOWN to stop

You want to stop and delete a deployment but you are unable to because the deployment status is “Unknown”. You have tried to submit a PATCH request as follows:

```
PATCH {{apiurl}}/lm/deployments/d4fec9c24c54f87e
```

However, you receive the following response:

```
{
    "error": {
        "code": "01010076"
        "message": "Invalid Request, Current status UNKNOWN cannot be changed..",
        "requestID": "e110820e-1cfe-456a-bb0e-77907b36422c",
        "target": "/apu/v2/deployments/d4fec9c24c54f87e"
    }
```

Complete the following steps:

1. Find out why your deployment status is “Unknown” by using the endpoint:
   `GET $AI_API_URL/v2/lm/deployments/<deploymentid>`
2. Delete the deployment without trying to stop it (stopping a deployment is necessary only when it's running):
   `DELETE $AI_API_URL/v2/lm/deployments/<deploymentid>`

Deployment remains in status PENDING

Check the following:

1. Check that the Docker Registry secret exists when using your private Docker Image.
2. Check that your Docker Image can be downloaded to your local system.

Deployment ID <abc> not found

This message appears when you have just started the deployment. Wait a few minutes and the message will resolve itself automatically.

Parent topic: Troubleshooting [page 175]

Related Information

Repository [page 176]
Configuration [page 177]
Artifacts [page 179]
Application [page 181]
Execution [page 186]
Docker [page 188]
Miscellaneous [page 190]

12.1.8 Miscellaneous

403 - forbidden: RBAC Access denied

When you submit a POST request for an execution or a configuration, you get the error `403 - forbidden: RBAC Access denied`.

Check the following:

1. Check that you are passing the correct `token` and `AI-Resource-Group header`.
2. Check your tenant provisioning.

You get a runtime adapter exception

You get the error: `"Runtime Adapter Exception; Failed to post deployments : {\n \"code\": \"400\",\n \"message\": \"Missing input parameter or artifacts, one or more placeholder values are not resolved in the serving spec and error is 'dict object' has no attribute ''.\"\n}\n"`

Check the following:

Check that your input artifact key doesn't contain any separators, such as “example-artifact-key”. If it does, rename the key (for example, “exampleArtifactKey”)

You have pushed your template to your GitHub repository but you can't see the executable created via the API

There may be an error in your template. Check the following:

- Serving templates always expect an input parameter to be configured. If you don't have a parameter in your Serving template, add a dummy parameter and create a configuration for it.
- Hyphens (-) are the only separator as permitted in the template name.

You have created a scenario in a template but you cannot see it in the AI API calls

Use the following solution:

Add a workflow template with the new `scenario_id` (not a serving template) to make the scenario visible.

Error: getaddrinfo ENOTFOUND

Check the following:

1. Check that all environment variables match your SAP AI Core service keys. Specifically, check:
   - `auth_url`
   - `client_id`
   - `client_secret`
   - `apiurl`
2. Submit a GET request to the endpoint `{{apiurl}}/v2/admin/repositories`
3. If the issue persists, contact SAP support as described at Monitoring and Troubleshooting [page 175].

Git repo doesn't synchronize with SAP AI Core instances.

When you try to synchronize your git repository with SAP AI Core, the response shows empty fields.

Check the following:

1. Check that you are using a GitHub personal access token and not your GitHub password.

If you are already using a personal access token, proceed as follows:

1. Delete all UNKNOWN applications from your SAP AI Core instance using the endpoint:
   DELETE `{{apiurl}}/v2/admin/applications/{{appName}}`
2. Offboard your GitHub repo from SAP AI Core by calling the endpoint:
   DELETE `{{apiurl}}/v2/admin/repositories/{{repositoryName}}`
3. Generate a GitHub personal access token and use it to onboard your git repo to SAP AI Core as before. For more information, see Creating a personal access token .

4. Sync your application again.

Parent topic: Troubleshooting [page 175]

Related Information

Repository [page 176]
Configuration [page 177]
Artifacts [page 179]
Application [page 181]
Execution [page 186]
Docker [page 188]
Deployment [page 189]

13 Support Process

Explore solutions to potential issues, and find out how to get support.

Getting Support

If you encounter an issue with this service, we recommend that you follow the procedure below.

Check Platform Status

Check the availability of the platform at SAP Trust Center.

For more information about platform availability, updates and notifications, see Platform Updates and Notifications in the Cloud Foundry Environment.

Contact SAP Support

You can report an incident or error through the SAP Support Portal.

Please use the following component(s) for your incident:

Component Name	Component Description
`CA-ML-AIC`	SAP AI Core

We recommend that you include the following information when you submit the incident:

- Region information
- Subaccount technical name
- URL of the page where the incident or error occurs
- Steps or clicks used to replicate the error
- Screenshots, videos, or the code entered

14 Service Offboarding

Tenant offboarding occurs when a customer deletes a subaccount. SAP AI Core polls for the subaccount deletion event and performs the necessary deprovisioning and deletion activities.

> ⓘ Note
>
> Data and resources are not deleted when a service instance is deleted (because we don't isolate based on the service instance). If you want to keep the subaccount but still deprovision SAP AI Core, create a **medium** support ticket on component `CA-ML-AIC` with the title **`Service Offboarding`** and request that your data and resources be deleted manually.

> → Remember
>
> When you delete a resource group, Manage mTLS Certificate Secrets in that group are deleted. Certificates that were issued before the deletion may remain valid until they expire. You are responsible for removing or revoking trust on any external services that rely on those certificates.

Important Disclaimers and Legal Information

Hyperlinks

Some links are classified by an icon and/or a mouseover text. These links provide additional information.
About the icons:

- Links with the icon : You are entering a Web site that is not hosted by SAP. By using such links, you agree (unless expressly stated otherwise in your agreements with SAP) to this:
  - The content of the linked-to site is not SAP documentation. You may not infer any product claims against SAP based on this information.
  - SAP does not agree or disagree with the content on the linked-to site, nor does SAP warrant the availability and correctness. SAP shall not be liable for any damages caused by the use of such content unless damages have been caused by SAP's gross negligence or willful misconduct.
- Links with the icon : You are leaving the documentation for that particular SAP product or service and are entering an SAP-hosted Web site. By using such links, you agree that (unless expressly stated otherwise in your agreements with SAP) you may not infer any product claims against SAP based on this information.

Videos Hosted on External Platforms

Some videos may point to third-party video hosting platforms. SAP cannot guarantee the future availability of videos stored on these platforms. Furthermore, any advertisements or other content hosted on these platforms (for example, suggested videos or by navigating to other videos hosted on the same site), are not within the control or responsibility of SAP.

Beta and Other Experimental Features

Experimental features are not part of the officially delivered scope that SAP guarantees for future releases. This means that experimental features may be changed by SAP at any time for any reason without notice. Experimental features are not for productive use. You may not demonstrate, test, examine, evaluate or otherwise use the experimental features in a live operating environment or with data that has not been sufficiently backed up.
The purpose of experimental features is to get feedback early on, allowing customers and partners to influence the future product accordingly. By providing your feedback (e.g. in the SAP Community), you accept that intellectual property rights of the contributions or derivative works shall remain the exclusive property of SAP.

Example Code

Any software coding and/or code snippets are examples. They are not for productive use. The example code is only intended to better explain and visualize the syntax and phrasing rules. SAP does not warrant the correctness and completeness of the example code. SAP shall not be liable for errors or damages caused by the use of example code unless damages have been caused by SAP's gross negligence or willful misconduct.

Bias-Free Language

SAP supports a culture of diversity and inclusion. Whenever possible, we use unbiased language in our documentation to refer to people of all cultures, ethnicities, genders, and abilities.

www.sap.com/contactsap

© 2026 SAP SE or an SAP affiliate company. All rights reserved.

No part of this publication may be reproduced or transmitted in any form or for any purpose without the express permission of SAP SE or an SAP affiliate company. The information contained herein may be changed without prior notice.

Some software products marketed by SAP SE and its distributors contain proprietary software components of other software vendors. National product specifications may vary.

These materials are provided by SAP SE or an SAP affiliate company for informational purposes only, without representation or warranty of any kind, and SAP or its affiliated companies shall not be liable for errors or omissions with respect to the materials. The only warranties for SAP or SAP affiliate company products and services are those that are set forth in the express warranty statements accompanying such products and services, if any. Nothing herein should be construed as constituting an additional warranty.

SAP and other SAP products and services mentioned herein as well as their respective logos are trademarks or registered trademarks of SAP SE (or an SAP affiliate company) in Germany and other countries. All other product and service names mentioned are the trademarks of their respective companies.

Please see https://www.sap.com/about/legal/trademark.html for additional trademark information and notices.